

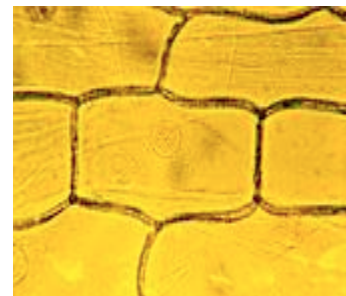
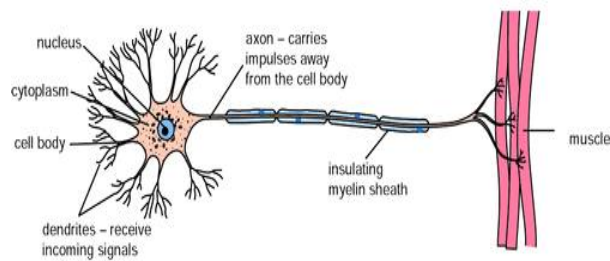
Name:

Form:

# ***MYP 1***

## ***Year 7***

### ***Cells and Reproduction***



| Statement of inquiry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Can we use advances in medicine and technology to improve the way organs and organs systems function?</p> <p>Organs work together as organ systems; each system carries out one life process, e.g. the digestive system breaks down and absorbs food while the circulatory system transports it.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                        |
| Inquiry questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Factual</b></p> <p>What are the features of living organisms?</p> <p>How does a microscope function?</p> <p>How can microscope slides be made easier to see?</p> <p>What are the levels of organisation in living organisms?</p> <p>What are the stages of development in cells (including examples of specialised cells)?</p> <p>What are the characteristics of stem cells?</p> <p>What are the stages involved in fertilization and pregnancy?</p> <p>Why do some couples have problems getting pregnant?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>A sperm fuses with an egg (fertilisation) to form a zygote; it divides into an embryo, implants in the uterus lining and develops into a fetus, born about 40 weeks later. Causes include too few or immobile sperm, eggs not released (hormone problem) or blocked oviducts. Treatments include fertility drugs, IVF and ICSI.</p> |
| <p><b>Conceptual</b></p> <p>How did the microscope enhance knowledge of cells and organs?</p> <p>How have cells evolved to fulfil specialised roles?</p> <p>How can we use our knowledge of unspecialised cells to treat disease?</p> <p>How might scientists apply knowledge of fertilisation to increase or decrease fertility rates?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Improved microscopes revealed cells and their structures, so scientists could link cell structure to the function of tissues, organs and organ systems, and discover microorganisms.</p>                                                                                                                                            |
| <p><b>Debatable</b></p> <p>What are the pros and cons of using stem cells?</p> <p>What factors should be considered before using stem cells therapy?</p> <p>What are the key factors to be considered when using reproductive technologies such as IVF?</p> <p>Living things carry out the seven life processes: movement, respiration, sensitivity, growth, reproduction, excretion and stimulation (MRS CGEN). The development of cells made to differentiate into the cells a patient needs, e.g. bone marrow cells for leukaemia. The development of cells made to differentiate into the cells a patient needs, e.g. bone marrow cells for leukaemia.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                        |
| Approaches to learning (ATL)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                        |
| <p>Destroyed, cells may be rejected or grow into tumours, and cost is high.</p> <p><b>ATL: CRITICAL THINKING SKILLS, Analysing and evaluating issues and ideas</b></p> <ul style="list-style-type: none"> <li>Consider ideas from multiple perspectives</li> <li>Develop contrary or opposing arguments</li> </ul> <p><b>ATL: COMMUNICATION SKILLS, Give and receive meaningful feedback</b></p> <p><b>ATL: COMMUNICATION, Take effective notes in class</b></p> <p><b>ATL: REFLECTION, Consider content</b></p> <p>Light passes through the specimen and the objective lens, which magnifies it; the eyepiece magnifies this image again to give the total magnification (eyepiece x objective).</p> <p>Use a very thin specimen, add a stain such as iodine or methylene blue, and lower the coverslip slowly so no air bubbles are trapped; adjust the light for contrast.</p> <p>Cell - tissue - organ - organ system - organism. Each level is made of the level below it, e.g. muscle tissue makes the heart, an organ of the circulatory system.</p> <p>A fertilised egg divides by mitosis to form many cells which then differentiate into specialised cells, e.g. red blood cells, nerve cells, muscle cells, ciliated cells and root hair cells.</p> <p>Stem cells are unspecialised cells that can divide many times by mitosis and can differentiate into other types of cell.</p> |                                                                                                                                                                                                                                                                                                                                        |

## 7P MYP Command Terms

| Term             | Definition                                                                                                                                                                                                                         |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analyse</b>   | Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.                                                                      |
| <b>Apply</b>     | Use knowledge and understanding in response to a given situation or real circumstances                                                                                                                                             |
| <b>Describe</b>  | Give a detailed account or picture of a situation, event, pattern or process                                                                                                                                                       |
| <b>Design</b>    | Produce a plan, simulation or model                                                                                                                                                                                                |
| <b>Discuss</b>   | Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence                                        |
| <b>Document</b>  | Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography |
| <b>Evaluate</b>  | Make an appraisal by weighing up the strengths and limitations                                                                                                                                                                     |
| <b>Explain</b>   | Give a detailed account                                                                                                                                                                                                            |
| <b>Formulate</b> | Express precisely and systematically the relevant concept(s) or argument(s)                                                                                                                                                        |
| <b>Interpret</b> | Use knowledge and understanding to recognize trends and draw conclusions from given information                                                                                                                                    |
| <b>Outline</b>   | Give a brief account                                                                                                                                                                                                               |
| <b>Present</b>   | Offer for display, observation, examination or consideration                                                                                                                                                                       |
| <b>Recall</b>    | Remember or recognize from prior learning experiences                                                                                                                                                                              |
| <b>Select</b>    | Choose from a list or group                                                                                                                                                                                                        |
| <b>Solve</b>     | Obtain the answer(s) using appropriate methods                                                                                                                                                                                     |
| <b>State</b>     | Give a specific name, value or other brief answer without explanation or calculation                                                                                                                                               |
| <b>Suggest</b>   | Propose a solution, hypothesis or other possible answer                                                                                                                                                                            |
| <b>Summarize</b> | Abstract a general theme or major point(s)                                                                                                                                                                                         |

## 7B MYP Glossary Cells & Reproduction

### Population Growth / On the slide / Building a life

| Word                  | Meaning                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Population            | The number of a given animal or plant living in a place (eg. The number of human in the world)                                              |
| Fertility Rate        | The number of children born per female                                                                                                      |
| cell                  | The basic unit which living things are made of.                                                                                             |
| cell surface membrane | Controls what goes into and out of a cell.                                                                                                  |
| cell wall             | Tough wall around <u>plant</u> cells. Helps to support the cell.                                                                            |
| chlorophyll           | Green substance found inside chloroplasts.                                                                                                  |
| chloroplast           | Green disc containing chlorophyll. Found in plant cells. Where the plant makes food by photosynthesis.                                      |
| coverslip             | Thin piece of glass used to hold a specimen in place on a slide.                                                                            |
| cytoplasm             | Jelly inside a cell where the cell's activities happen.                                                                                     |
| magnification         | How much bigger a microscope makes something appear.                                                                                        |
| microscope            | Used to magnify small things.                                                                                                               |
| nucleus               | Controls what a cell does.                                                                                                                  |
| photosynthesis        | Process that plants use to make their own food. It needs light to work. Carbon dioxide and water are used up. Food and oxygen are produced. |
| slide                 | Glass sheet that a specimen is put on.                                                                                                      |
| specimen              | What you look at down a microscope.                                                                                                         |
| stain                 | Dye used to colour parts of a cell to make them easier to see.                                                                              |
| Large vacuole         | Storage space in plant cells.                                                                                                               |

**Population/Cells/Human organs/Organ parts/Seeing is believing**

| <b>Word</b>      | <b>Meaning</b>                                                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| brain            | Organ that controls what the body does.                                                                                                   |
| eyepiece lens    | Part of the microscope you look down.                                                                                                     |
| focusing wheel   | Wheel on a microscope that moves parts of the microscope to get the image into focus.                                                     |
| heart            | Organ that pumps blood.                                                                                                                   |
| image            | What you see down a microscope.                                                                                                           |
| intestine        | The small intestine is an organ used to digest and absorb food.<br>The large intestine is an organ that removes water from unwanted food. |
| kidneys          | Organs used to clean the blood and make urine.                                                                                            |
| leaf             | Plant organ used to make food by photosynthesis.                                                                                          |
| liver            | Organ used to make and destroy substances in our bodies.                                                                                  |
| lungs            | Organs used to take oxygen out of the air and put waste carbon dioxide into the air.                                                      |
| magnification    | How much bigger a microscope makes something appear.                                                                                      |
| microscope       | Used to magnify small things.                                                                                                             |
| objective lens   | Part of the microscope that is closest to what you are looking at.                                                                        |
| organ            | A large part of a plant or animal that does a very important job.                                                                         |
| photosynthesis   | Process that plants use to make their own food. It needs light to work.                                                                   |
| root             | Plant organ used to take water out of the soil.                                                                                           |
| root hair tissue | Found in roots. Takes in water from the soil.                                                                                             |
| skin             | Organ used for protection and feeling.                                                                                                    |
| slide            | Glass sheet that a specimen is put on.                                                                                                    |
| specimen         | What you look at down a microscope.                                                                                                       |
| stage            | Part of the microscope. You put slides on it.                                                                                             |
| stem             | Plant organ used to take water to the leaves and to support the leaves.                                                                   |
| stomach          | Organ used to store and break up food.                                                                                                    |
| tissue           | A group of cells of the same type working together to do a job. Organs are made of different tissues.                                     |
| xylem tissue     | Found in roots, stems and leaves. Transports water.                                                                                       |

**On the slide/Building a life**

| <b>Word</b>           | <b>Meaning</b>                                                                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| cell                  | The basic unit which living things are made of.                                                                                             |
| cell surface membrane | Controls what goes into and out of a cell.                                                                                                  |
| cell wall             | Tough wall around plant cells. Helps to support the cell.                                                                                   |
| chlorophyll           | Green substance found inside chloroplasts.                                                                                                  |
| chloroplast           | Green disc containing chlorophyll. Found in plant cells. Where the plant makes food by photosynthesis.                                      |
| coverslip             | Thin piece of glass used to hold a specimen in place on a slide.                                                                            |
| cytoplasm             | Jelly inside a cell where the cell's activities happen.                                                                                     |
| magnification         | How much bigger a microscope makes something appear.                                                                                        |
| microscope            | Used to magnify small things.                                                                                                               |
| nucleus               | Controls what a cell does.                                                                                                                  |
| photosynthesis        | Process that plants use to make their own food. It needs light to work. Carbon dioxide and water are used up. Food and oxygen are produced. |
| slide                 | Glass sheet that a specimen is put on.                                                                                                      |
| specimen              | What you look at down a microscope.                                                                                                         |
| stain                 | Dye used to colour parts of a cell to make them easier to see.                                                                              |
| vacuole               | Storage space in plant cells.                                                                                                               |

**Shaped for the job**

| <b>Word</b>              | <b>Meaning</b>                                                                                                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| adapted                  | When something has certain features to help it do a particular job. When the features of a cell help it do its job, the cell is said to be 'adapted' to its job. |
| cilia                    | Small hairs on some cells.                                                                                                                                       |
| ciliated                 | Having cilia.                                                                                                                                                    |
| ciliated epithelial cell | Cell with cilia found in the lungs.                                                                                                                              |
| muscle cell              | Cell that can change its length and so help us to move.                                                                                                          |
| nerve cell               | Cell that carries messages around the body.                                                                                                                      |
| neurone                  | Another name for a nerve cell.                                                                                                                                   |
| palisade cell            | Cell found in leaves which contains many chloroplasts.                                                                                                           |
| root hair cell           | Cell found in roots. It has a large surface area to help the cell absorb water quickly.                                                                          |
| tissue                   | A group of the same cells all doing the same job.                                                                                                                |
| xylem tube               | Hollow tube formed from xylem cells and used to carry water up a plant.                                                                                          |

**Reproduction**

| Keyword       | Definition                                                                                                                                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sperm cell    | <b>Sperm</b> is the male reproductive <b>cell</b>                                                                                                                                             |
| Egg cell      | Female reproductive cell.                                                                                                                                                                     |
| DNA           | the fundamental and distinctive characteristics or qualities of someone or something, especially when regarded as unchangeable.                                                               |
| Mitosis       | a type of cell division that results in two daughter cells each having the same number and kind of chromosomes as the parent nucleus, typical of ordinary tissue growth                       |
| Vagina        | the muscular tube leading from the external genitals to the cervix of the uterus in women and most female mammals.                                                                            |
| Cervix        | the narrow neck-like passage forming the lower end of the uterus.                                                                                                                             |
| Uterus        | the organ in the lower body of a woman or female mammal where offspring are conceived and in which they gestate before birth; the womb.                                                       |
| Ovary         | a female reproductive organ in which ova or eggs are produced, present in humans and other vertebrates as a pair.                                                                             |
| Oviduct       | the tube through which an ovum or egg passes from an ovary.                                                                                                                                   |
| Penis         | the male genital organ of higher vertebrates, carrying the duct for the transfer of sperm during copulation. In humans and most other mam                                                     |
| Fertilisation | the action or process of fertilizing an egg or a female animal or plant, involving the fusion of male and female gametes to form a zygote.                                                    |
| Implantation  | (in a mammal) the attachment of the fertilized egg or blastocyst to the wall of the uterus at the start of pregnancy.                                                                         |
| Gestation     | he process or period of developing inside the womb between conception and birth.                                                                                                              |
| IVF           | a medical procedure whereby an egg is fertilized by sperm in a test tube or elsewhere outside the body.                                                                                       |
| Puberty       | the period during which adolescents reach sexual maturity and become capable of reproduction.                                                                                                 |
| Adolescence   | the period following the onset of puberty during which a young person develops from a child into an adult.                                                                                    |
| Menstruation  | the process in a woman of discharging blood and other material from the lining of the uterus at intervals of about one lunar month from puberty until the menopause, except during pregnancy. |
| Condom        | a thin rubber sheath worn on a man's penis during sexual intercourse as a contraceptive or as a protection against infection.                                                                 |

|                    |                                                                                                                                                                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sterilisation      | surgery to make a person or animal unable to produce offspring.                                                                                                                                                                                        |
| Surrogacy          | the process of giving birth as a surrogate mother or of arranging such a birth.                                                                                                                                                                        |
| Caesarean section  | is a surgical procedure used to deliver a baby through incisions in the mother's abdomen and uterus                                                                                                                                                    |
| Placenta           | a flattened circular organ in the uterus of pregnant eutherian mammals, nourishing and maintaining the fetus through the umbilical cord.                                                                                                               |
| Umbilical cord     | a flexible cord-like structure containing blood vessels and attaching a human or other mammalian fetus to the placenta during gestation.                                                                                                               |
| Amniotic fluid     | the fluid surrounding a fetus within the amnion.                                                                                                                                                                                                       |
| Contraceptive pill | Combination <b>birth control pills</b> , also known as the <b>pill</b> , are oral <b>contraceptives</b> that contain estrogen and a progestin. Combination <b>birth control pills</b> suppress ovulation — keeping your ovaries from releasing an egg. |

7B 1a Can you name and define the 7 things all organisms have in common?



|             |             |                    |
|-------------|-------------|--------------------|
| 1. <b>M</b> |             |                    |
| 2. <b>R</b> | RESPIRATION |                    |
| 3. <b>S</b> |             |                    |
| 4. <b>G</b> |             | Increasing in size |
| 5. <b>R</b> |             |                    |
| 6. <b>E</b> |             |                    |
| 7. <b>N</b> |             |                    |



## 7B1b Can you deduce whether they are alive or not?



Elephant

- ☐ Movement
- ☐ Respiration
- ☐ Sensitivity
- ☐ Growth
- ☐ Reproduction
- ☐ Excretion
- ☐ Nutrition

ALIVE/NOT ALIVE



iPhone

- ☐ Movement
- ☐ Respiration
- ☐ Sensitivity
- ☐ Growth
- ☐ Reproduction
- ☐ Excretion
- ☐ Nutrition

ALIVE/NOT ALIVE



Fire

- ☐ Movement
- ☐ Respiration
- ☐ Sensitivity
- ☐ Growth
- ☐ Reproduction
- ☐ Excretion
- ☐ Nutrition

ALIVE/NOT ALIVE



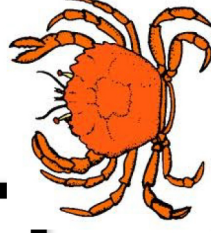
Zombie

- ☐ Movement
- ☐ Respiration
- ☐ Sensitivity
- ☐ Growth
- ☐ Reproduction
- ☐ Excretion
- ☐ Nutrition

ALIVE/NOT ALIVE



## 7B1b Can you deduce whether they are alive or not?



Elephant



iPhone



Fire



Zombie

The elephant is  
alive/ not alive  
because

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

The iPhone is  
alive/ not alive  
because

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

The fire is  
alive/ not alive  
because

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

The Zombie is  
alive/ not alive  
because

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_



7B1c Plants ARE alive too!

Can you prove it by explaining how they carry out MRS GREN



Sunflower

Movement =

Respiration =

Sensitivity =

Growth =

Reproduction =

Excretion =

Nutrition =



## 7B 1d Robert Hooke

### Activity Help

You are going to read the description of the life and work of Robert Hooke and then answer some questions.

Robert Hooke was one of the best scientists of the 17th century. One of his most famous inventions was a new kind of microscope.

It was different from earlier microscopes because it had three glass lenses to magnify the object, rather than two. This means that you get a clearer image of the object. It was called the **compound microscope**.

Although this microscope looked very different from modern ones, being over half a metre tall, the microscope in your classroom is based on the same design.

This new microscope allowed Robert Hooke to observe things people had never seen before. He looked at many things including insects, other tiny living things and feathers.

He is most famous for his description of a slice of cork which he noticed was divided into small sections which he called 'cells', because they reminded him of the rooms or cells in a monastery.

Robert Hooke recorded his observations in a book called *Micrographia*.

His observations of cork were very important because until then scientists had not realised that by looking at things closely you could see new things that could not be seen at a distance, such as cells.

- ① ~~Robert~~..... ~~Hooke~~..... invented the compound microscope.
- ② It was different from earlier microscopes because it had  
~~three glass lenses~~..... instead of .....
- ③ Robert Hooke looked at lots of things including  
~~insects, other tiny living things and feathers~~.....
- ④ He called the structures he saw in cork ~~cells~~..... .
- ⑤ His observations of cork were important because he noticed  
~~that by looking at things closely you could see new things that could not be seen at a distance, such as cells~~

## 7B 1e (Extension)- Robert Hooke and Cell Theory

Although Robert Hooke had discovered 'cells' in 1665, he did not understand the importance of his discovery. At that time, scientists knew that organisms were made of tissues and organs but they thought that these were not living. It was believed that only a whole organism was living, and that the parts that made it were not!

By the end of the eighteenth century, microscopes had improved so much that scientists were able to see cells in plants. Plant cells were much easier to see than animal cells because of their thick cell walls. Cell surface membranes, separating individual animal cells, were too thin to be seen. Microscopes continued to become more powerful and, in 1831, Robert Brown discovered a small, dark structure in each plant cell which he called a nucleus (after the Latin for 'small nut', which is what he thought it looked like).

In 1838 the importance of cells became clearer, when Mattias Schleiden came to the conclusion that all plants were made of cells. The next year another scientist, Theodor Schwann, discovered that all animals were also made from cells and that all living things started from one cell. However, he thought that new plant cells sprouted off from other cells. It was not until 1875 that Walther Fleming disproved this idea and discovered how cell division occurs, with a cell splitting into two, down the middle. Walther Fleming also developed the idea of staining cells so that he could see them better.

This century, scientists have discovered a great deal about cells and how they work, using the electron microscope which can see things in much more detail ever before.

- 1 In which year did Robert Hooke discover cells?

1665

- 2 a Explain what the word 'tissue' means today. A group of cells of the same type working together to do a job; organs are made of different tissues.

- b How does your answer to part a differ from what scientists thought in the seventeenth century?

In the 17th century scientists thought tissues and organs were not living, and that only the whole organism was living. Today we know tissue is a group of cells of the same type working together.

- 3 At the start of the second paragraph it says 'By the end of the eighteenth century, microscopes had improved so much ...' In what way do you think they 'improved'?

They had stronger lenses (higher magnification), so they could magnify more and see finer detail, such as individual plant cells.

- 4 Why did scientists find cells in plants before finding them in animals?

Plant cells have thick cell walls which were easy to see; animal cells are separated only by very thin cell surface membranes, which were too thin to be seen.

- 5 What job of cell surface membranes is referred to in the text?

Separating individual animal cells.

- 6 Who discovered the nucleus? Robert Brown.

- 7 When did Theodor Schwann say that animals were made up of cells? 1839.

- 8 Draw a diagram to show how Theodor Schwann thought a new plant cell would be made.

Draw a diagram showing a new plant cell sprouting off from a parent cell (Schwann thought new plant cells grew out of other cells).



- 9 Name a stain that Walther Fleming might have used. Iodine solution (or any stain such as methylene blue).

## 7B 2a What's inside an animal cell?

1. Use these words to draw arrows to label the correct part (proteins, membrane, nucleus, DNA, reactions, energy)

Mitochondria:

Power stations for the cell  is made here.

Ribosome: P

are made here.

N

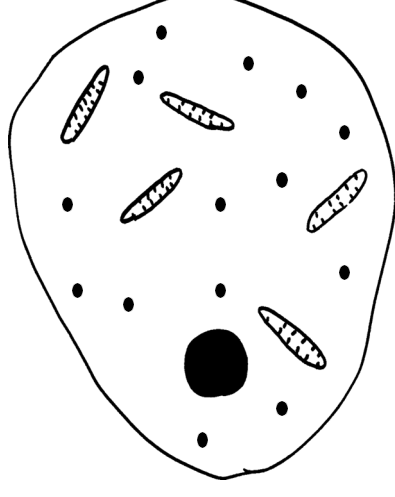
: This controls the cell's activities.

Cell m

membrane

: Controls what moves in and out of the cell.

· Draw arrows from each word (proteins, membrane, nucleus, DNA, reactions, energy) to the correct labelled part of the animal cell diagram.



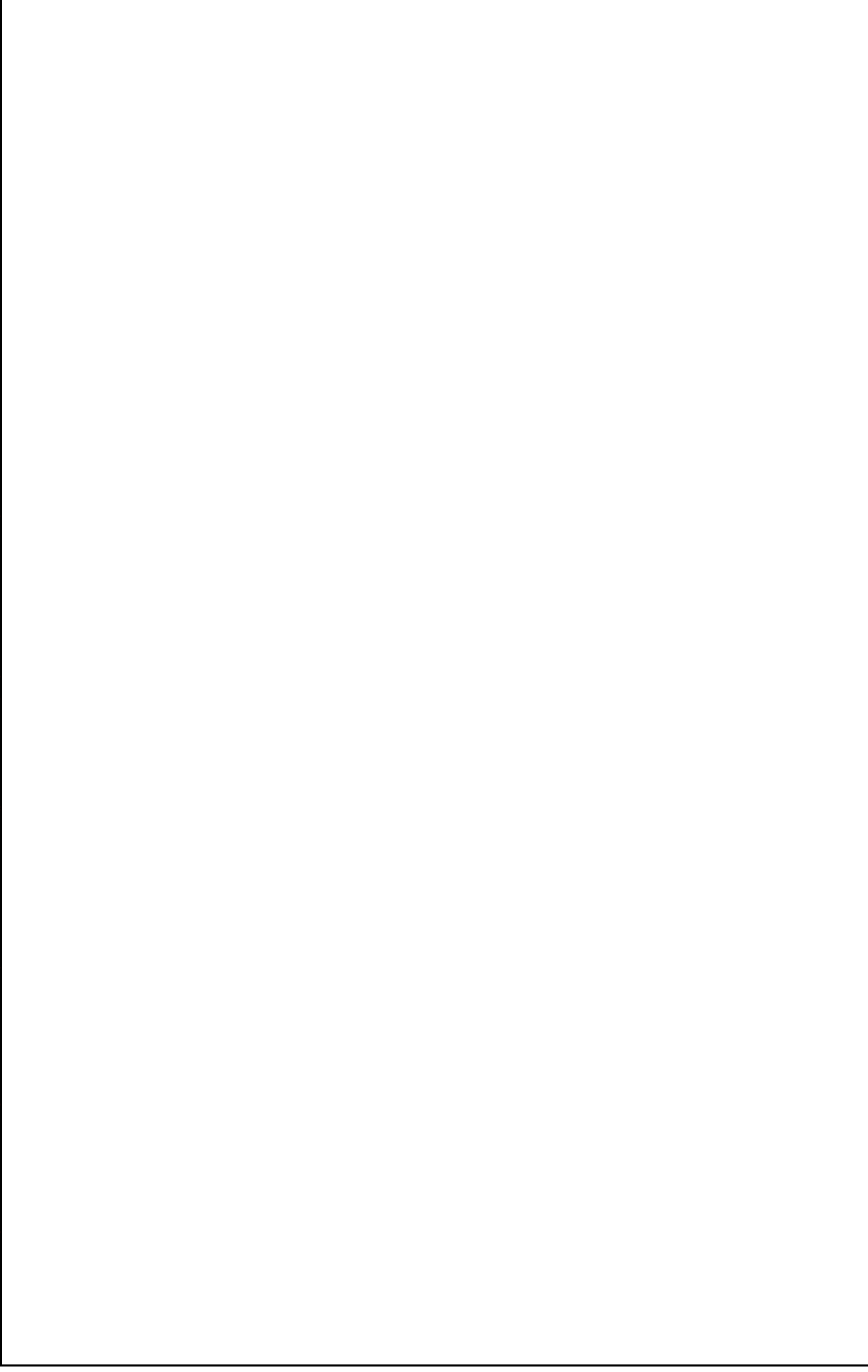
Cytoplasm: Where chemical r

reactions

: Controls what moves in and out of the cell.

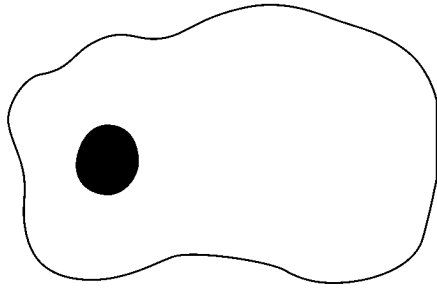
nucleus

2. Now draw and label a plant cell adding a cell wall, chloroplasts and a vacuole. (Please use the template below.)

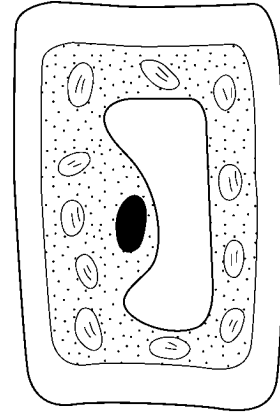


## 7B 2b Plant and Animal Cells

- 1 Draw lines from the words to the correct part of each cell. The words can be used more than once or not at all.



cell membrane  
cell wall  
chloroplasts  
cytoplasm  
nucleus  
large vacuole



· Draw lines from each word (cell membrane, cell wall, chloroplasts, cytoplasm, nucleus, large vacuole) to the correct part of each cell; words can be used more than once or not at all.

This is a diagram of an animal cell

This is a diagram of a plant cell

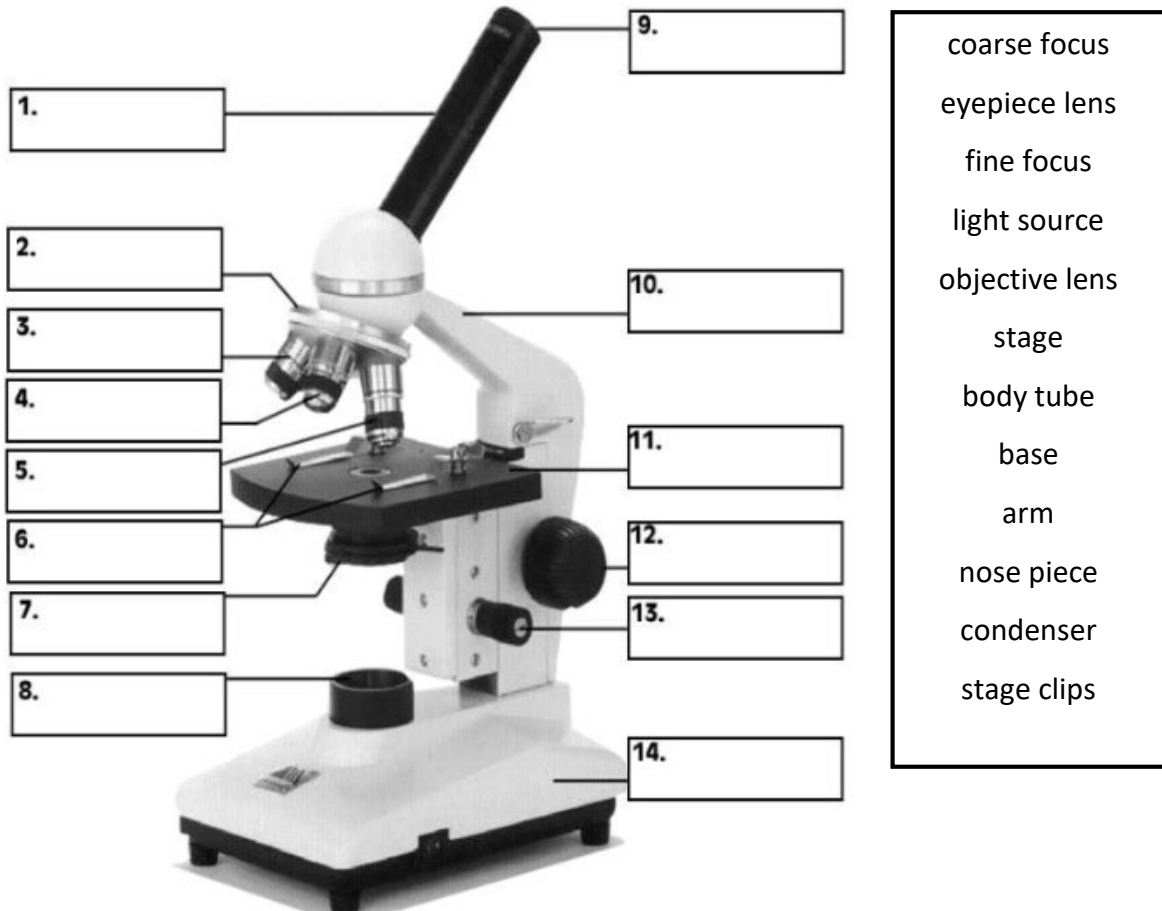
- 2 In the blank space in the sentence underneath each diagram, fill in the word **plant** or the word **animal**.
3. Draw lines from the labels above to the cells. **Watch out! Not all the cells have all those organelles!**
- 3 Draw lines from the parts of cells to their functions as listed below. The first one has been done.

· Draw lines from each cell part to its correct function, using the first completed line as an example.

| PART OF CELL          | FUNCTION                                                                 |
|-----------------------|--------------------------------------------------------------------------|
| Cell surface membrane | Tells the cell what to do.                                               |
| Chloroplasts          | Keeps the cell together and controls what goes into and out of the cell. |
| Nucleus               | A jelly-like substance. Many of the cell's activities happen here.       |
| Cell wall             | A storage space filled with cell sap.                                    |
| Cytoplasm             | Green discs that allow the plant to make food by photosynthesis.         |
| Vacuole               | Supports the cell.                                                       |

## 7B 2c The Microscope

1 Label the parts using words from the box.



· Label the microscope diagram using the words from the box: coarse focus, eyepiece lens, fine focus, light source, objective lens, stage, body tube, base, arm, nose piece, condenser, stage clips.

2 Complete the sentences;

a) We use a microscope to make objects look

(larger/ smaller).

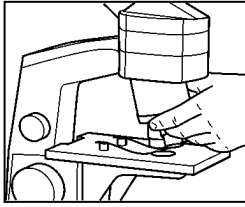
b) Why do we need a mirror or a light when using a microscope?

lenses.

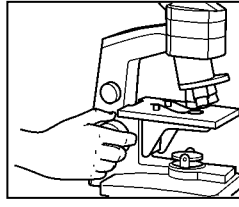
c) How do you calculate the magnification of a microscope?

Multiply the magnification of the eyepiece lens by the magnification of the objective lens.

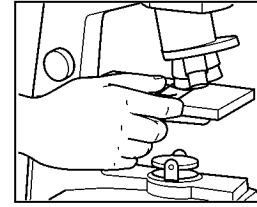
## 7B 2d Using a Microscope – the basics



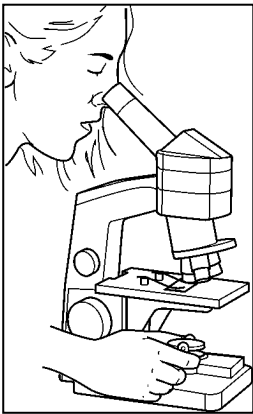
- 1 Put the smallest **objective lens** so that it is over the hole in the **stage**.



- 2 Turn the **coarse focusing wheel** to make the gap between the **stage** and the **objective lens** as small as possible.



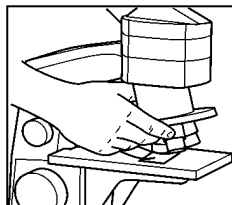
- 3 Place a **slide** on the **stage** and place the clips on the **slide** to hold it in place.



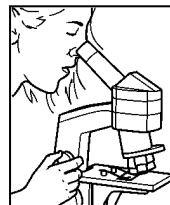
- 4 Adjust the **light source**. Either turn on the lamp or turn your mirror so that light is reflected up into the hole in the stage. *Never* point the mirror directly at the Sun. This can permanently damage your eyesight.
- 5 Look into the **eyepiece lens**. It is best to look down your microscope with both eyes open! If you do this your eyes will not get tired and you will see more.



- 6 Turn the **coarse focusing wheel** so that the gap between the **stage** and the **objective lens** gets bigger. Keep turning until what you see is clear (**in focus**).



- 7 To see a bigger image, place the next largest **objective lens** over your **specimen**.



- 8 Look down the microscope again and turn the **fine focusing wheel** a small amount one way to get your **image in focus** again. If this doesn't work, turn the **fine focusing wheel** a small amount the other way. Do *not* use the coarse focusing wheel. If you can't see anything, go back to a lower **magnification**.

### Looking after the microscope

- Always cover it up once you have used it. The glass in the lenses scratches easily – even dust will scratch it!
- Always use the 'fine focusing wheel' with the larger objective lenses, otherwise you may break your slide and damage the objective lens.
- Always place the smallest objective lens over the hole in the stage when changing slides.
- Always carry the microscope in two hands.
- Always carry the microscope upright or the eyepiece lens can fall out and break.

## 7B 2e Inside Cells – Extension

The modern microscope was invented in about 1590 by a Dutch spectacles maker called Zacharias Janssen. Since that time the microscope has helped scientists to discover what plants and animals are made up of and how cells work. A light microscope, like the ones most often used, can magnify things up to about x1500. Electron microscopes can magnify things up to about x1,500,000.

Using electron microscopes, scientists have been able to see what cytoplasm is made up of. It's not just jelly! It contains lots of very small parts called **organelles**. One of these organelles is the **mitochondrion**. Respiration happens inside the mitochondria. Other organelles are used to make new chemicals.

To measure the sizes of very small things, scientists have to use units that are smaller than millimetres. A **micrometre** (written ' $\mu\text{m}$ ') is 1/1000th of a millimetre; that is  $1\ \mu\text{m} = 0.001\ \text{mm}$ . Animal cells are generally between 10 and 30  $\mu\text{m}$  wide and plant cells are between 10 and 100  $\mu\text{m}$ .

- 1 What sort of microscopes are most often used in schools and universities?

Light microscopes.

- 2 What would be the best type of microscope to use to look at:

- a whole water fleas which are 3 mm long

Light microscope (a whole water flea is large enough to see with a light microscope).

- b cell membranes?

Electron microscope (cell membranes are far too small for a light microscope).

- 3 Where are 'organelles' found?

In the cytoplasm of a cell.

- 4 a Name one type of 'organelle'.

A mitochondrion (mitochondria).

- b Say what this organelle does in the cell.

Respiration happens inside it - it releases energy for the cell.

5 a How big is 1 mm in micrometres?

1000 micrometres (1000  $\mu\text{m}$ ).

b What is the symbol for a micrometre?  $\mu\text{m}$

6 In millimetres, what range of sizes do animal cells come in?

0.01 mm to 0.03 mm.

7 If a cell which is 20  $\mu\text{m}$  wide is looked at with a magnification of  $\times 1500$ , how wide will it appear to be? Give your answer in the most convenient units.

$20 \times 1500 = 30\,000 \mu\text{m} = 30 \text{ mm}$ .

## 7B 2f Microscope and Cells - Summary

- 1 Here are some instructions on using a microscope. Put the number '1' next to the instruction you should do first. Put the number '2' next to the instruction you should do next and so on.

|                      |                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------|
| <input type="text"/> | Look into the eyepiece lens. 5                                                               |
| <input type="text"/> | Place the smallest objective lens directly above the stage. 1                                |
| <input type="text"/> | Place the slide on the stage. 3                                                              |
| <input type="text"/> | Turn the coarse focusing wheel until the slide comes into focus.                             |
| <input type="text"/> | Turn the coarse focusing wheel to move the objective lens as close to the stage as possible. |
| <input type="text"/> | Adjust the light source. 4                                                                   |
| <input type="text"/> | 6                                                                                            |
| <input type="text"/> | 2                                                                                            |

- 2 Fill in the missing words in the sentences below. Use the words in the box.

eyepiece   magnified   microscope   objective   slide   specimen

Cells need to be  to be seen clearly. We can use a

to do this. When using a microscope, the thing we are looking at is

called the .

It is placed on a piece of glass called a .

The lens of the microscope that is closest to the object we want to look at is called the

lens. The other lens is the  lens.

- 3 Each of the following sentences has at least one mistake in it. Rewrite each, correcting the mistakes.

a The vacuole contains a substance called cytoplasm

The vacuole contains a substance called cell sap.

b Animal cells have a cell wall, a nucleus and cytoplasm.

Animal cells have a cell membrane, a nucleus and cytoplasm (they do not have a cell wall).

c Chloroform is a green substance found in chloroplasts.

Chlorophyll is a green substance found in chloroplasts.

d Microscopes have two lenses, the eyepiece lens and the objective lens.

This sentence is already correct: microscopes have two lenses, the eyepiece lens and the objective lens.

e The magnification of a microscope is worked out by adding the magnifying power of the two lenses together.

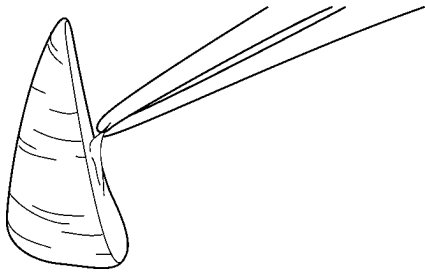
The magnification of a microscope is worked out by multiplying the magnifying power of the two lenses together.

- 4 You are given a microscope which has three objective lenses: x4, x10 and x40. The microscope has eyepiece lenses: x10. Make a table like the one below to show all the possible magnifications you could get.

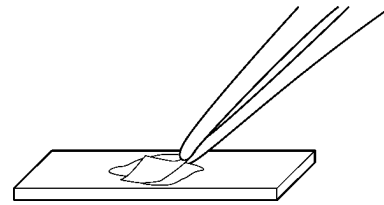
· Total magnification = eyepiece x objective. Row 1: x40, Row 2: x100, Row 3: x400.

| Magnification of eyepiece lens | Magnification of objective lens | Total magnification |
|--------------------------------|---------------------------------|---------------------|
| x10                            | x4                              | x40                 |
|                                | x10                             | x100                |
|                                | x40                             | x400                |

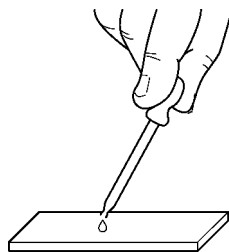
## 7B 3a Making a Slide of Onion Cells



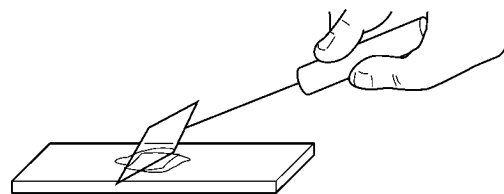
- 1 Using some forceps, a scalpel or your fingernails, peel off a thin layer of onion cells. Remember – the sample needs to be thin otherwise light will not go through it and you will not see anything.



- 2 Place your sample (your **specimen**) onto the middle of your slide.



- 3 Add a drop of iodine solution to your sample. The iodine solution is a **stain** to make the cells more visible.

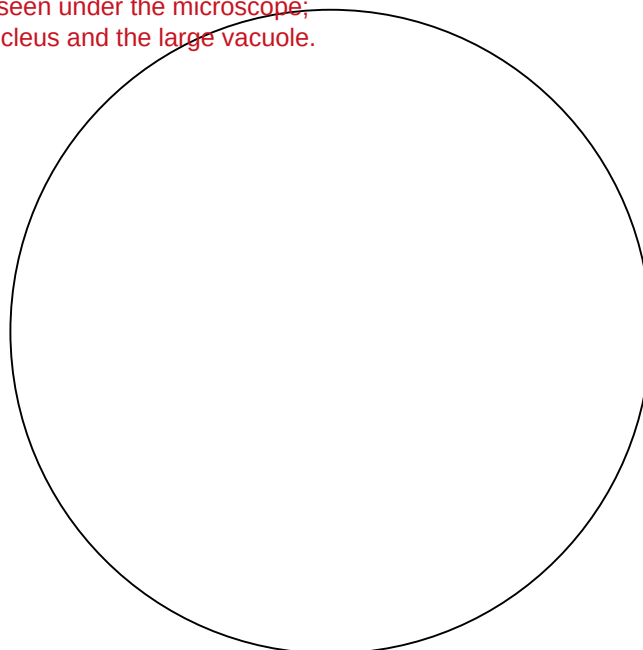


- 4 Lower a coverslip onto your specimen using a mounted needle. If you do this carefully and slowly you will not get any air bubbles.

**Results:** Draw what you see in the space here.

Try and label the **cell walls** and the **nucleus** and the **large vacuole**.

· Draw the onion cells as seen under the microscope; label the cell walls, the nucleus and the large vacuole.



**Conclusion/Analysis**

1 What does a stain do?

It colours parts of the cell to make them easier to see.

2 Why do we use coverslips?

To hold the specimen flat and in place, to stop it drying out, and to protect the microscope lens.

3 Why is the coverslip lowered gently using a mounted needle?

To avoid trapping air bubbles under the coverslip.

4 What things should you do to stay safe while preparing a slide?

Wear goggles; handle the scalpel, forceps and mounted needle carefully; never point the mirror at the Sun; take care with the iodine stain; wash your hands afterwards.

5. From the next set of sentences, circle the three that are the reasons we use a coverslip.

- a To flatten the specimen.
- b To stain the specimen.
- c To magnify the specimen.
- d To hold the specimen in place.
- e To stop the specimen drying out.
- f To heat up the specimen.

ans:

## 7B 3b Observing Cheek Cells

All living things are made of cells. You are made up of many millions of cells. Cells are very small, so small you cannot see them with your naked eye. Instead we use a microscope to observe cells. You are going to see some cells using a light microscope.

### THINK ABOUT

- How do you prepare cells to see them through a microscope?
- How do you focus a microscope safely to see cells?
- How do you draw what you see under a microscope?
- How can you measure what you see under a microscope?

### TASK

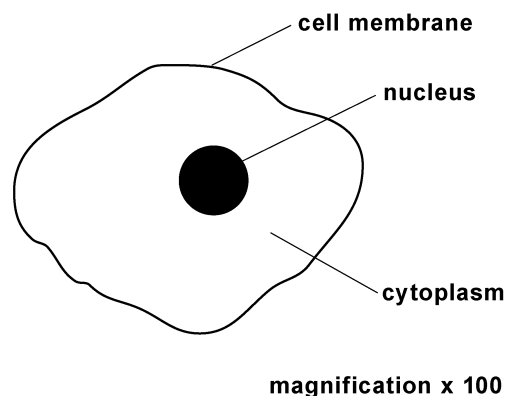
Your teacher will show you how to prepare a slide carefully and use a microscope to observe some cells. Focus your microscope at low and high magnification carefully and draw what you can see. Draw a diagram carefully using the tips below.

(total magnification = magnification of eye piece x magnification of objective lens)

#### Tips for microscope drawings:

- Use a sharp pencil on plain paper.
- Decide which part of the image you want to draw.
- Use continuous lines to draw your diagram; do not sketch.
- Label the parts of the image you can identify.
- Write a title for your diagram, stating the name of the cell.
- Write down the magnification at which you observed the cell.

#### A human cheek cell



#### KEY WORDS

cell, cell membrane, cell wall, cover slip, focus, lens, magnification, magnify, microscope, slide, stain, cytoplasm, eyepiece, objective, nucleus, vacuole

Place a thin sample on a clean slide, add a drop of water or stain, then lower a coverslip at an angle so that no air bubbles are trapped.

Start on the lowest power objective, lower the stage while watching from the side, focus with the coarse knob until the image appears, then sharpen with the fine knob.

Use a sharp pencil and clear single lines (no shading), keep the proportions correct, add labels with straight lines that do not cross, and state the magnification.

Measure the specimen in mm or eyepiece units and divide by the total magnification to estimate its actual size, or compare it with a scale bar or graticule.

## 7B 3c Microscope Crossword

- 1 Using the clues below, fill in this crossword.



· Crossword answers - Across: 2 slide, 3 coverslip, 6 nucleus, 8 glass. Down: 1 needle, 2 specimen, 4 lenses, 5 cells, 7 stage.

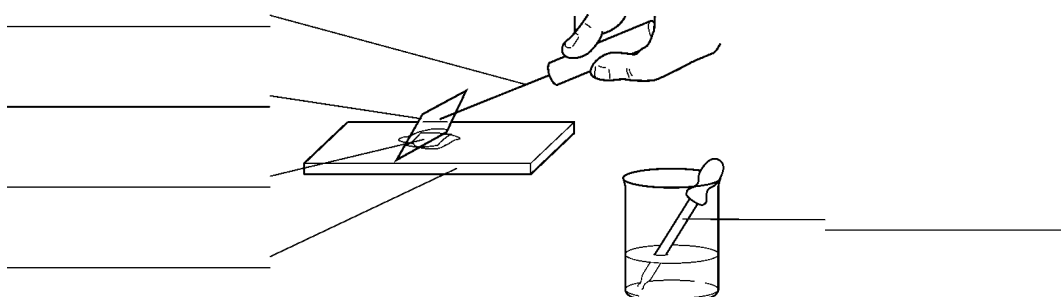
| Across                                                | Down                                           |
|-------------------------------------------------------|------------------------------------------------|
| 2 The specimen is put on this.                        | 1 We should lower a coverslip using a mounted. |
| 3 The thin piece of glass put on top of the specimen. | 2 What we look at using a microscope.          |
| 6 All cells have one of these.                        | 4 A microscope has two of these.               |
| 8 Slides and coverslips are made of this.             | 5 All living things are made of these.         |
|                                                       | 7 Part of a theatre and part of a microscope.  |

- 2 From the crossword, write down the letters in the shaded squares. Now rearrange these letters to make a word. The word is something that all plant cells have.

The word is vacuole

- 3 Look at the diagram below. It shows somebody making a slide. Label the parts on the diagram using the words in the box.

Label the diagram using the words in the box: coverslip, mounted needle, pipette, slide, specimen, slide, specimen.



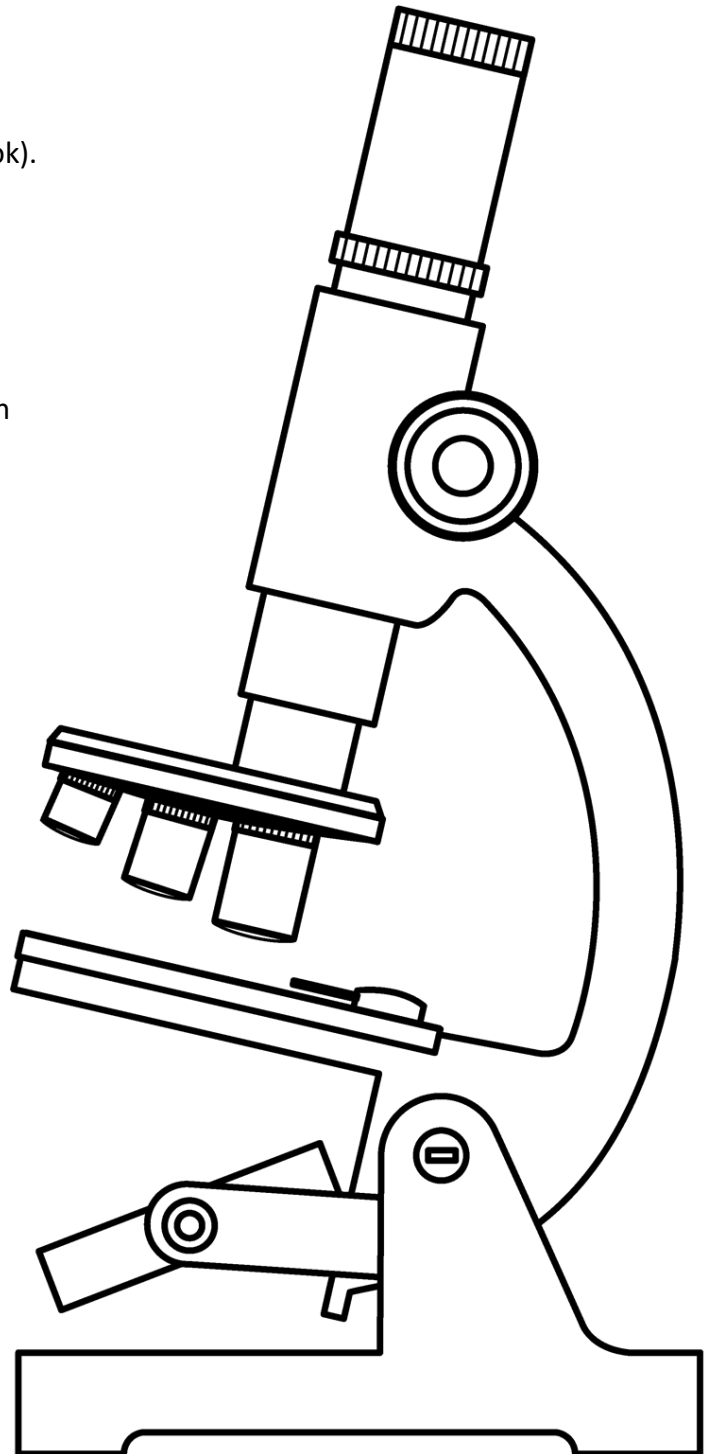
## 7B 3d Making A Poster of a Cell

Imagine the Science Museum has asked you to make a poster of a cell for a display. Make a POSTER of one of the following:

- an animal cell
- a plant cell
- a specialised cell (research in the text book).

Use the key words below to label each part of the cell clearly. Make an information card, perhaps using a key, to describe what cells are and the structures they contain. Include the function of the different structures.

Your teacher may ask you to give a short presentation about your model.



### KEY WORDS

cell, cell membrane, cell wall, chloroplast, cytoplasm, nucleus, mitochondria, vacuole

## 7B 4a Specialized Animal Cells

1 Fill in the missing words in the sentences below. Use the words from the box. Each word can be used more than once.

**adapted**  
**move**  
**muscle**  
**tissue**

a Some cells have special jobs to do. They often have special shapes to help them with their function. We say that these cells are **adapted**

for their function.

b A group of the same type of cells, all working together, is called a **tissue**

c A group of **muscle cells** all grouped together is called

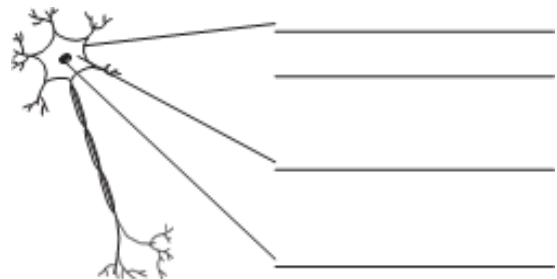
**tissue**

Muscle cells help us to

**move**

2 Here is a drawing of a **nerve cell**.

a Write in the words **cell surface membrane**, **cytoplasm** and **nucleus** on the correct lines.



· Label the nerve cell diagram with cell surface membrane, cytoplasm and nucleus.

b How is its shape adapted to carry messages around the body?

It is long and thin (has a long axon) so it can carry messages over long distances; branches let it connect with many other cells.

3 Here is a drawing of a **ciliated epithelial cell**.  
How is it adapted to remove dirt from the lungs?



It has cilia (tiny hairs) that wave and sweep mucus with trapped dirt up and out of the lungs.

## 7B 4b Specialized Plant Cells

1 Fill in the missing words in the sentences below.

Use the words from the box.

Plants have some specialized cells.

**a** Some cells help to carry water around the plant.

These are called **xylem** cells.

**b** Some plant cells help to take water out of the ground.

These are called **root hair** cells.

**c** A group of the same type of cells, all working together, is called a **tissue**

**hair**

**root**

**tissue**

**xylem**

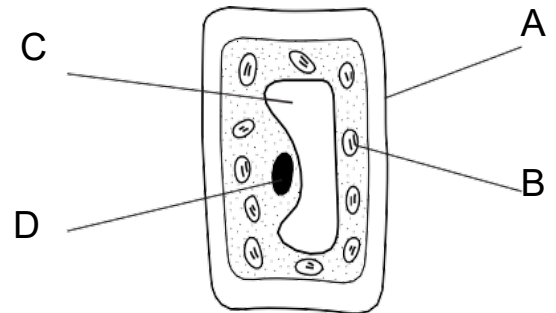
**2** Here is a palisade cell.

Palisade cells are in the upper parts of the leaves.

It helps the plant to make food.

Which part of the cell does this?

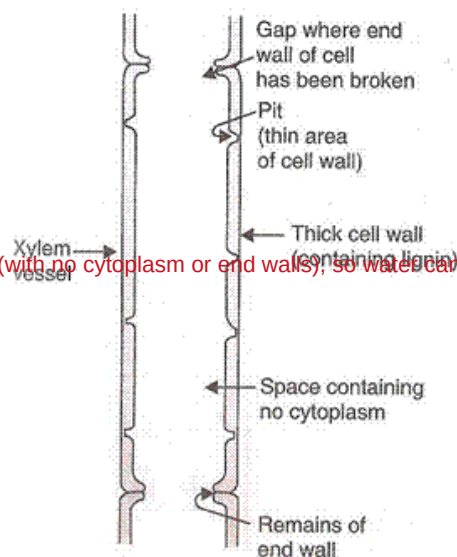
Chloroplasts - they contain chlorophyll and carry out photosynthesis to make food for the plant.



**3** Here is a xylem cell.

Which feature of the cell helps it carry water around the plant?

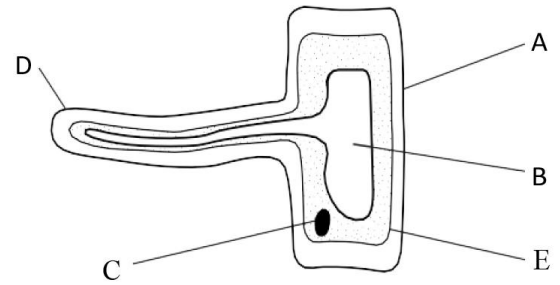
It is a long, hollow tube (with no cytoplasm or end walls), so water can flow through easily.



The structure of xylem vessels

**4** Here is a root hair cell.

**a** Which part of the cell helps it do its job?



The long, thin root hair - it increases the surface area so the cell can absorb water and minerals from the soil.

**b** In this drawing, what are the names of the parts labelled A, B and C?

**i** A is the root hair

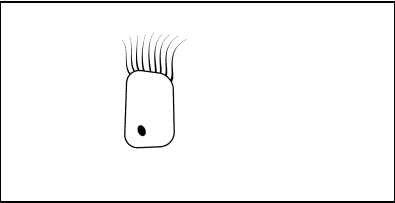
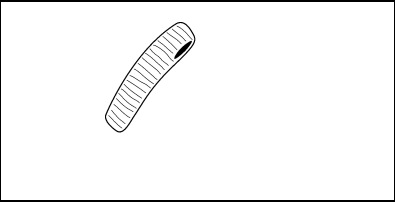
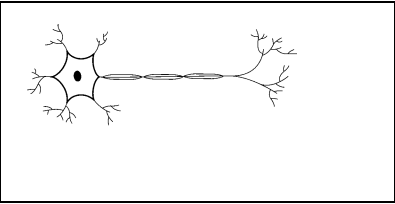
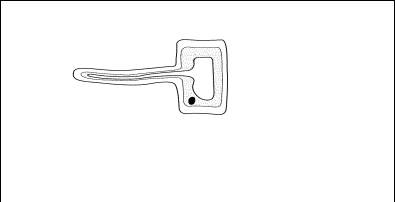
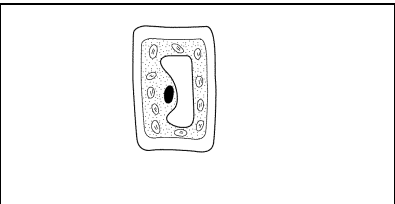
**ii** B is the nucleus

**iii** C is the vacuole

## 7B 4c Specialized Cells

Cells are different shapes because they do different jobs.

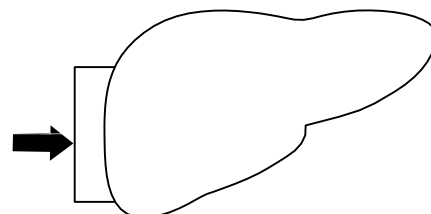
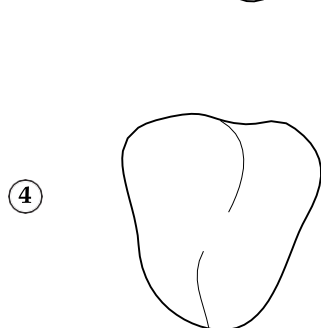
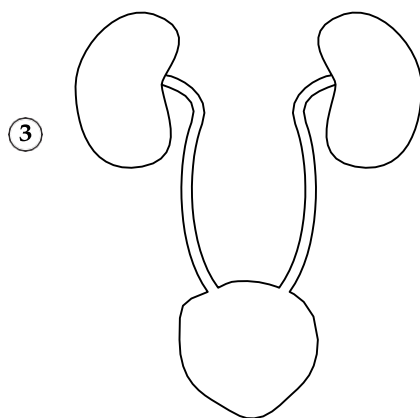
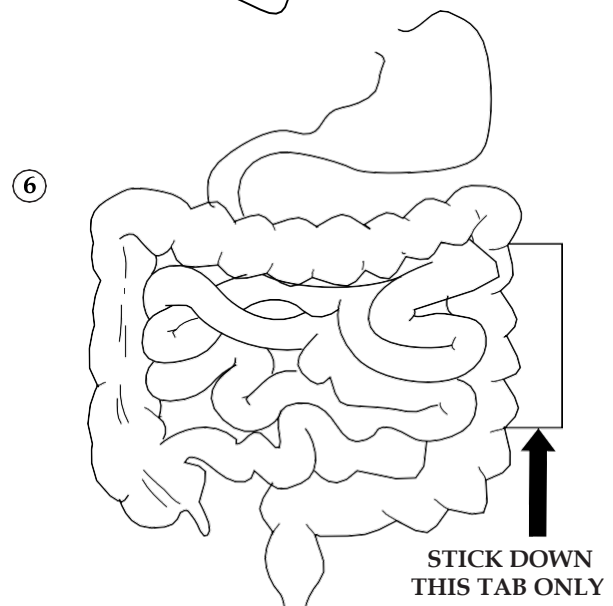
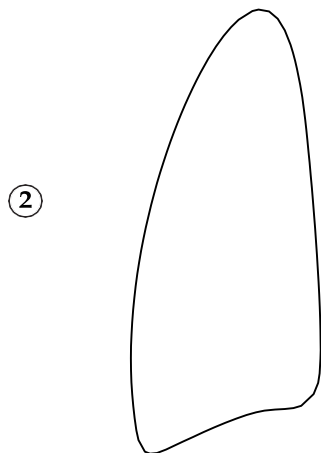
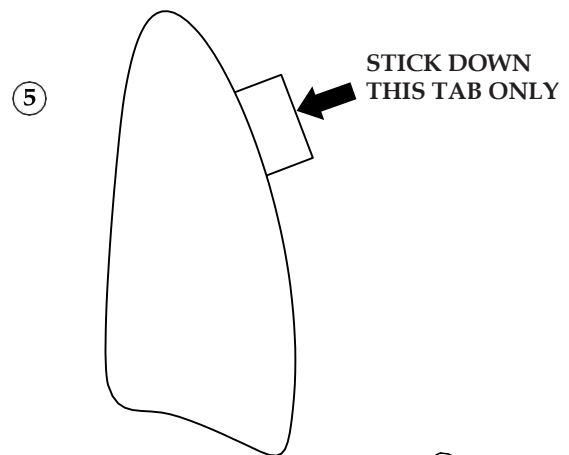
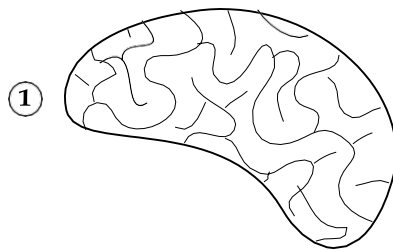
Match up the picture with the name of the cell and its job.

| cell                                                                                | name                                     | job                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------|
|    | muscle cell<br>muscle cell               | it can change length<br>strands wave about to move things                      |
|    | root hair cell<br>root hair cell         | it takes water from the soil<br>it can change length                           |
|   | cell with cilia<br>cell with cilia       | strands wave about to move things<br>it takes water from the soil              |
|  | palisade leaf cell<br>palisade leaf cell | it has lots of chloroplasts to help the plant make food<br>it carries messages |
|  | nerve cell                               | it has lots of chloroplasts to help the plant make food                        |

· Nerve cell row: name 'nerve cell', job 'it carries messages'.

## 7B 4d Human Organs (1)

7  
A  
a



Cut out and stick the organs on the human outline on the next page. Stick them on in order. Some have tabs that you need to cut around – do not cut these off!

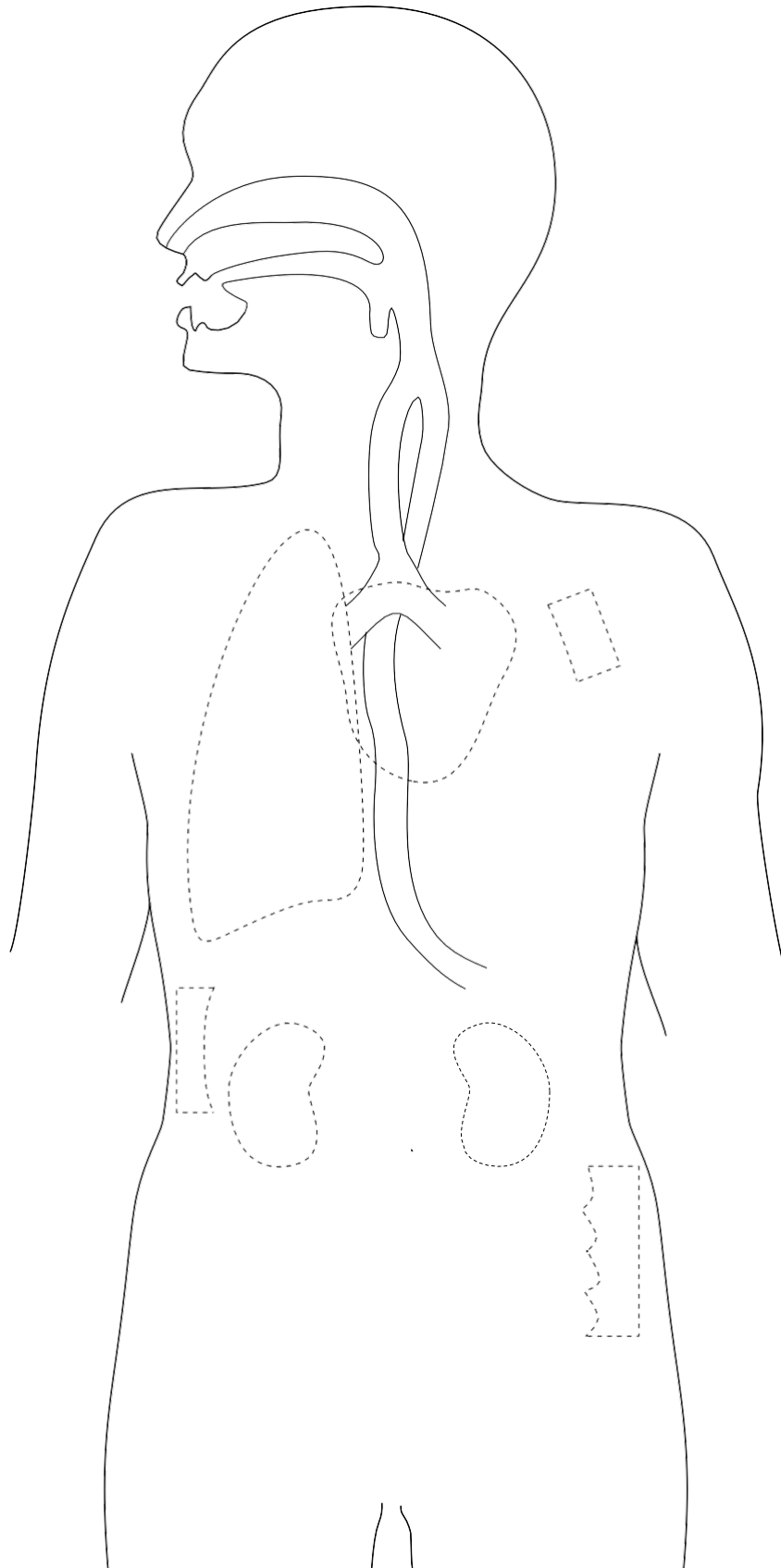
· Cut out the organs, keeping the tabs on, and stick them onto the human outline on the next page in the correct positions.



- Stick the cut-out organs onto the human outline in the correct positions (brain in head, lungs and heart in chest, etc.).

## 7B 4d Human Organs (2)

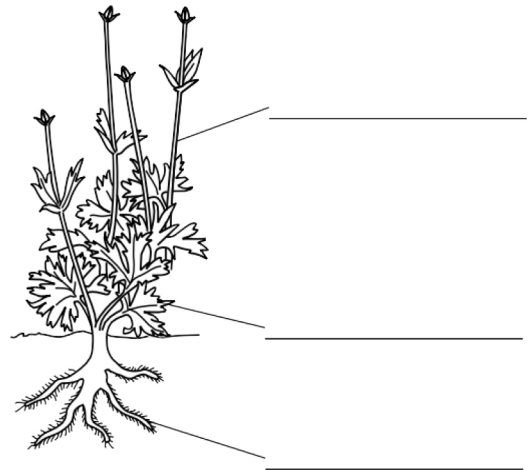
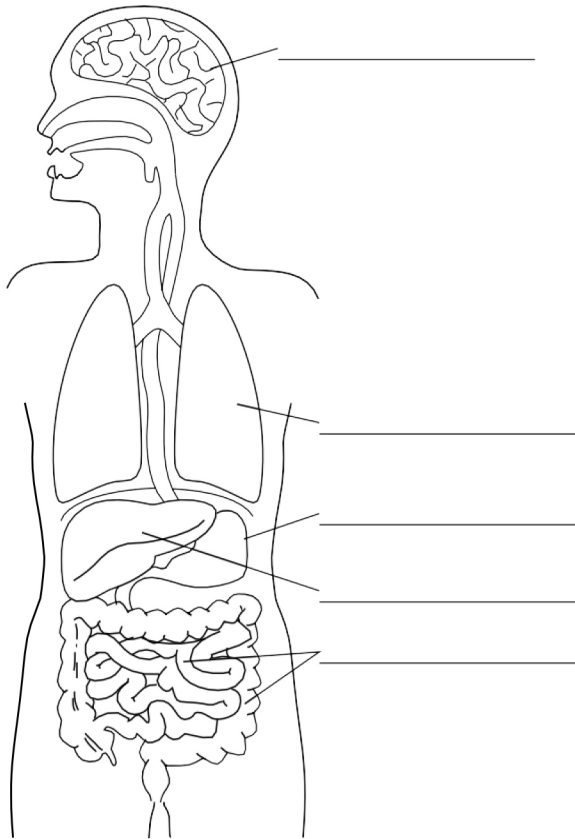
· Place/stick the organs on the outline in the correct order - brain, lungs, heart, liver, stomach, intestines.



## 7B 4e Where the organs are

- 1 On the diagrams below,  
write the names of the organs on the lines provided.  
Use the words from the box.

|                   |
|-------------------|
| Brain intestines  |
| Leaf liver lung   |
| Root stem stomach |



- 2 On the diagram of the human, draw in the position of the heart.

- 3 Select two of the human organs and explain what they do.

a:

b:

· Human organs to label (from the box): brain, lungs, liver, stomach, intestines.

- 4 Select two of the plant organs and explain what they do.

a:

b:

· Plant organs to label (from the box): leaf, root, stem.

· Draw the heart in the middle of the chest, slightly to the left, between the lungs.

Heart: it pumps blood around the body so that oxygen and nutrients can reach all the cells.

Lungs: they take oxygen from the air into the blood and remove carbon dioxide.

Leaf: it makes food (glucose) by photosynthesis using sunlight, water and carbon dioxide.

Root: it absorbs water and minerals from the soil and anchors the plant in the ground.

## 7B 4f Organ Systems

1 Fill in the correct word in each of the sentences.

Use the words in the box.

a A group of organs working together is called an **organ system**  
organ

b Plants take water in through their **roots**

. The water travels up

the **stem**  . Some of the water is

used to make food using **photosynthesis**

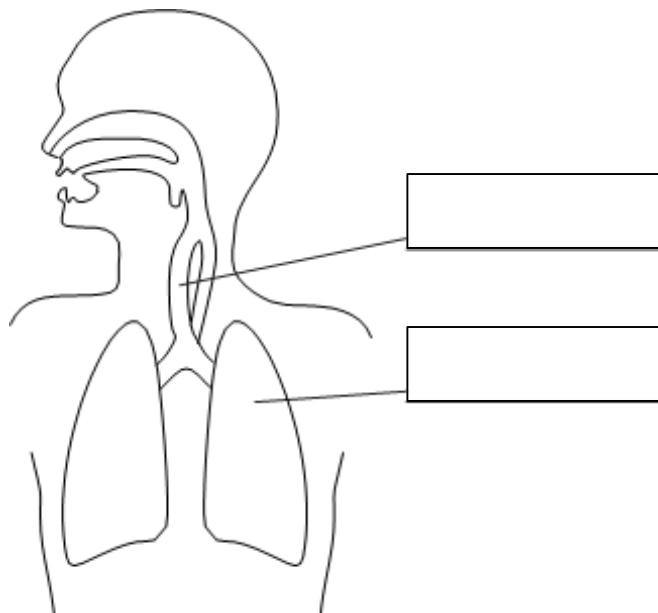
A lot of the water is lost by the **leaves**

c food is digested in the **digestive** system.

d The heart is part of the **circulatory** system.

e The brain is part of the **nervous** system.

2 On the diagram below, label the parts of the breathing system.



· Label the parts of the breathing system: trachea (windpipe), bronchi, lungs and diaphragm.

**circulatory**  
**digestive leaves**  
**nervous**  
**photosynthesis**  
**roots**  
**stem**  
**system**

## 7B 5a Levels of Organization Review

Look at the levels next to the questions and decide where you should start.  
If you know the answers to your target level questions, move on to the next level.

☐ Write a cell organisation paragraph using these keywords: work, cells, heart, together, tissue, organ. organ system

☐ Define cell, tissue and organ  
☐ Draw and label the organs onto the body in the correct positions

☐ Draw and label the organs onto the body in the correct positions  
☐ Write down their functions  
☐ Understand which systems are necessary for life and which continue life

Low

• **Recall** the difference between cell, tissue and organ, organ system.

Mid

• **Research** the roles and components of various organ systems.

High

• **Describe** the position and functions of various organs within our body.

**Key Words** - Tissue, organ, organ system, cell

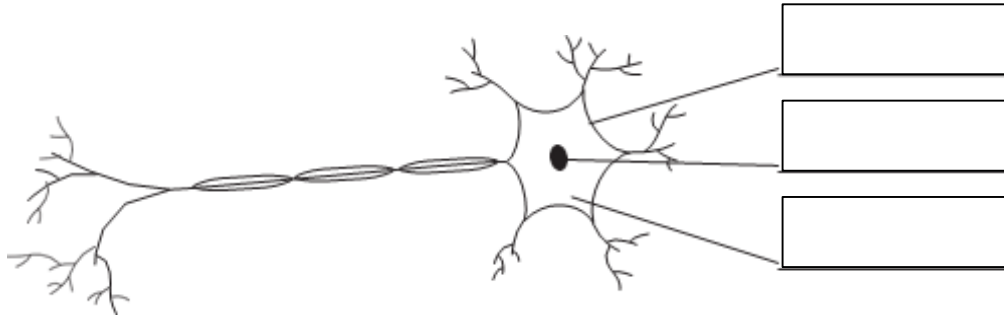
Cells are the basic unit of life. Tissues are groups of similar cells working together. Organs are groups of tissues working together to do a job. Organ systems are groups of organs working together to form organs, such as the heart. Organs work together to form

Cell: the basic unit of life. Tissue: a group of similar cells working together. Organ: a group of tissues working together to do a job. Organ system: a group of organs working together.

• Draw and label the organs (brain, lungs, heart, liver, stomach, intestines) on the body outline in the correct positions.

# 7B 5b Cells, Organs & Organ Systems Revision

1 a Label the parts of this cell.



b Which type of **tissue** does this cell form?

c Which **organ** might this tissue be found in?

muscle tissue

d Which **organ system** would this organ be found in?

the heart

the circulatory system

2 Name the organ system which:

a carries oxygen and food around the body?

b breaks down food?

circulatory system

c takes oxygen out of the air?

digestive system

3 In the following sentences, **one** ~~organ is not correct~~ <sup>organ is not correct</sup>. Put a cross through the wrong organ and write the name of the correct one in the space.

a The breathing system contains the lungs and intestine.

Cross through 'little toe'; the correct organ is the spinal cord.

b The nervous system contains the brain and the little toe.

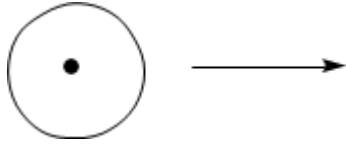
Cross through 'liver'; the correct organ is the heart.

c The circulatory system contains the liver and blood vessels

- Label the three parts of the cell shown: cytoplasm, nucleus and cell membrane.
- Cross through 'intestine'; the correct organ is the trachea (windpipe).

4 The drawing shows an animal cell.

- a Draw two or three more diagrams to show the steps taken when this cell splits into two.



~~Draw two or three more diagrams to show the steps taken when this cell splits into two.~~  
Draw two or three more diagrams to show the steps taken when this cell splits into two. In the first stage, the nucleus divides, the cytoplasm splits, then the cell pinches in.

- b What is the name of this process?

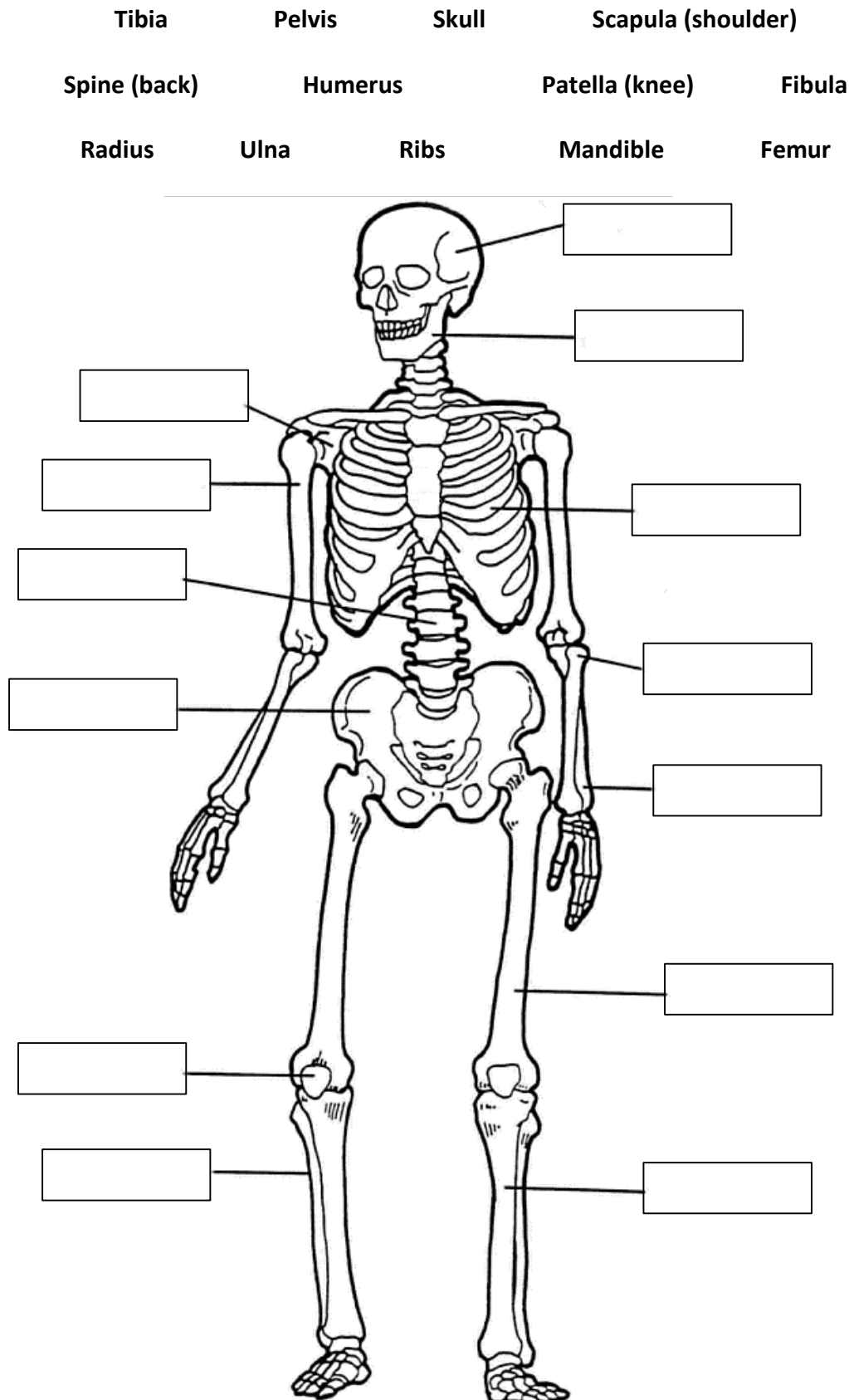
mitosis (cell division)

- c What are the two new cells called?

daughter cells

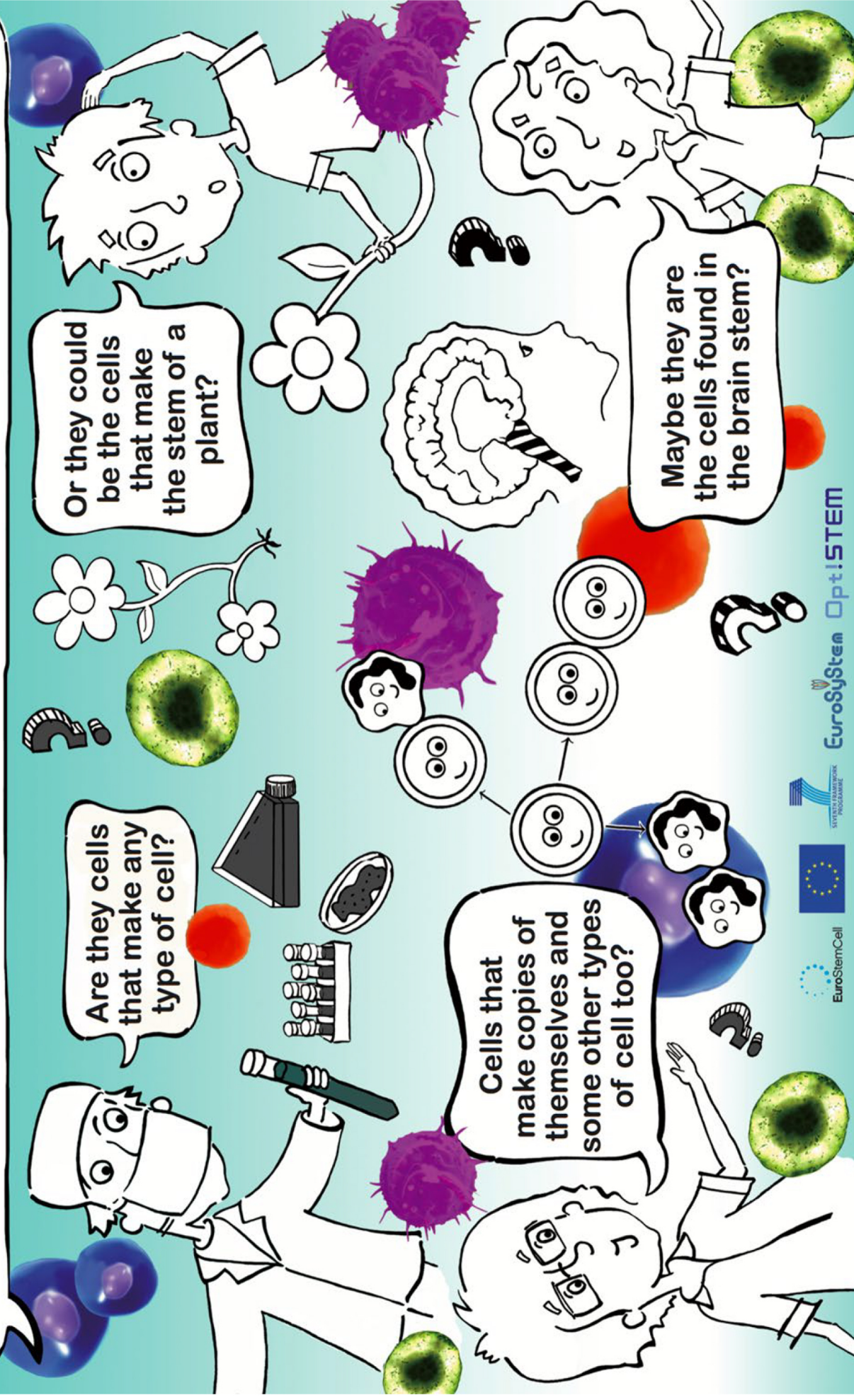
## 7B5c The Skeletal System

Use the bone names below to label the human skeleton.



· Label all 13 bones: skull (head), mandible (jaw), spine (back), scapula (shoulder), ribs (chest), humerus (upper arm), radius and ulna (forearm), pelvis (hip), femur (thigh), patella (knee), tibia and fibula (lower leg).

# What are stem cells?



# What is a stem cell?

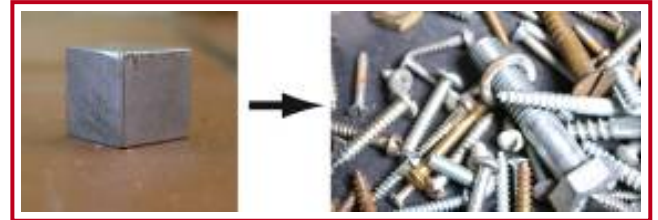
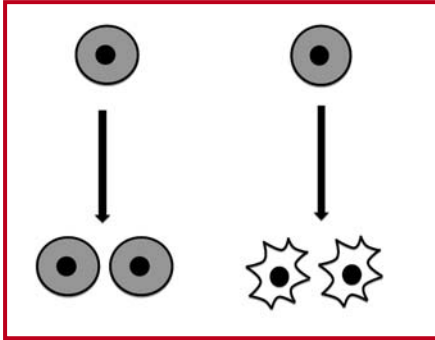
11-14 worksheet  
All about stem cells

## What stem cells can do

Stem cells can divide

AND differentiate

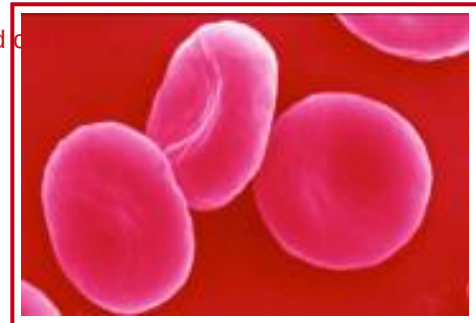
Differentiation = the process by which a cell becomes specialised to do a particular job, e.g. becoming a muscle or nerve cell



## Your body needs stem cells

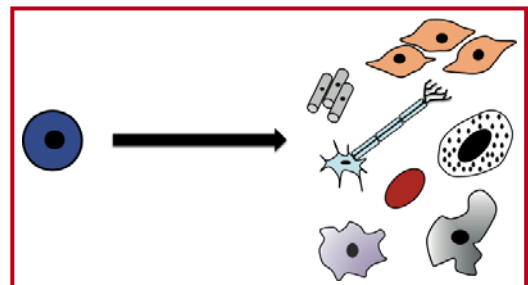
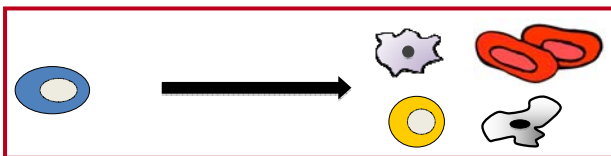
Stem cells make new, healthy cells to replace dead or damaged cells

Some cells in your body are replaced a lot.  
Red blood cells only live for about 120 days



Stem cells also divide to make new stem cells  
so that your body never runs out of stem cells.

## Two kinds of stem cells



can make ALL types of cell in your body Embryonic stem cells

can make a few types of specialized cells Adult (tissue) stem cells



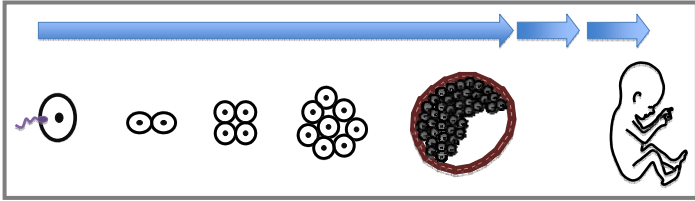
# Where do embryonic stem cells come from?

11-14 worksheet  
All about stem cells

## Embryonic stem cells and human development

A sperm fertilizes an egg.

The fertilized egg divides  
to make 2 cells, then 4 cells, then 8...



A blastocyst is a ball of about 100 cells.

Embryonic stem cells come from the inner cell mass of the blastocyst (early embryo)

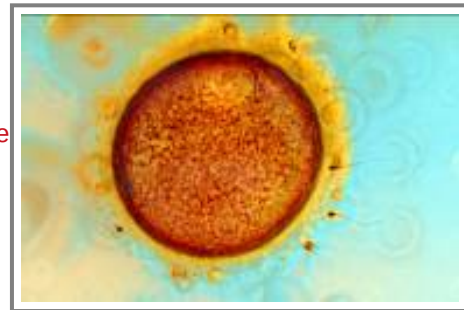


## IVF and stem cells

IVF stands for in vitro fertilisation

It is used to make embryos for people who cannot conceive easily.

Sometimes the mother and father let scientists use some of the embryos for stem cell research.



## Scientists grow cells in the laboratory

Some cells are put in a flask with the food they need. divide and grow (multiply)

The cells \_\_\_\_\_

Scientists can grow a lot of cells like this.

Growing lots of cells in the laboratory is called cell culture



# Why bother with stem cells?

11-14 worksheet

All about stem cells

Stem cells can differentiate, so they can replace damaged or lost cells and may treat diseases that have no other cure, e.g. leukaemia and Parkinson's disease.

## Treating patients

Stem cells are used to grow new skin for people with bad burns. To do this we:

1. \_\_\_\_\_
2. Take stem cells from the patient's own body
3. (or a donor).

In the future, we might use stem cells to treat diseases (e.g. diabetes, Parkinson's) by growing new healthy cells or replacement organs

Grow the stem cells in the laboratory to make lots of new skin cells.

Graft (transplant) the new skin onto the burned areas.



## Understanding disease

Scientists could use stem cells to learn about diseases that are caused by faulty (damaged) cells

They could:

- Grow specialized cells from the stem cells, for example nerve cells, blood cells or liver cells
- Investigate diseases and how they develop



## Testing drugs



Stem cells could be used to test new medicines(drugs).

Scientists would:

- Use stem cells to grow a lot of specialised cells, e.g. nerve cells, blood cells or liver cells.
- Find out what the new drugs do to these cells.

# Making stem cells

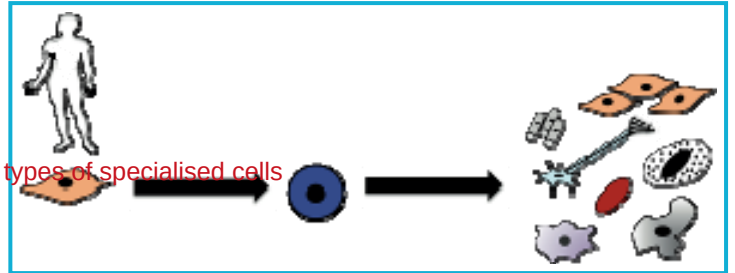
11-14 worksheet  
All about stem cells

## Turning skin cells into stem cells

iPS stands for **induced pluripotent stem cells**

\_\_\_\_\_

iPS cells are pluripotent. That means they can **all types of specialised cells**  
make \_\_\_\_\_



iPS cells are made by adding **genes**  
to specialized cells such as skin cells.

The process used to make iPS cells is called **reprogramming**  
genetic \_\_\_\_\_



## iPS cells in medicine

iPS cells might be used to treat diseases such as **diabetes and Parkinson's disease**

\_\_\_\_\_

Scientists would take skin cells from the patient  
and use them to make iPS cells.

Then they would use the iPS cells to grow **new healthy cells**

\_\_\_\_\_

for the patient.

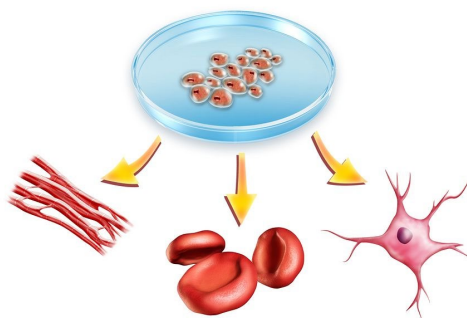


# 7B6/7a Stem Cells Assessment Task Research Template

| Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Notes                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What is a stem cell?</b><br>1. What is a stem cell?<br>(Hint: Explain what 2 things stem cells can do.)<br>2. Why do our bodies need stem cells?<br>3. What different kinds of stem cells are there?<br>(Hint: Name 2 kinds of stem cell and explain what each kind can do.)                                                                                                                                                                                                                                                                                                          | To replace dead or damaged cells and repair tissues, e.g. making new skin, blood and lining cells.<br>Embryonic stem cells (make ALL types of cell) and adult/tissue stem cells (make just a few specialised cell types). |
| An unspecialised cell that can divide to make more of itself and differentiate into specialised cells (e.g. blood, muscle or nerve cells).                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |
| <b>Where do embryonic stem cells come from?</b><br>4. Where are embryonic stem cells found?<br>(Hint: Think about the stages of human development.)<br>5. Where can scientists get embryonic stem cells for research?                                                                                                                                                                                                                                                                                                                                                                    | In the inner cell mass of a blastocyst - an early embryo of about 100 cells, a few days after fertilisation.<br><br>From spare embryos donated by couples having IVF treatment, with their permission.                    |
| <b>Why are stem cells useful?</b><br>6. How are stem cells used to treat patients now?<br>7. How might stem cells be used to treat patients in the future?<br>8. How else could stem cells be used?<br>(Hint: Give 2 ways that are described in the information you have been given. You can add more ideas of your own if you wish.)<br>9. Why might some people object to using stem cells?<br>(Hint: You could consider different viewpoints based on culture, religion, background, personal opinions. You could also think about where they come from and the risks of using them.) |                                                                                                                                                                                                                           |
| Some people believe embryos are human life, so using embryonic stem cells raises moral/religious objections; there are also risks                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                           |
| <b>Making stem cells (hardest)</b><br>10. What is an iPS cell?<br>(Hint: Explain how iPS cells are made and what they can do. What other kind of stem cell are they like?)<br>11. How could iPS cells be used?                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                           |
| To grow new healthy cells for the patient from their own cells - no rejection and fewer ethical problems - e.g. nerve or blood cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                           |
| <b>Additional notes/diagrams</b><br>Growing new skin for burn patients and bone-marrow transplants for blood diseases such as leukaemia.<br>To grow replacement organs/tissues or healthy cells to treat diseases such as diabetes and Parkinson's.<br>To learn how diseases develop by growing specialised cells, and to test new drugs on cells before patients use them.<br>An induced pluripotent stem cell: a skin (or other specialised) cell reprogrammed by adding genes so it can make all cell types, like an embryonic stem cell.                                             |                                                                                                                                                                                                                           |



**MYP 1: Year 7 Science**  
**Criterion D Assessment**  
**Reflecting on the Impacts of Science**



|                                   |                                                                                                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT:</b>                      | Cells and Reproduction                                                                                                                                                                                                                                          |
| <b>ASSESSMENT TITLE:</b>          | Stem Cells Explainer Clip                                                                                                                                                                                                                                       |
| <b>Statement of Inquiry:</b>      | Can we use advances in medicine and technology to improve the way organs and organs systems function?                                                                                                                                                           |
| <b>Key Concept:</b>               | Systems                                                                                                                                                                                                                                                         |
| <b>Related Concepts:</b>          | Consequences, Function                                                                                                                                                                                                                                          |
| <b>ATL Focus:</b>                 | <b>ATL: CRITICAL THINKING SKILLS, Analysing and evaluating issues and ideas</b> <ul style="list-style-type: none"> <li>Consider ideas from multiple perspectives</li> <li>Develop contrary or opposing arguments</li> </ul>                                     |
| <b>Key Scientific Vocabulary:</b> | stem cells, tissue, unspecialised, specialised, pluripotent, embryonic, progenitor<br><br>Organs work together as organ systems. Each system carries out a life process such as digestion or gas exchange, and a failure in one organ affects the whole system. |

## Reflecting on the use of stem cells

### Assignment Instructions (GRASPS)

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal</b>                            | Your goal is to provide information on stem cell therapy. This is being used to treat some diseases, but there is still a lot of research going on to investigate their full potential.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Role</b>                            | You work for the Hospital Authority. You are responsible for producing patient information literature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Audience</b>                        | Individuals (and their families) who are interested in or considering using stem cell technology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Situation</b>                       | Different stem cell therapies are becoming more common. Before individuals decide to use this therapy, they need to be informed on how it works and the positive and negative impacts associated. In particular, you should focus on the different types of stem cells (adult/tissue and embryonic stem cells) and their pros and cons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Product</b>                         | You will produce a short video of you presenting a slideshow on stem cells. This would be shown in a hospital or clinic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Success Criteria for Assessment</b> | <p>The leaflet should be 2 sides of A4 and include: (these get harder as you move down the list)</p> <p><b>What is a stem cell?</b></p> <ol style="list-style-type: none"> <li>1. What is a stem cell?<br/>(Hint: Explain what 2 things stem cells can do.)</li> <li>2. Why do our bodies need stem cells?</li> <li>3. What different kinds of stem cells are there?<br/>(Hint: Name 2 kinds of stem cell and explain what each kind can do.)</li> </ol> <p><b>Where do embryonic stem cells come from?</b></p> <ol style="list-style-type: none"> <li>4. Where are embryonic stem cells found?<br/>(Hint: Think about the stages of human development.)</li> <li>5. Where can scientists get embryonic stem cells for research?</li> </ol> <p><b>Why are stem cells useful?</b></p> <ol style="list-style-type: none"> <li>6. How are stem cells used to treat patients now?</li> <li>7. How might stem cells be used to treat patients in the future?</li> <li>8. How else could stem cells be used?<br/>(Hint: Give 2 ways that are described in the information you have been given. You can add more ideas of your own if you wish.)</li> <li>9. Why might some people object to using stem cells?<br/>(Hint: You could consider different viewpoints based on culture, religion, background, personal opinions. You could also think about where they come from and the risks of using them.)</li> </ol> <p><b>Making stem cells (hardest)</b></p> <ol style="list-style-type: none"> <li>10. What is an iPS cell?<br/>(Hint: Explain how iPS cells are made and what they can do. What other kind of stem cell are they like?)</li> <li>11. How could iPS cells be used?</li> </ol> <p>Source: <a href="http://www.eurostemcell.org">www.eurostemcell.org</a></p> <p>Write clearly and concisely. Scientific key language should be used accurately. The leaflet needs to include a clear bibliography, with at least five sources cited for the top marking bank..</p> <p>A stem cell is an un specialised cell that can divide repeatedly by mitosis and can differentiate into other, specialised types of cell.</p> |

**Key Scientific Vocabulary / Definitions:**

|                      | <b><u>MYP definition on top</u></b><br><b><u>Our definition underneath</u></b>                                                                                                                                                                                                                                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Apply</b>         | <p><u>MYP definition:</u> Use knowledge and understanding in response to a given situation or real circumstance.</p> <p><u>Our definition:</u> Use what you already know in a new and/or different situation.</p>                                                                                                                                                                                                |
| <b>Cultural</b>      | <p><u>MYP definition:</u> Patterns of knowledge, behavior, beliefs, shared attitudes, values, goals and practices that characterize groups of people</p> <p><u>Our definition:</u> The shared beliefs and values of different groups of people.</p>                                                                                                                                                              |
| <b>Describe</b>      | <p><u>MYP definition:</u> Give a detailed account or picture of a situation, event, pattern or process.</p> <p><u>Our definition:</u> Tell how something happened (i.e. “first this, then that, then the next thing”). In experiments, identify the minimum and maximum values of data in experimental results and recount the overall trends between variables.</p>                                             |
| <b>Discuss</b>       | <p><u>MYP definition:</u> Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence.</p> <p><u>Our definition:</u> Look at both sides of an argument or an issue, weighing the strengths and limitations of each side. Conclusions should be based on scientific evidence.</p> |
| <b>Document</b>      | <p><u>MYP definition:</u> Credit sources of information used by referencing or citing, following the MLA system. References should be included in the text and also at the end of the piece of work in a bibliography.</p> <p><u>Our definition:</u> Use in-text references and a bibliography to show where you found the ideas and information in a piece of work.</p>                                         |
| <b>Economic</b>      | <p><u>MYP definition:</u> Production, distribution, and use of income, wealth, and commodities</p> <p><u>Our definition:</u> Anything having to do with money, either personally or as a society</p>                                                                                                                                                                                                             |
| <b>Environmental</b> | <p><u>MYP definition:</u> Circumstances, objects, or conditions by which one is surrounded</p> <p><u>Our definition:</u> Our surroundings, particularly the natural (not built) world</p>                                                                                                                                                                                                                        |
| <b>Ethical</b>       | <p><u>MYP definition:</u> Process of rational inquiry to decide on issues as right or wrong, as applied to the people and their actions</p> <p><u>Our definition:</u> Determining whether or not an action is right or wrong</p>                                                                                                                                                                                 |

|                  |                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Evaluate</b>  | <p><u>MYP definition:</u> Make an appraisal by weighing the strengths versus the limitations.</p> <p><u>Our definition:</u> Look at how well something worked, whether it's a hypothesis, method, or collection of information.</p> |
| <b>Explain</b>   | <p><u>MYP definition:</u> Give a detailed account.</p> <p><u>Our definition:</u> Tell how and why something happened and justify it with scientific evidence and/or reasons.</p>                                                    |
| <b>Moral</b>     | <p><u>MYP definition:</u> Principles of right or wrong behaviour derived from a particular society</p> <p><u>Our definition:</u> Relating ethical decisions to culture</p>                                                          |
| <b>Outline</b>   | <p><u>MYP definition:</u> Give a brief account.</p> <p><u>Our definition:</u> Briefly describe the major points or concepts.</p>                                                                                                    |
| <b>Political</b> | <p><u>MYP definition:</u> Relates to government or public affairs</p> <p><u>Our definition:</u> How different groups or countries have power under the law</p>                                                                      |
| <b>Social</b>    | <p><u>MYP definition:</u> Interactions between groups of people involving issues such as welfare, safety, rights, justice or class</p> <p><u>Our definition:</u> The ways in which people interact with one another</p>             |
| <b>Summarize</b> | <p><u>MYP definition:</u> Abstract a general theme or major points.</p> <p><u>Our definition:</u> Distill a lot of information down to the key essential points.</p>                                                                |

## Level Descriptors and Task Specific Clarifications for Criterion D:

| Level | Level Descriptor                                                                                                                                                                                                                                                                                                                                                                                     | Task Specific Clarification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | The student does not reach a standard described by any of the descriptors below.                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1-2   | <p>The student is able to, <b>with limited success</b>:</p> <p>i. <b>state</b> the ways in which science is used to address a specific problem or issue</p> <p>ii. <b>state</b> the implications of using science to solve a specific problem or issue, interacting with a factor</p> <p>iii. <b>apply</b> scientific language to communicate understanding</p> <p>iv. document sources.</p>         | <p>i. <b>State</b> a disease that stem cells could be used to treat.</p> <p>ii. <b>State</b> at least <b>one impact (positive or negative)</b> that stem cells have on the society, focusing on <b>one of the factors</b> (moral, economic, political or cultural), but with inaccuracies.</p> <p>iii. Some <b>clear and concise language is applied</b> but this is rare. <b>Scientific vocabulary</b> generally not applied.</p> <p>iv. <b>Limited</b> source (one source) included in the bibliography, not in any particular format..</p>    |
| 3-4   | <p>The student is able to:</p> <p>i. <b>State</b> the ways in which science is used to address a specific problem or issue</p> <p>ii. <b>State</b> the implications of using science to solve a specific problem or issue, interacting with a factor</p> <p>iii. <b>sometimes apply</b> scientific language to communicate understanding</p> <p>iv. <b>sometimes</b> document sources correctly.</p> | <p>i. <b>State</b> a disease which could be treated by stem cells. What are the symptoms?</p> <p>ii. <b>State</b> at least <b>one impact (positive or negative)</b> that stem cells have on the society, focusing on <b>one of the factors</b> (moral, economic, political or cultural).</p> <p>iii. Sometimes use <b>clear and concise language</b>. <b>Scientific vocabulary</b> sometimes applied. Limited <b>images / graphs</b> used to illustrate key points.</p> <p>iv. <b>Some</b> sources (2 sources) included in the bibliography.</p> |

| Level | Level Descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Task Specific Clarification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-6   | <p>The student is able to:</p> <p>i. <b>Outline</b> the ways in which science is used to address a specific problem or issue</p> <p>ii. <b>Outline</b> the implications of using science to solve a specific problem or issue, interacting with a factor</p> <p>iii. <b>usually apply</b> scientific language to communicate understanding <b>clearly and precisely</b></p> <p>iv. <b>usually</b> document sources correctly.</p>                                               | <p>i. <b>Outline</b> 2-3 diseases that stem cell therapy could be used to treat. Refer to the different types of stem cells.</p> <p>ii. <b>Outline</b> at least <b>one positive and one negative impact</b> that stem cells have on society, focusing on <b>one of the factors</b> (moral, economic, political or cultural). <b>Support</b> these arguments with <b>evidence</b>. <b>Outline</b> the main ideas and impacts.</p> <p>iii. Usually use <b>clear and concise language</b>. <b>Scientific vocabulary</b> generally applied. Some <b>images / graphs</b> used to illustrate key points. Leaflet is 2 sides of A4.</p> <p>iv. <b>Adequate</b> sources (3 sources) included in the bibliography. Sources generally reliable. Bibliography presented in a clear format.</p>                                                                                                                                                                  |
| 7-8   | <p>The student is able to:</p> <p>i. <b>Summarize</b> the ways in which science is applied and used to address a specific problem or issue</p> <p>ii. <b>Describe and summarize</b> the implications of using science and its application to solve a specific problem or issue, interacting with a factor</p> <p>iii. <b>consistently apply</b> scientific language to communicate understanding <b>clearly and precisely</b></p> <p>iv. document sources <b>completely</b></p> | <p>i. <b>Summarize</b> 4 diseases that stem cell therapy could be used to treat. Refer to the different types of stem cells and the specialised cell which is affected. <b>Summarize</b> how stem cells would solve the problem.</p> <p>ii. <b>Describe</b> at least <b>two positive and two negative impacts</b> that this technology has on the society, focusing on <b>one of the factors</b> (moral, economic, political or cultural). <b>Support</b> these arguments with reliable <b>evidence</b>. <b>Summarize</b> the main ideas and impacts.</p> <p>iii. Use <b>clear and concise language</b> throughout. Appropriate <b>scientific vocabulary</b> generally applied. Relevant <b>images / graphs</b> used to illustrate key points. Leaflet is 2 sides of A4.</p> <p>iv. <b>Numerous</b> reliable sources (4-5 sources) included in the bibliography. Bibliography presented in a clear APA format. <b>Image references</b> included.</p> |

**Student Task Planning Notes:**

## 7B8a Components of Blood

Platelets help blood clot by forming a platelet plug

Platelets are formed in the bone marrow

150,000 to 400,000 platelets in each microlitre of blood (average = 250,000)

# Platelets



Platelets contain chemicals that stimulate blood vessel repair

Platelets last for an average of 10 days,

Photo credit: [visualsonline.cancer.gov](https://visualsonline.cancer.gov)

## 7B8a Components of blood

# Red blood cells

People who do not get enough iron in their diet have small, pale red blood cells and feel tired all the time.

People who lose a lot of blood have normal red blood cells but too few of them.

Stem cells in the centre of bones make about 2 million new ones per minute.

Healthy adults have about 20,000,000,000,000 red blood cells.

Full of a red chemical called haemoglobin that contains iron.

Haemoglobin picks up oxygen in the lungs and releases it in other parts of the body.

The thin shape means that it can bend to fit through tiny blood vessels and oxygen can get in and out the cell easily.

New red blood cells have a nucleus but as they mature in bone marrow they break down their nucleus to make space for more haemoglobin.

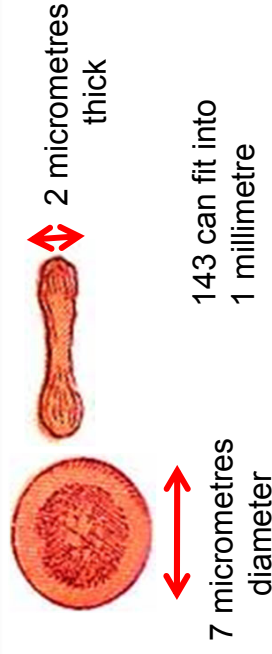
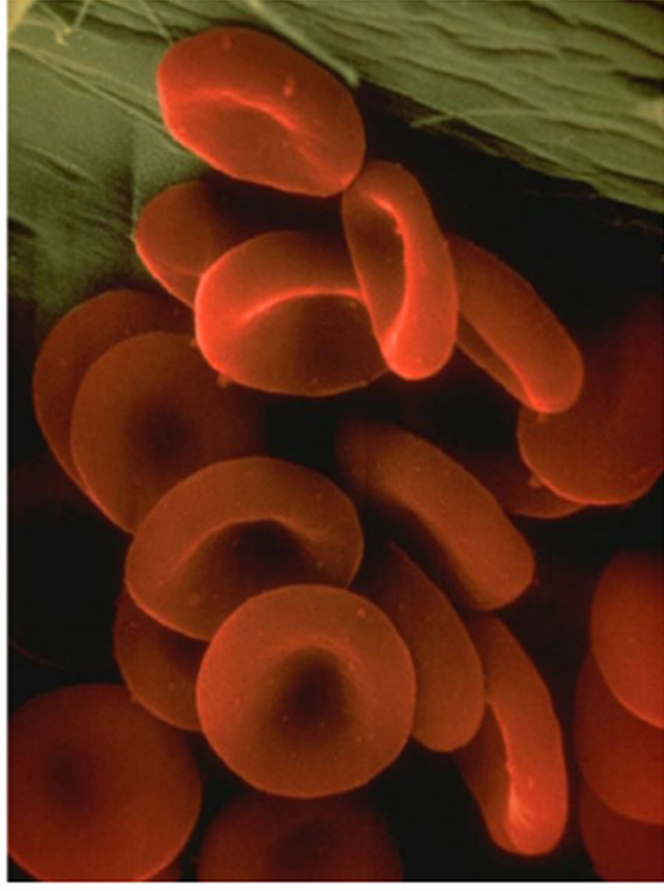


Photo credit: [visualsonline.cancer.gov](https://visualsonline.cancer.gov)

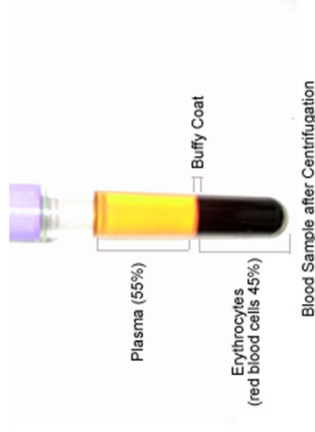
## 7B 8a Components of blood

# Plasma

Plasma is the liquid part of blood.

Plasma is yellow in colour and is mostly water (93%)

Plasma contains dissolved substances around the body.



Plasma makes up 55% of blood.

If someone has eaten a lot of fat, their plasma can appear milky!

Plasma contains lots of different substances, including glucose, hormones, minerals and carbon dioxide.



Photo credit: [visualsonline.cancer.gov](https://visualsonline.cancer.gov)

## 7B 8a Components of blood

# White blood cells

More of these are produced when you have an infection.

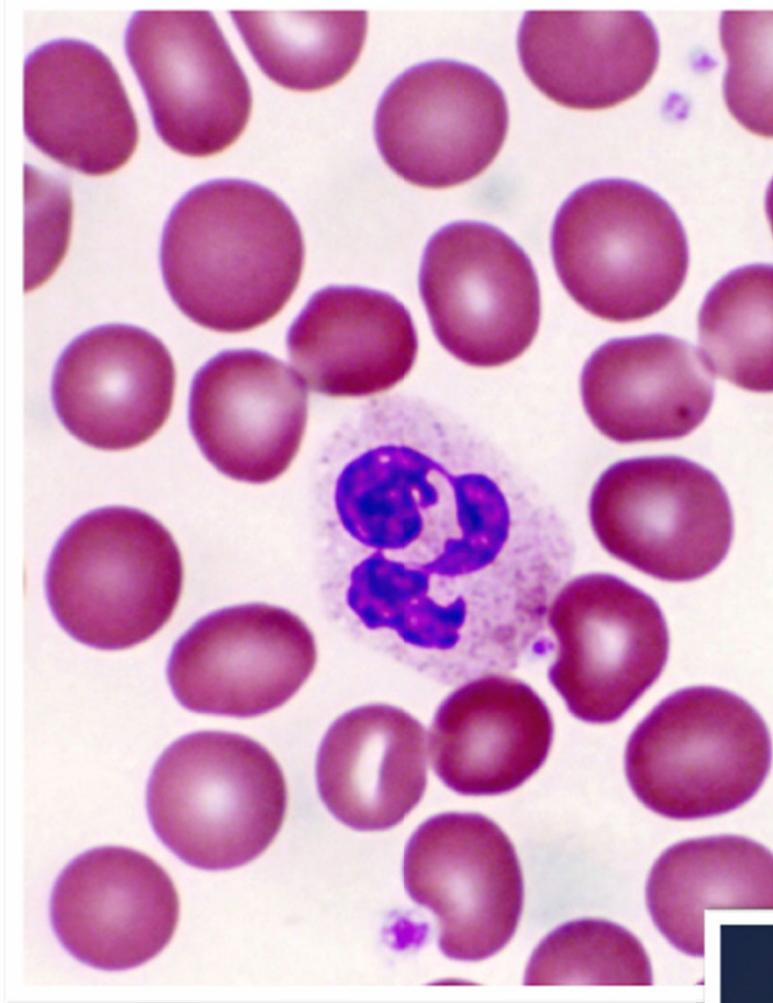
They have unusual shaped nuclei.

They move through blood vessels until they find microbes.

When they catch a microbe they change shape, wrap themselves around it and take it into their cytoplasm.

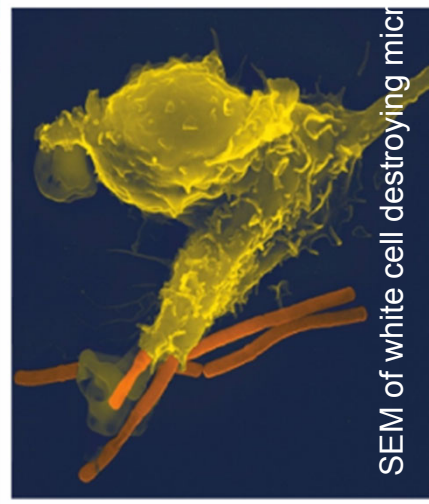
Inside their cytoplasm are little bags of chemicals that can destroy most microbes.

They can squeeze out of blood vessels to attack infections in other tissues.



13 micrometres

77 can fit into 1 millimetre



SEM of white cell destroying microbe

## 7B 8b Analysing a 'blood' sample

Your teacher will give you the samples of each patient's blood.

The 4 different colours represent the different components of the blood.

- red beads = red blood cells to carry oxygen
- white beads = white blood cells to fight infection
- blue beads = platelets to help in blood clotting
- yellow beads = plasma, the liquid part of blood



Analyze the 'blood' samples and use the diagnosis cards below to determine if any of the patients are healthy or have a particular medical condition,

• Count the coloured beads in each sample and compare the percentages with the healthy values (red blood cells about 40%, white blood cells about 1%, platelets about 4%).

|                                                                                                                                                                                      |                                                                                                                                                                                            |                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Healthy person</b></p> <p>average <b>red blood cell</b> count:<br/>is approximately 40%</p>                                                                                    | <p><b>Healthy person</b></p> <p>average <b>white blood cell</b> count:<br/>is approximately 1%</p>                                                                                         | <p><b>Healthy person</b></p> <p>average <b>platelet</b> count:<br/>is approximately 4%</p>                                                                                                  |
| <p><b>Anaemia</b></p> <p>Does the <b>red blood cell</b> count in your sample match the normal value?</p> <p>Could your patient have anaemia? (reduced number of red blood cells)</p> | <p><b>The flu</b></p> <p>Does the <b>white blood cell</b> count in your sample match the normal value?</p> <p>Could your patient have the flu? (increased number of white blood cells)</p> | <p><b>Haemophilia</b></p> <p>Does the <b>platelet</b> count in your sample match the normal value?</p> <p>Could your patient have haemophilia? (large reduction in number of platelets)</p> |

No - the red blood cell count is well below the healthy about 40%, so the patient has anaemia (too few red blood cells).

Write up your results using the MYP Practical Report template.

Remember to make your results **RELIABLE!**

• Write up the practical using the MYP report template (aim, method, results, analysis, conclusion, evaluation) and repeat counts to make results reliable.

No - the white blood cell count is well above the healthy about 1%, so the patient has the flu (more white blood cells fighting infection).

No - the platelet count is far below the healthy about 4%, so the patient has haemophilia (too few platelets, blood cannot clot).

## 7B 8c MYP Criterion C Assessment Criteria (optional)

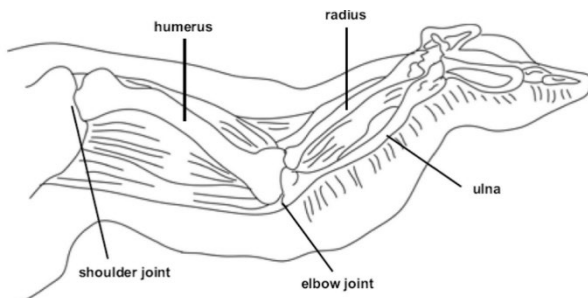
Maximum score: 8 points

| Assessment Level | Level Descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-8              | <p>i. <b>correctly collect, organize, transform and present</b> data in numerical and/ or visual forms<br/> <b>Data table is organized, correct headings, units and repeats. Percentages are used for blood cell counts.</b><br/> <b>Bar chart/pie chart or similar correctly labelled axes.</b></p> <p>ii. <b>accurately interpret data</b> and <b>outline</b> results <b>using correct scientific reasoning</b><br/> <b>All 4 blood samples are correctly categorized and explained using the given diagnosis information.</b></p> <p>iii. <b>discuss</b> the validity of a prediction based on the outcome of a scientific investigation <b>N/A</b></p> <p>iv. <b>discuss</b> the validity of the method based on the outcome of a scientific investigation<br/> <b>Any reasonable justification, that the method achieved reliable results or not.</b></p> <p>v. <b>describe</b> improvements or extensions to the method that would benefit the scientific investigation.<br/> <b>At least 2 improvements described in reasonably detail (more repeats, wider range of blood samples/people, larger samples (more blood cells to count))</b></p> |
| 5-6              | <p>i. <b>correctly collect, organize and present</b> data in numerical and/or visual forms</p> <p>ii. <b>accurately interpret</b> data and <b>outline</b> results <b>using scientific reasoning</b></p> <p>iii. <b>outline</b> the validity of a prediction based on the outcome of a scientific investigation</p> <p>iv. <b>outline</b> the validity of the method based on the outcome of a scientific investigation</p> <p>v. <b>outline</b> improvements or extensions to the method that would benefit the scientific investigation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3-4              | <p>i. correctly collect and present data in numerical and/or visual forms</p> <p>ii. accurately interpret data and outline results</p> <p>iii. state the validity of a prediction based on the outcome of a scientific investigation</p> <p>iv. state the validity of the method based on the outcome of a scientific investigation</p> <p>v. state improvements or extensions to the method that would benefit the scientific investigation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1-2              | <p>i. collect and present data in numerical and/or visual forms</p> <p>ii. interpret data</p> <p>iii. state the validity of a prediction based on the outcome of a scientific investigation, with limited success</p> <p>iv. state the validity of the method based on the outcome of a scientific investigation, with limited success</p> <p>v. state improvements or extensions to the method that would benefit the scientific investigation, with limited success.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0                | The student does not reach a standard described by any of the descriptors below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## 7B 9a Chicken Wing Dissection

**INTRODUCTION:** the chicken wing in some respects is similar to the human arm. It is composed of numerous different cells, tissues and organs. These body parts interact and cooperate to allow the chicken to perform a variety of movements.

### Bones of the Chicken



The **upper wing** consists of a **humerus**, which is at one end, and the **ulna** and the **radius** at the lower wing. These bones connect at the elbow joint. The rest of the wing is composed of modified hand bones.

### MATERIALS:

- Chicken wing
- Blunt and sharp dissecting probes
- Scalpel
- Paper Towel
- Forceps
- Dissecting tray
- Scissors

**CAUTION: Scalpels and scissors are very sharp.**

**Use extreme care. Only cut downward and/or away from your body.**

### PROCEDURE:

a) Examine the chicken wing, and compare it with the picture below. Do you have the left or right wing? (Circle one)

b) Identify the upper wing, the lower wing, and the wingtip.

c) Examine the wing at the point where it was removed from the body. Depending on the way the wing is cut, you might see cartilage and bone marrow.



d) Using scissors cut down the middle of the skin, starting at the top end of the upper wing. Try not to cut through the muscles below the skin. Do this by piercing the skin and then slipping the scissors between the skin layer and the muscle.

e) Cut down the sides of the skin to make a T-shaped cut. Start at the first cut and cut away from it in both directions. Peel the skin and cut to loosen it. (Note: chicken skin can be very difficult to remove.

**Take your time peeling it back so as not to** damage the tissues underneath it.)

• Circle 'left' or 'right' to match the wing being examined.

The upper wing (shoulder end) is the thickest part with the most muscle; the lower wing is the thinner middle section; the third part is the pointed wing tip.

**OBSERVATIONS Complete Table 1 and answer the questions as you carry out the dissection**

**Fat** - Look for yellowish tissue clumped together beneath the skin. This is fat tissue, made of fat cells. Where do you find the most fat tissue? Describe its location.

**Muscles** - Observe the muscles in the wing. Describe what you see – focus on colour and texture *The muscles are pinkish-red, soft and slightly stringy or fibrous in texture.*

- Find two muscles in the wing that bend and straighten the elbow joint. Each muscle pulls on the lower wing bones in one direction (the flexor bends the joint). Since the flexor cannot lengthen by itself to push the bone back to straighten the joint, another muscle pulls the bone in the opposite direction (extensor).
- Hold the wing down at the shoulder and alternately pull on each muscle. Observe what happens.

**Tendons** – These are shiny white tissues at the ends of the muscles that attach muscles to bones.

- Find as many tendons as you can on the chicken wing. Describe their locations carefully. *Tendons are found at the ends of the muscles, where the muscle joins a bone near the elbow joint.*
- Pull on a tendon to see how it helps the chicken move its wing. What do tendons do? *Tendons attach muscle to bone and pull on the bone when the muscle contracts, moving the wing.*

**Joints and Ligaments** - Two bones come together at a joint. Bend and straighten the elbow joint and observe how the bones fit together. Ligaments connect bones to other bones at joints. They look like a shiny white covering of the joint surfaces.

- Closely examine the elbow joint between the upper wing and the lower wing and identify the ligaments. Describe their location and appearance.

**Cartilage** - Between the bones is another shiny white material that is slippery. This is cartilage. What do you think its function is?

**Wing** - Move the wing again. Explore how the muscles, tendons, ligaments, and cartilage play roles in the wing's movement.

**EXTENSION** - Can you find any other examples of tissues? Add these to your table and describe them in as much detail as you can. (HINT: there should be at least three other tissue types to find – look for string-like structures, hollow tubes and more....)

When you have finished observing the wing and writing your notes, tidy up the equipment as directed by your teacher and wash your hands WITH SOAP.

*Most fat tissue is found just beneath the skin, clumped in yellowish lumps around the upper wing.*

*Ligaments are shiny white bands on the outside of the elbow joint, connecting the upper wing bones to the lower wing bones.*

*Cartilage reduces friction and acts as a cushion or shock absorber between the bones at the joint, stopping them grinding.*

*Muscles contract to pull bones and move the wing; tendons pass the pull to the bones; ligaments hold the bones together; cartilage lets the joint move smoothly.*

**Table 1**

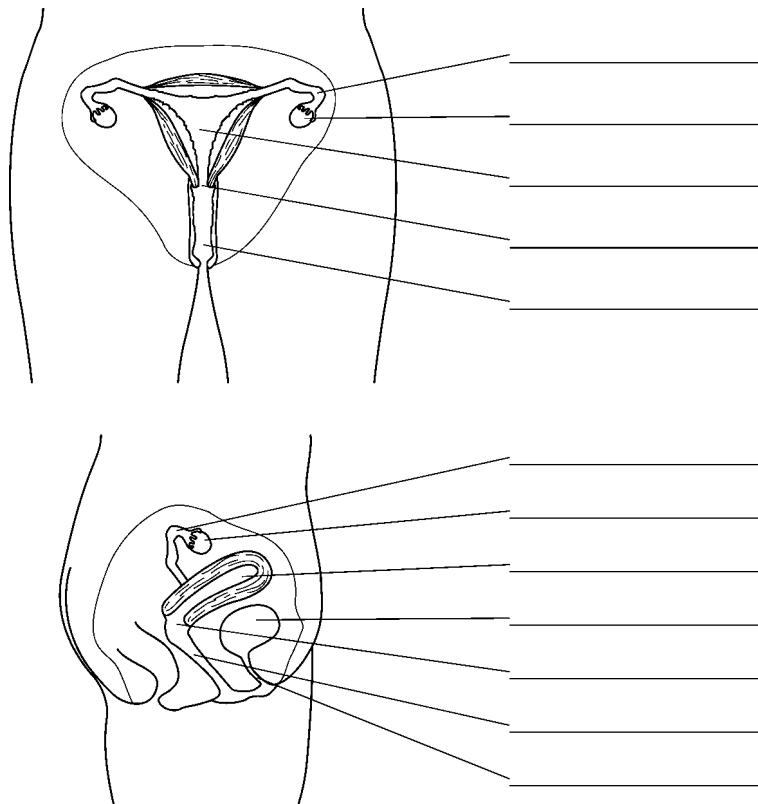
| <b>Tissue</b>    | <b>Description (colour, texture etc.)</b>                     | <b>Tissue it attaches to</b>                              |
|------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| <b>Skin</b>      | Pinkish, tough, elastic layer covering the whole wing.        | Muscle and fat lying beneath it.                          |
| <b>Fat</b>       | Yellowish, soft and clumped together in lumps.                | Skin and muscle under it.                                 |
| <b>Muscle</b>    | Pinkish-red, soft, stringy fibres.                            | Bones (through tendons) and skin.                         |
| <b>Tendon</b>    | Shiny white cords at the ends of the muscles.                 | Muscle to bone.                                           |
| <b>Ligament</b>  | Shiny white bands covering the joint surfaces.                | Bone to bone at the joint.                                |
| <b>Cartilage</b> | Smooth, shiny, slippery white material between the bones.     | Bone to bone (covers the ends of the bones at the joint). |
| <b>Blood</b>     | Red liquid that carries oxygen and nutrients around the body. | Runs inside blood vessels passing through all tissues.    |
| <b>Bone</b>      | Hard, white, rigid and strong.                                | Muscle (via tendons) and other bones at the joints.       |
| <b>Nerves</b>    | Thin white string-like strands carrying electrical signals.   | Skin and muscle (spread through the whole wing).          |

**HOMEWORK**

Your homework is to write a comprehensive conclusion about what you've learned. Start by explaining what a tissue is. Try to include all the words in bold from the table.

## 7B 10a The Female Reproductive System

- 1 Label the parts of the female reproductive system on the diagrams. Use the words in the box below. Each word can be used more than once.



- 2 Complete these sentences using the words in the box. Each word can be used more than once.

**bladder   cervix   oviducts   ovaries   urethra   uterus   vagina**

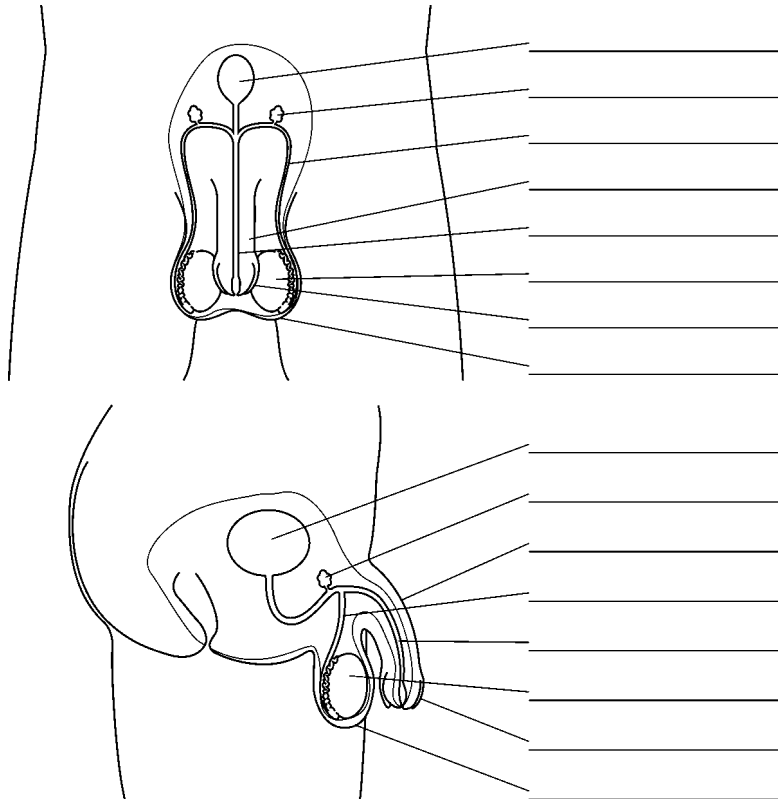
Label both diagrams using the words (each may be used more than once): bladder, cervix, oviducts, ovaries, urethra, uterus, vagina.

- a The egg cells are released from the **ovaries**
- b Egg cells travel down the **oviducts** on their way to the **uterus**
- c When a woman is pregnant the **cervix** helps to hold the baby in place.
- d A baby develops in the **uterus**

**Questions:** If you have any questions about the female reproductive system, you may want to write them down below and show this to your teacher.

## 7B 10b The Male Reproductive System

- 1 Label the parts of the male reproductive system on the diagrams.  
Use the words in the box below. Each word can be used more than once.



- 2 Complete these sentences using words in the box. Each word can be used more than once.

· Label both diagrams using the words (each may be used more than once): bladder, foreskin, glands, penis, scrotum, sperm duct, testes, urethra.

**bladder   foreskin   glands   penis   scrotum   sperm duct   testes   urethra**

- a Sperm cells are made in the

testes

- b The tubes that sperm cells travel down on their way out of the body are the

sperm duct

and

urethra

- c The head of the penis is protected by the

foreskin

- d

add special liquids to the sperm cells to make semen.

glands

**Questions:** If you have any questions about the male reproductive system, you may want to write them down below and show this to your teacher.

## 7B 10c Reproductive Organs

- 1 On the diagrams below, label the parts described in the sentences with the letters. The first one has been done for you.

**A** where female sex cells are stored

**D** where male sex cells are made

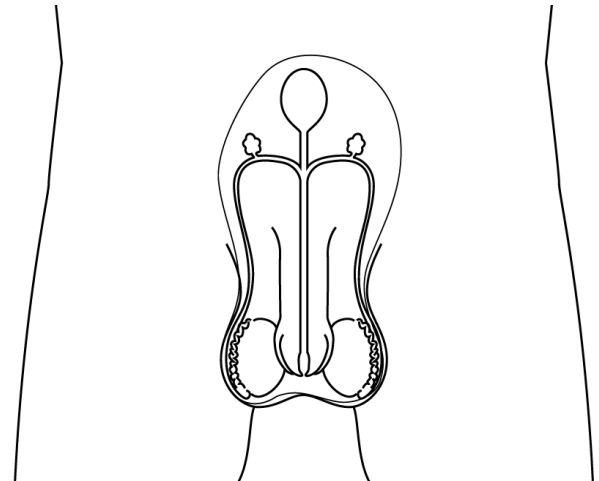
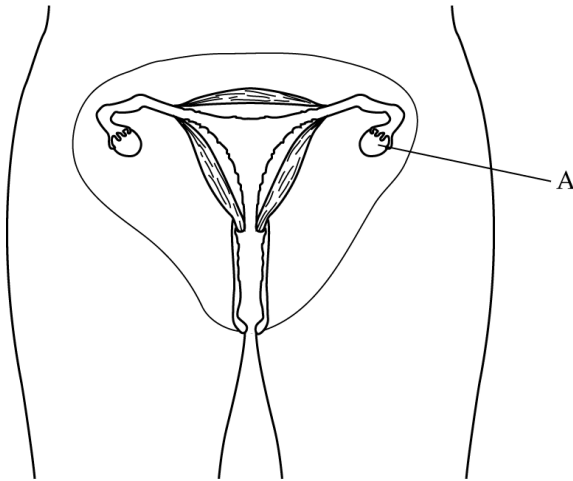
**B** where a baby could develop

**E** a tube containing cells with cilia (hairs)

**C** adds special fluids to sperm cells

**F** protects the head of the penis.

Put the letters on the diagrams: A on an ovary (female), B on the uterus (female), C on the glands (male), D on a testis (male), E on the oviduct (female), F on the foreskin (male).



- 2 Answer the following questions by circling the right answer.

a How often is an egg cell released from an ovary?

once a year

once a day

**once a month**

once a decade

never

b What does the cervix do?

**holds a baby in place**

produces egg cells

produces sperm cells

moves the egg cells

c What is the name given to the mixture of sperm cells and fluids?

sperm

oviduct liquid

scrotum

urine

**semen**

- 3 List the names of the tubes that sperm cells pass through on their way out of the body.

sperm duct and urethra

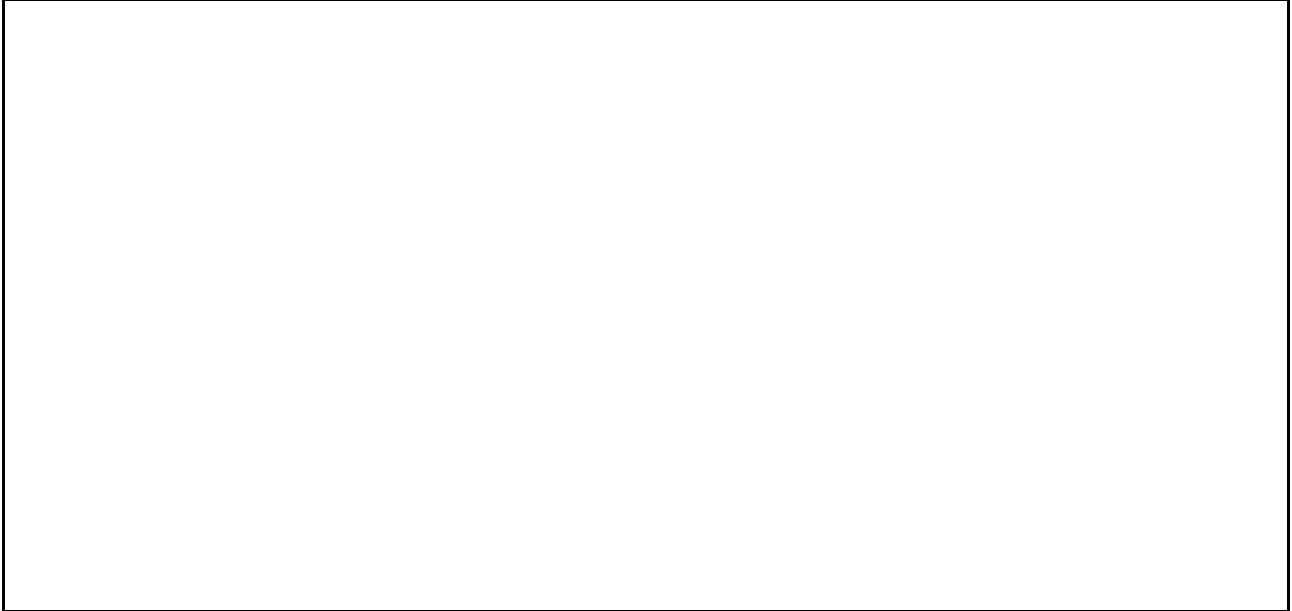
- 4 Explain why sperm cells are made in a part of the body that hangs outside.

The scrotum hangs outside the body so the testes stay cooler than body temperature; sperm cells need the lower temperature to develop properly and survive.

**Extension:** Research about sperm cells.

What makes them special? How many are produced? How is this different for egg cells.

Sperm cells are tiny and have a tail to swim; millions are produced each day (millions in one ejaculation), while only one egg cell is released each month.



**Extension:** Anne's ovaries are not releasing egg cells.

Why not? Think of as many reasons as you can.

She may be pregnant, or too young (before puberty) or too old (after menopause); her hormones may be imbalanced, so ovulation and the menstrual cycle do not happen.



## 7B 10d Reproductive System Word Search

- 1 Find the following words in the grid below. Draw a coloured line through the words that you find. If the word is something found in the female reproductive system, draw a red line through it. If it is something found in the male reproductive system, draw a blue line through it.\* If it is found in both, draw two lines (one red and one blue) through the word.

|         |         |            |            |          |       |
|---------|---------|------------|------------|----------|-------|
| bladder | cervix  | egg cell   | oviduct    | sex cell | ovary |
| penis   | scrotum | sperm cell | sperm duct | testis   |       |
|         | urethra | uterus     | vagina     |          |       |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P | A | M | O | U | S | E | E | N | F | G | F | G | R |
| F | T | R | O | T | U | F | D | B | O | V | A | R | Y |
| J | K | U | D | E | G | P | G | T | H | Q | L | Z | V |
| F | W | R | U | R | U | E | G | G | C | E | L | L | T |
| B | C | E | L | U | B | N | I | G | H | T | O | L | B |
| B | O | T | E | S | T | I | S | I | L | T | V | A | L |
| S | T | H | E | S | C | S | N | V | A | G | I | N | A |
| C | E | R | V | I | X | I | G | R | E | E | D | N | D |
| R | E | A | L | L | K | L | A | D | I | N | U | F | D |
| O | R | M | I | S | E | X | C | E | L | L | C | A | E |
| T | A | R | E | R | Y | A | E | C | E | L | T | S | R |
| U | A | R | S | P | E | R | T | S | U | C | B | T | I |
| M | O | F | A | L | S | P | E | R | M | C | E | L | L |
| F | S | P | E | R | M | D | U | C | T | O | P | I | A |

\* Draw a line through each listed word in the grid: red for words found in the female system, blue for the male system, and one red and one blue line for words in both.

- 2 a Take the first nine letters of the first row of the grid and rearrange them to make a word.

menopause

- b Explain what this word means.

Menopause is when a woman's periods stop permanently, usually around age 45-55, because her ovaries run out of eggs and stop releasing them.

## 7B 11a Teen Dreams Video Worksheet

1. a) What is the male sex hormone called? **Testosterone**  
  
b) What does it do to the male brain and body during puberty? List at least 4 effects.  
**Voice deepens, facial and pubic hair grows, shoulders broaden and muscles get bigger, the genitals grow and sperm production begins, and moods or sex drive change.**
  
2. a) What is the female sex hormone called? **Oestrogen**  
  
b) What does it do to the female brain and body during puberty? List at least 4 effects.  
**Breasts develop, hips widen, pubic hair grows, periods begin, egg cells mature and are released each month, and moods change.**
  
3. Why do males have some of the female sex hormone?  
**Males make a little oestrogen too; it is needed for healthy bones and normal body functioning, not just for reproduction.**
  
4. Why do females have some of the male sex hormone?  
**Females make a little testosterone too; it helps with growth, muscle strength, energy and sex drive.**
  
5. What can happen as a drawback to rapid growth during puberty?  
**Growth can be uneven, so arms and legs may grow faster than the body; teenagers can feel clumsy or unbalanced and may get stretch marks or tiredness.**

## 7B 11b Adolescence

1. Complete the table below. The first example has been done for you.

| Physical change                            | Boy          | Girl |
|--------------------------------------------|--------------|------|
| A<br>Hair grows on face                    |              |      |
| B<br>Underarm hair grows                   | Boy and girl |      |
| C<br>Voice deepens (breaks)                | Boy          |      |
| D<br>Hips widen                            | Girl         |      |
| E<br>Pubic hair grows                      | Boy and girl |      |
| F<br>Breasts develop                       | Girl         |      |
| G<br>Shoulders broaden and muscles develop | Boy          |      |
| H<br>Periods begin                         | Girl         |      |
| I<br>Sperm production begins               | Boy          |      |

Other changes that are not physical happen during adolescence.

What changes are these?

Emotional and social changes: mood swings, new feelings for other people, wanting more independence, and changing relationships with friends and family.

2. Write a definition for the term “adolescence”. Adolescence is the stage of life between childhood and adulthood (roughly ages 10-19) when the body and mind change from a child to an adult.

Explain how puberty is different from adolescence.

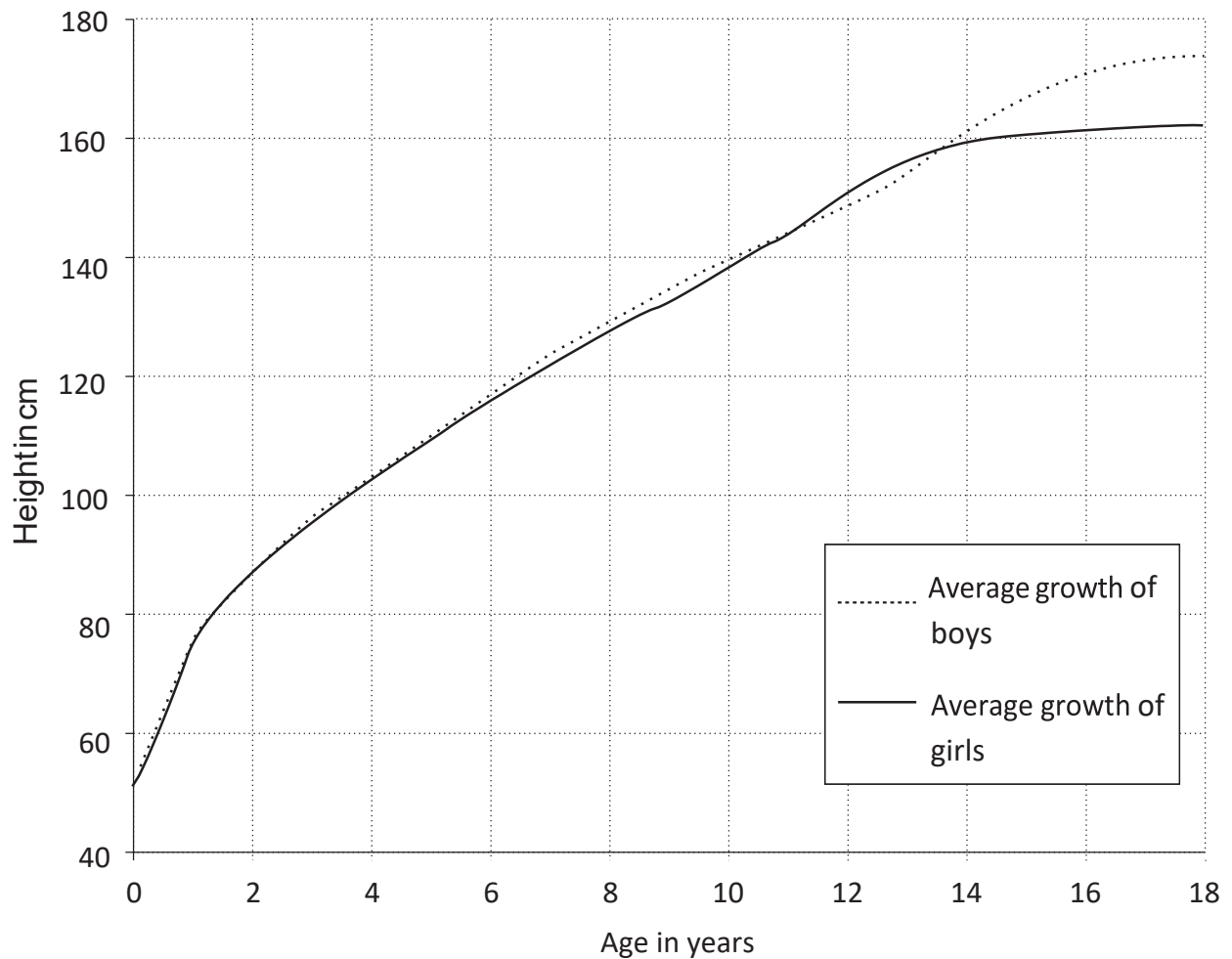
Puberty is only the physical changes of the body (growth and sexual development); adolescence is the whole period and also includes emotional and social changes.

3. Your body consists of lots of cells made by cell division. Each cell lives for a certain period of time, then dies. Cell division can happen faster, slower or at the same speed as cell death. This controls how much we grow.

Explain how the rates of cell division and cell death compare during adolescence

During adolescence cell division happens faster than cell death, so the body gains new cells and grows.

4



**a** Look at the graph. Who grows faster, boys or girls:

i. At age 12?

ii. At age 15?

**b** How does this difference relate to the start of puberty in boys and girls?

Girls grow faster than boys at age 12 because girls start their growth spurt earlier. Girls start puberty earlier (around age 10-11) so their growth spurt comes first; boys start later (around 12-13) so their growth spurt and rapid growth come later.

Boys grow faster than girls at age 15 because boys' growth spurt starts later.

**EXTENSION**

**5 a** Testosterone and oestrogen are very important body chemicals.

What is the general name for these chemicals?

They are hormones - chemical messengers made by glands and carried in the blood, e.g. oestrogen, progesterone, FSH and LH which control the menstrual cycle.

Hormones.

FSH and LH which control the menstrual cycle.

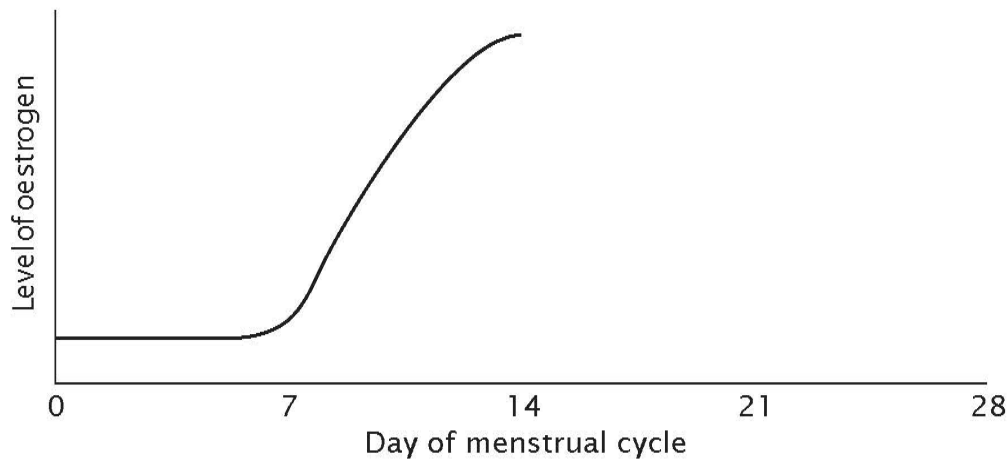
**b** Write a paragraph to explain how these two chemicals are linked to boys and girls, and why they are important.

Testosterone is the main male hormone and oestrogen the main female hormone; they cause the puberty changes, control the menstrual cycle and reproduction, and are needed for growth.

**6** Read the following information about hormones and the menstrual cycle.

The menstrual cycle is controlled by the hormones called oestrogen and progesterone. While an egg is developing in the ovary, ready to be released into the oviduct, oestrogen makes the lining of the uterus grow thicker, ready to receive a fertilised egg.

The graph shows how the oestrogen levels change during the first half of the menstrual cycle.



**a** Copy the graph and continue the curve to show what you think happens to the oestrogen level after day 14.

Copy the graph and continue the curve: after day 14 the oestrogen level falls back towards its starting low level (it peaks around day 14 just before ovulation, then drops).

**b** Explain why you think this happens.

Ovulation happens about day 14, so after the egg is released oestrogen falls; the low level at the end of the cycle lets the uterus lining break down, so the period starts.

- c** Mark an X on the curve to show when ovulation takes place.



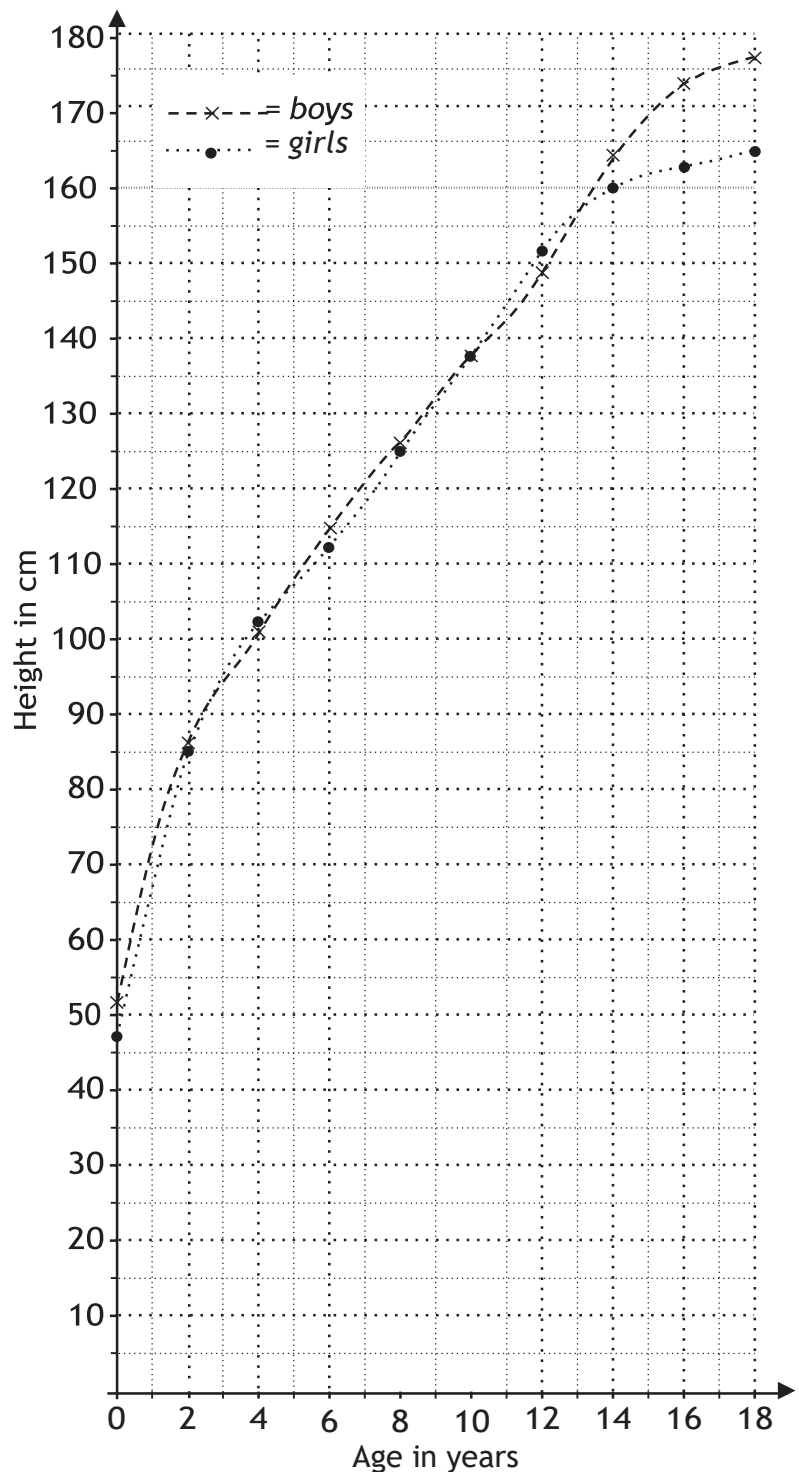
- d** Circle the days on the horizontal axis when the woman is most likely to have her period.

Circle the days at the end of the cycle (about days 26-28) on the horizontal axis, when the period is most likely.

## 7B11c Growth spurts

Children are growing all the time, but as teenagers you grow faster for a couple of years. This fast growth is called a growthspurt. You are going to analyse a graph showing growth of children.

The graph below shows the average height of girls and boys today. A graph can tell you how fast or slow something is happening. A steep line shows faster growth. A less steep line shows slower growth.



This data was obtained from Davidson and Passmore Human Nutrition and Dietetics [Passmore and Eastwood]

1 Between what ages are girls taller than boys?

Between about 10 and 12 years old.

2 At age 12 girls are taller than boys. At what age will boys be taller than girls again?

About 14 years old, when boys' growth spurt has made them catch up with and pass the girls.

3 At age 6, how much taller are boys than girls? Boys are about 1-2 cm taller than girls at age 6.

4 At age 18, how much taller are boys than girls? Boys are about 12-13 cm taller than girls at age 18.

5 How much do girls grow between ages 16 and 18? Very little - about 1-2 cm, because girls have almost finished growing by age 16.

6 How much do boys grow between ages 16 and 18? About 4-6 cm, because boys are still in or just finishing their growth spurt.

7 Who grows the most between ages 16 and 18? Boys grow the most between ages 16 and 18.

8 At age 10 ½ are girls or boys growing faster? Girls are growing faster at age 10 1/2, because their growth spurt has already started.

9 At age 13 ½ are girls or boys growing faster? Boys are growing faster at age 13 1/2, because their growth spurt has started.

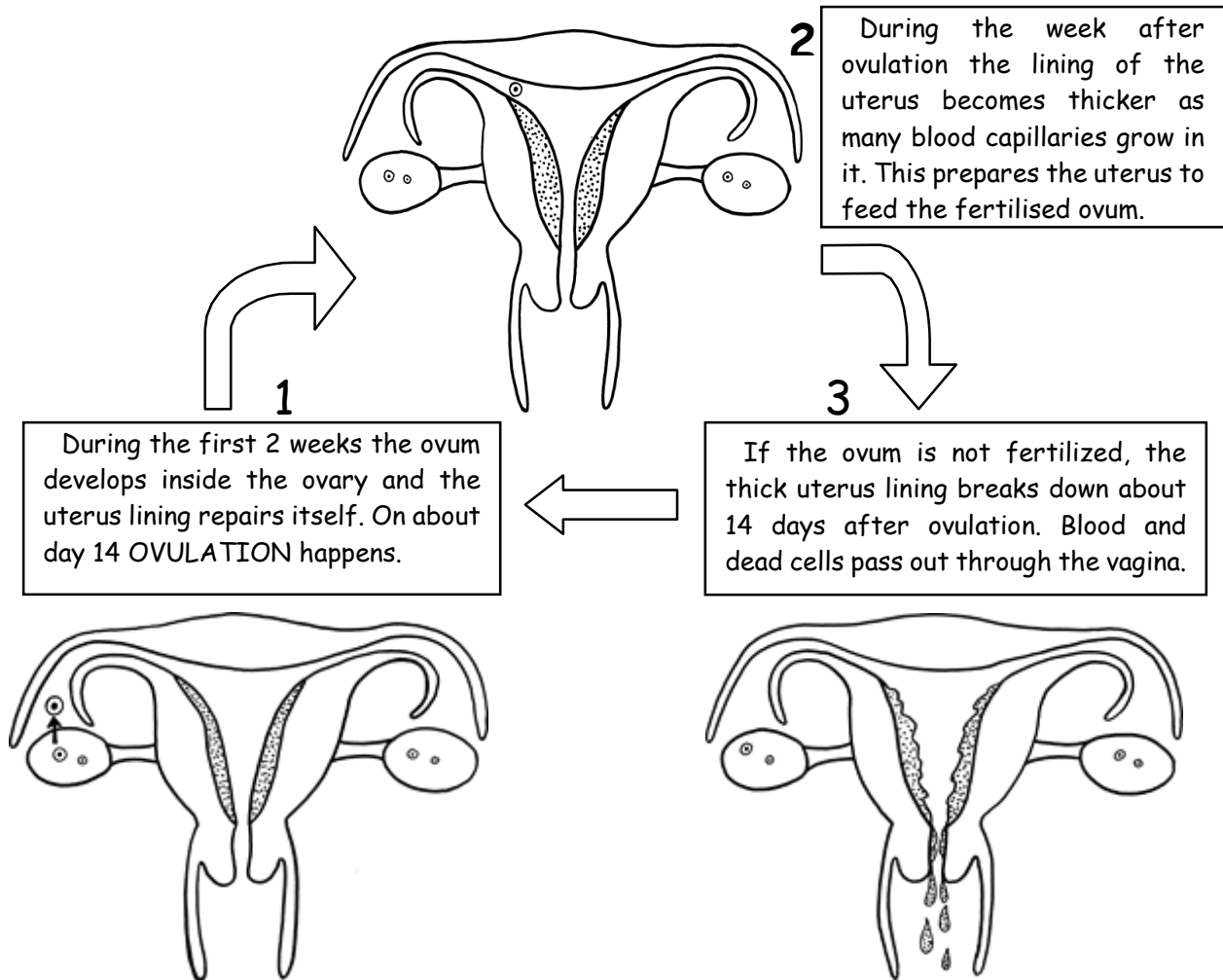
10 At what age between 0 and 18 do girls and boys grow fastest? Girls grow fastest at about age 12 and boys fastest at about age 14.

11 The estimated average height data for a man aged 18 years in the 19th century is 165 cm. Why do you think this differs from the recently collected data above?

People today are taller because of better diet, nutrition and healthcare than in the 19th century.

## 7B 12a The Menstrual Cycle

Once every month a woman's body releases an ovum (egg cell) into the oviduct (Fallopian tubes). Usually the ovum is not fertilised and it dies. The woman has her period when the lining of the uterus breaks down and blood and dead cells pass out through the vagina. The diagram below shows what happens during a woman's monthly cycle.



Exercise - Complete the sentences below.

1) Only one ovum is released every

month

2) The release of an ovum from the ovary is called

ovulation

3) Ovulation happens after about

14

days.

4) The uterus lining

feeds

the fertilised ovum..

5) If the ovum is not fertilised it will

die

6) A woman has her period when the

uterus

lining breaks down.

## 7B 12b Menstrual Cycle Questions

1. Add these words into the spaces in the sentences below.

**period egg menstrual vagina hormones uterus pregnant**

A woman's monthly cycle is called the **menstrual** cycle.

It is controlled by **hormones** that move in her bloodstream.

The cycle begins with her **period**, which is when blood

and dead cells pass out of her **vagina**,

which had thickened ready for a fertilised **egg**

A woman has her period only if she is not **pregnant**

2 Copy and complete these sentences by choosing from the words in **bold**.

When an egg splits into two after fertilisation, the twins born are

**identical** / **non-identical**. They look **different** / **the same**

When two eggs are fertilised at the same time, the twins born are

**identical** / **non-identical**. They look **different** / **the same**.

3 You will need a calendar for these questions.

**a** Lucie begins her period on 3rd April. She has a regular 28-day menstrual cycle.

i. On which date would ovulation happen? **About 16th April (day 14 of her cycle: 3rd April + 13 days).**

ii. When would Lucie expect her next period to begin? **1st May (3rd April + 28 days).**

iii. On which date, in March, did her last period start? **6th March (3rd April - 28 days).**

**b** With a 28-day menstrual cycle, how many periods should Lucie expect to have every year? **About 13 periods a year ( $365 / 28 = 13.04$ ).**

**c** Lucie had her first period on her thirteenth birthday, and had 481 eggs in her ovaries. At what age would she stop having periods if she never became pregnant? **About 50 years old ( $481 / 13 = \text{about } 37$  years of periods, starting at age 13).**

**EXTENSION**

**4** Use this information to answer the questions.

A gerbil is fertile and can become pregnant again within a couple of days of giving birth. The gestation period for a gerbil is 24 days. If the male gerbil is not removed immediately the babies are born, he will mate again and the female will be pregnant within seven days. Typically, a gerbil has five or six babies in each litter. If another litter is born within a month, the mother will abandon the first litter and the babies will die.

- a** How many babies could a female gerbil have in a year if the male is not removed when a litter is born? Show your working.

Working: the female can mate again straight after giving birth; gestation is about 24 days, so about 15 litters a year, each with about 5 babies = about 75 babies in a year.

- b** A gerbil typically lives for 2½ to 3 years and the female is fertile from about 2 months old to 18 months old. Why do you think she reproduces so frequently in this time?

Her fertile life and lifespan are short, so she must produce many offspring quickly so that some survive and her genes are passed on.

- c** If you are a gerbil breeder you usually keep a male with four or five females. Suggest why this might help the babies to survive.

The females give birth at different times, so there is always an adult to care for the young and each mother has time to recover, so more babies survive.

- 5** When a woman reaches around 50 years of age she begins her menopause. This is when her ovaries contain very few eggs. Explain why this causes her periods to stop.

Her ovaries contain very few eggs, so an egg is no longer released each month; without ovulation the cycle's hormones do not trigger a period, so periods stop.

## 7B 12c The Menstrual Cycle (extra questions)

- 1 The diagram shows how the thickness of the lining of the uterus changes. Write these labels in the spaces above the diagram.

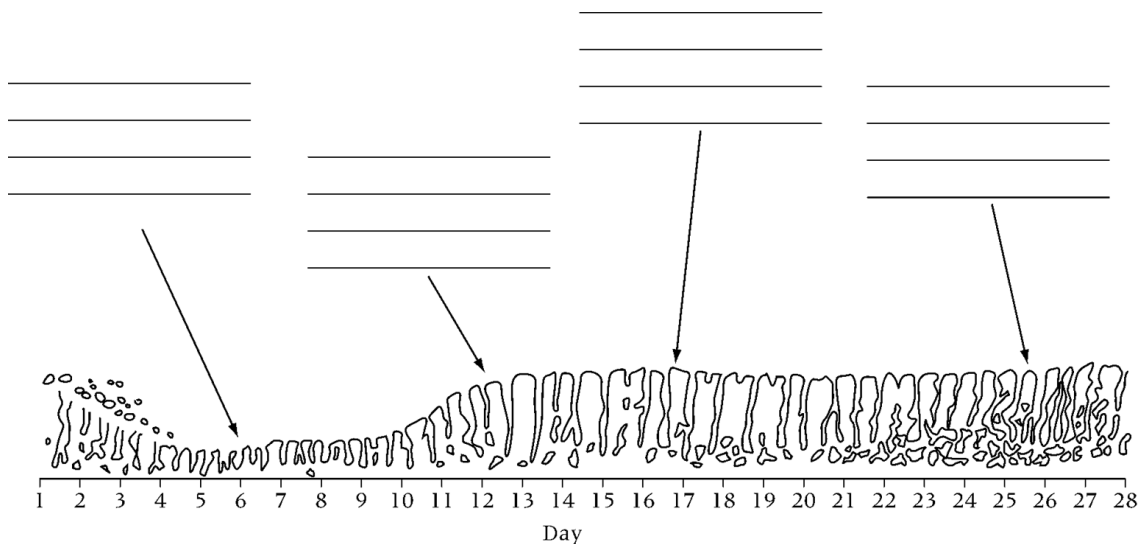
The lining gets thicker

An egg is released

The lining breaks down

The lining stays thick

· Label the four stages on the diagram in cycle order: the lining breaks down (period), the lining gets thicker, an egg is released (ovulation), the lining stays thick.



- 2 Now complete these sentences using the words in the box.

**blood fertilised menstruation ovaries ovulation period vagina**

a Egg cells are released from the **ovaries**. The time when they are

released is called **ovulation**. If the egg cell meets a sperm cell it

will be **fertilised**. If it does not, it will die.

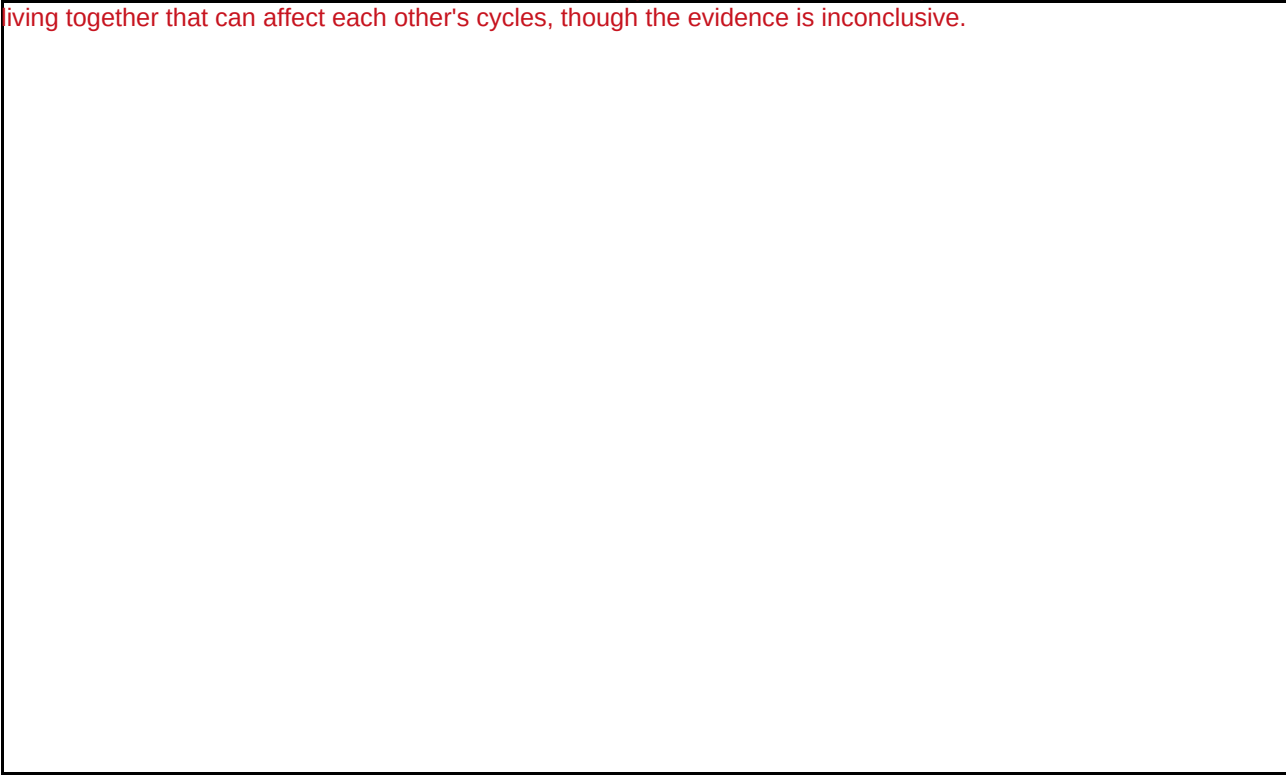
b The time when the uterus lining breaks apart is called **menstruation**

Another name for this is 'having a **period**'. During this time the uterus

lining and a little **blood** passes out of the **vagina**

**Extension:** It seems strange that women living in the same home, often have their menstrual cycle at the same time of the month. Is this actually true? If so, try and find out some reasons why.

This is called menstrual synchrony; it may be caused by pheromones, chemical signals between women living together that can affect each other's cycles, though the evidence is inconclusive.



## 7B 13a Sex Cells (Gametes)

- 1 The male and female sex cells have different shapes to help them do their jobs. They are said to be **adapted** to their jobs.

Here is a list of how cells can be adapted. In the space next to each sentence write the words sperm cell, egg cell, both or neither.

a It has a tail to help it to swim. sperm cell

b It has a food store. egg cell

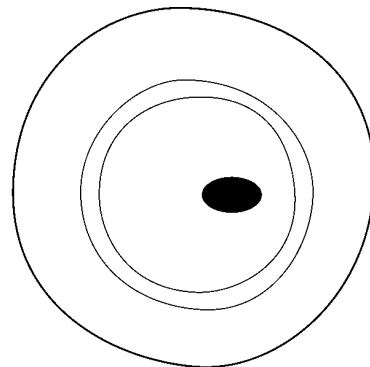
c It has special hairs on it to help it push things along. neither

d It contains half the information needed to make a new life. both

e It has special chemicals on the tip of its head to help it get into things. sperm cell

f It has only a little cytoplasm so that it can have a thin, streamlined shape. sperm cell

- 2 a On the diagrams below label where the **adaptations** in question 1 are found. Just use the letters for the labels.



A sperm cell at a magnification of x1000.  
magnification of x500.

An egg cell at a

- b Look at the drawings of the sex cells above. On the lines below, write down how long or wide each one is in real life.

A sperm cell is  mm long. Its head is  mm wide.

· Label the sperm diagram with a (tail), e (chemicals on the head tip) and f (little cytoplasm, streamlined); label the egg diagram with b (food store) and d (nucleus with half the information).

An egg cell is  mm wide.  
0.1  
0.05

0.005

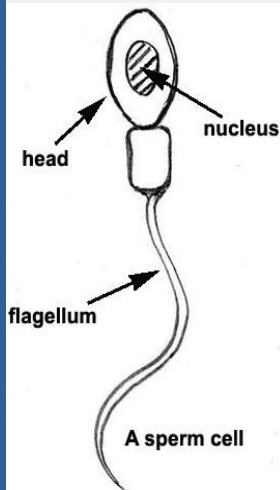
**Extension:** Explain how identical twins may form compared to non-identical twins.

Identical twins form when one fertilised egg splits into two, so they share the same genes and are the same sex; non-identical twins form when two separate eggs are fertilised.

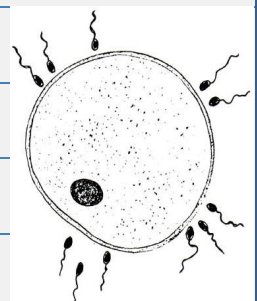
## 7B 13b The journey of a sperm cell

### Keywords

Testicles    Sperm duct    Urethra    penis    Vagina    Cervix  
Uterus    Fallopian tube    Egg cell    Fertilise

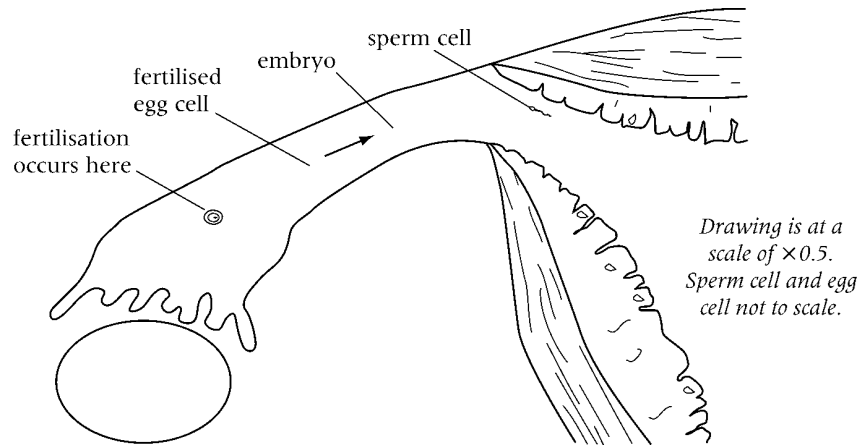


· Write the keywords at the correct places on the diagram: testes (sperm made) to sperm duct to urethra to penis to vagina to cervix to uterus to fallopian tube, meeting the egg cell and fertilising it.



## 7B 13c Making an Embryo

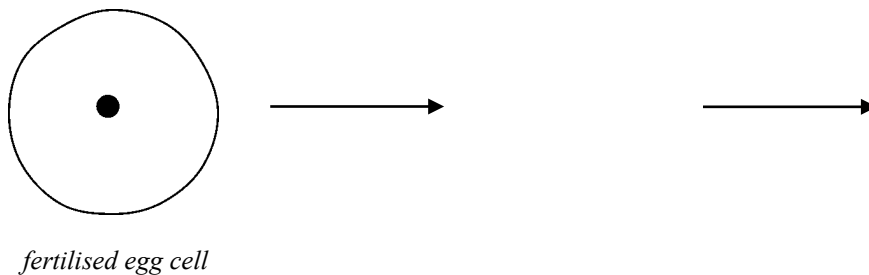
- 1 The place where fertilisation normally occurs is marked on the diagram. Sperm cells start swimming when they get to the oviduct. A sperm cell is marked. Sperm cells swim **about 1 mm in every minute**.



- a Work out how long it would take for the sperm cell to get to the egg cell (use a ruler to measure). Show your working.

Measure the distance from the sperm to the egg with a ruler in mm, then divide by 1 mm per minute (time = distance / speed); e.g. 5 mm / 1 mm per min = 5 minutes.

- b In the places marked on the diagram draw in the fertilised egg cell and the embryo.  
c The fertilised egg cell turns into an embryo by **cell division**. The fertilised egg cell divides into two new cells. These two new cells divide into two more cells each. Draw two more diagrams to show what happens.



- d Look at the last drawing you did. If each of these cells divides into two, how many cells will the embryo now have?

16 cells ( $8 \times 2 = 16$ ).

· Draw a fertilised egg cell in the oviduct where fertilisation happens, and an embryo further along the oviduct towards the uterus.

- e What is **implantation**?

Implantation is when the embryo attaches (beds) into the lining of the uterus, where it continues to grow.

· Draw two more diagrams to show: 1 cell to 2 cells to 4 cells to 8 cells, because every cell divides into two.

- f On the diagram, label the place where implantation occurs.

· Label implantation at the uterus, where the embryo attaches to the uterus lining.

2 These sentences are about twins. Write 'true' or 'false' by each sentence.

a Identical twins happen when a fertilised egg cell divides in two, but the two cells split apart. true

b Identical twins will always be different sexes. false

c Non-identical twins happen when two egg cells are fertilised. true

d Non-identical twins can be the same sex or different sexes. true

e Identical twins happen when two egg cells are fertilised. false

f Non-identical twins happen when egg cells turn into embryos on their own. false

**Extension:** Sometimes some couples find it difficult to get pregnant. Do some research and find out about different types of **fertility treatment**. How do these improve the chances of successful fertilization?

Fertility drugs make more eggs be released, and IVF fertilises eggs outside the body before they are put back in the uterus; both improve the chance of fertilisation.

## 7B 13d Fertilisation

Put the steps of fertilisation in the correct order

1 sperm into vagina; 2 semen to top of vagina; 3 uterus via cervix to oviduct; 4 breaks egg lining, joins; 5 divides into ball of cells; 6 embryo attaches to uterus lining, becomes foetus.

During ejaculation,  
the man's penis  
releases sperm  
into the vagina

Sperm breaks  
through the egg's  
outer lining and  
they join together

Sperm enters uterus  
through the cervix  
and travels to the  
egg/fallopian tubes

Fertilised egg divides  
to form a ball of cells  
called an embryo

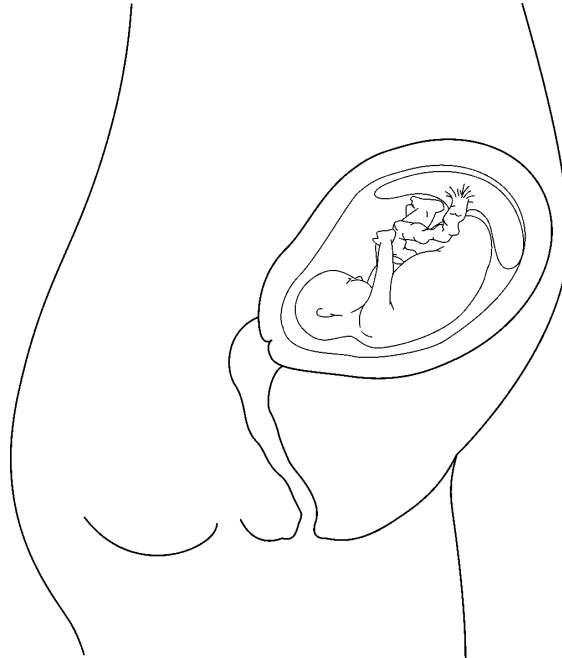
Embryo attaches to  
the lining of the  
uterus and develops  
into a foetus

The sperm travels in  
semen to the top of  
the vagina

## 7B 14a The Developing Fetus

- 1 a Label the following parts on the drawing below.

amniotic fluid    umbilical cord    fetus    placenta    vagina    uterus



- b Add an arrow in pen to show the direction food and oxygen takes between the placenta and umbilical cord.
- c Add an arrow in pencil to show the direction carbon dioxide and waste takes between the placenta and umbilical cord.

· Label on the diagram: fetus (inside the uterus), amniotic fluid (around the fetus), placenta (on the uterus wall), umbilical cord (fetus to placenta), uterus, vagina.

- 2 Fill in the table to show what each part does.

|                |                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part           | Draw a pen arrow from the placenta towards the fetus to show food and oxygen travelling through the umbilical cord to the fetus.                                                                                        |
| Amniotic fluid | Cushions the fetus and protects it from bumps and shocks.                                                                                                                                                               |
| Umbilical cord | Carries food and oxygen from the placenta to the fetus, and waste back. Draw a pencil arrow from the fetus towards the placenta to show carbon dioxide and waste travelling through the umbilical cord to the placenta. |
| Placenta       | Food, oxygen and antibodies pass from the mother's blood into the fetus's blood; waste passes the other way.                                                                                                            |

- 3 What do you think happens to the uterus as the fetus grows?

The uterus stretches and expands (gets bigger) so it can hold the growing fetus.

**Extension:** Find out why it is dangerous for a pregnant woman to drink alcohol, smoke or take drugs.

Alcohol, smoke and drugs cross the placenta into the fetus's blood and can damage its development, causing low birth weight, miscarriage or birth defects.



## 7B 14b Stages of Pregnancy

Try to match each diagram with the correct description.

### 3 weeks

The fertilised egg is dividing rapidly. It has formed a ball of cells. It is too small to be seen without a microscope.

### 12 weeks

The fetus is about 50 mm long. It can kick its legs and move its fingers. Its head is out of proportion with the rest of its body. Most of the organs inside its body are working.

### 6 weeks

The ball of cells now looks like a tadpole. There is a 'head end' and a 'tail end' and a bulge where the heart is forming.

### 24 weeks

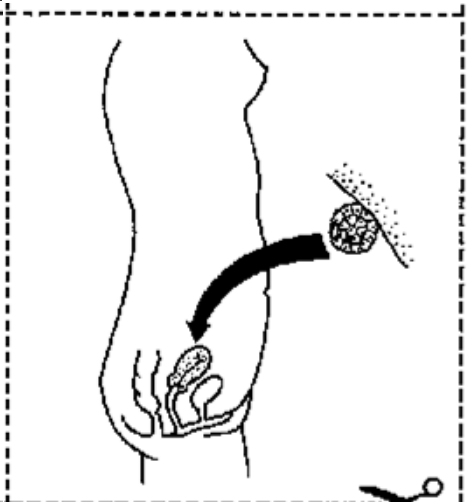
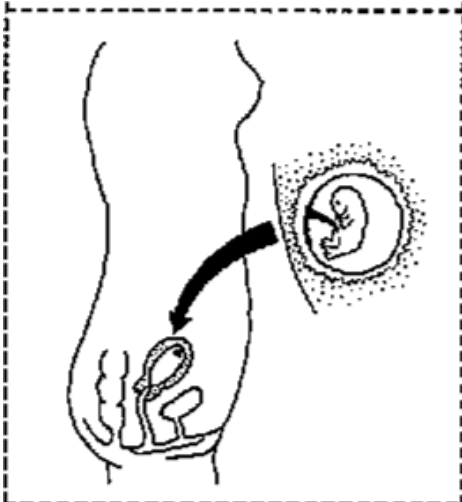
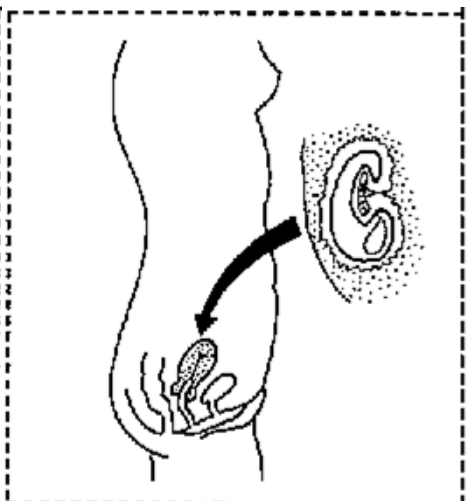
The fetus has grown so much that the mother looks pregnant. She can feel her baby moving inside her. Its heart can be heard beating.

### 8 weeks

You can now see the eyes. The arms and legs have formed with tiny fingers and toes. At this stage the baby is called a fetus.

### 30 weeks

The baby is almost ready to be born. Its head is just above the cervix. When the muscles of the mother's uterus contract, it will be born head first.



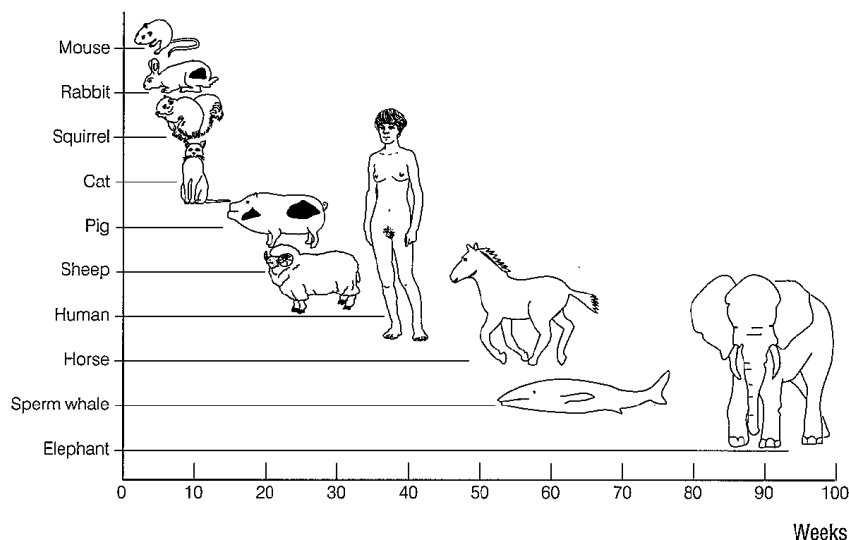
· Match each diagram to the correct description, in order: fertilised egg cell -> ball of cells (embryo) -> embryo implants in the uterus lining -> embryo develops -> fetus -> baby ready to be born (about 38 weeks).



## 7B 14c How long in the womb?

**Gestation** is the length of time a pregnancy lasts. It is the time from fertilisation until birth.

Look at the chart below, which shows the gestation time for different mammals.



1 What can you say about the differences in gestation time between a rabbit and a pig?

A pig's gestation is much longer than a rabbit's (about 10 weeks compared with about 4 weeks). The length of time in the womb is the gestation period - about 9 months (38-40 weeks) for humans; because a pig is larger than a rabbit.

2 What is the gestation time of a) a sheep? About 147 days (about 5 months).

b) a horse? About 336 days (about 11 months).

3 Do your answers to question 2 fit in with your ideas about a rabbit and a pig?

If so, explain how.

Yes. Sheep and horses are bigger than rabbits and pigs and have longer gestation times, so the pattern 'bigger mammal = longer gestation' still fits.

4 Can you see any pattern in the gestation times?

The bigger the mammal, the longer its gestation time: small mammals have short pregnancies and large mammals have long ones.

5 a) Can you name an animal that does not fit into this pattern? A kangaroo (a marsupial).

b) Describe how it does not fit the pattern.

It is a large mammal but its gestation is very short (about 33 days) - its young are born tiny and finish developing in the mother's pouch.

### Mark scheme

You will receive 2 marks for each question correctly answered.

## 7B 14d Birth and After

Match the half-sentences from the left-hand side (these start with a number) with those from the right-hand side.

|                                                                     |
|---------------------------------------------------------------------|
| <b>1</b> A baby is ready to be born                                 |
| <b>2</b> Usually it is lying head downwards                         |
| <b>3</b> The start of birth is when the mother                      |
| <b>4</b> These 'squeezes' are called contractions                   |
| <b>5</b> The mother is now                                          |
| <b>6</b> The baby is surrounded by a bag                            |
| <b>7</b> As the uterus slowly pushes the baby out                   |
| <b>8</b> Contractions now become stronger                           |
| <b>9</b> When the baby is born it is still attached to the placenta |
| <b>10</b> Later the rest of the cord and the placenta come out      |

|                                                     |
|-----------------------------------------------------|
| by the cord which must be cut and tied.             |
| when it is 9 months old.                            |
| and this is called the after-birth.                 |
| the sac bursts letting out the fluid.               |
| feels the muscles of her uterus start to 'squeeze'. |
| since this position makes it easier for birth.      |
| and at first they happen every 30 minutes.          |
| in labour.                                          |
| and come every 3 minutes.                           |
| or sac containing fluid.                            |

· Match each numbered half-sentence with its ending: the uterus contracts -> pushing the baby out through the cervix and vagina; the cord is cut -> leaving a scar called the navel; the placenta comes out -> as the afterbirth; the baby is fed on milk -> from the mammary glands.

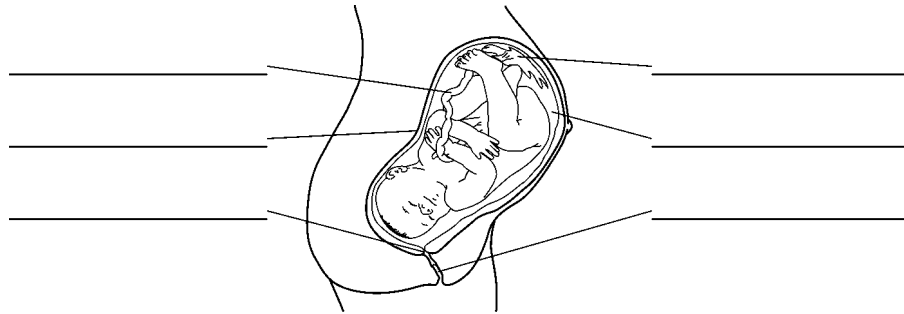
## 7B 14e Being Born

- 1 Write these words in order of size, smallest first: baby, fertilised egg cell, fetus, embryo.

fertilised egg cell, embryo, fetus, baby

- 2 a Label the drawing below with the following words:

uterus    amnion    cervix    vagina    placenta    cord



- b Which parts does the baby go through on its way out of the uterus?

The cervix and then the vagina.

Labels on the drawing: uterus (the muscular bag), amnion (the bag of fluid around the baby), cervix (the neck of the uterus), vagina (the birth canal), placenta (attached to the uterus lining), cord (joining the baby to the placenta).

- c Which part breaks when the baby is ready to be born?

The amnion - the bag of fluid around the baby breaks ('the waters break').

- d Which part needs to be cut after the baby has been born?

The cord (umbilical cord).

- e Which part comes out as the **afterbirth**?

The placenta.

- f How does a woman know that she is about to give birth?

The uterus starts to contract and she goes into labour - she feels regular painful contractions and the waters may break.

- g What is a **navel**? The navel is the scar left on the baby's abdomen where the cord was cut.

- 3 a How is the baby protected inside the uterus?  
It floats in amniotic fluid inside the amnion, which cushions it against knocks, and it is kept warm and safe inside the uterus.
- b How is it protected after birth? Its parents look after it and feed it milk, keep it warm and clean, and the antibodies in the milk destroy microbes that cause disease.
- 4 a How does the baby get food and oxygen in the uterus? Food and oxygen pass from the mother's blood through the placenta and along the cord into the baby's blood.
- b How does it get these after birth? It breathes air into its lungs to get oxygen and feeds on milk through its mouth to get food.

**Extension:** Research and explain how the **placenta** works.

The placenta is joined to the uterus lining. The mother's blood brings oxygen and food to it; these pass across into the baby's blood along the cord, and waste (carbon dioxide) passes back the other way. The two blood supplies never mix.

# IVF – The Facts

- Look through the facts about IVF below.
- Highlight the facts you already know in yellow.
- Highlight the facts you are most surprised by in pink.



## Scientific

Infertility is the inability to get pregnant (conceive).

- The fertility rate in HK is 1.19 (this means that on average, most couple in HK have 1 child).
- IVF means ‘in vitro’ fertilization.
- The first IVF baby was born in the UK in July 1978. Her name is Louise Brown.
- In 1997 the IVF pregnancy rate was 2%
- The IVF pregnancy rate in HK was 33% per cycle in 2018.

## Legal

- In HK ONLY heterosexual married couple are eligible for IVF treatment (single women and unmarried couples do not qualify for IVF)
- Choosing the sex (gender) of a baby is illegal.
- If one partner removes consent for use of eggs/sperm, embryos must be destroyed.

## Political

- ONLY permanent HK residents are eligible for subsidized IVF treatment in HK.
- Women have to be younger than 40 years old and have NO biological children.

## Economic

- The hospital authority in HK offer subsidized IVF treatment at \$20 000 per cycle.
- The average waiting time through the public system is 3 years.
- Couples may choose to opt for private treatment which reduces waiting time but costs between \$80 000 – 100 000 per cycle.

## Social

- Some pro-life supporters disagree with destroying embryos that are not used in treatment.
- Some religious leaders believe that we shouldn't interfere with natural reproduction.
- Many women are waiting until they are older to have children, when their fertility is lower and it is more difficult to conceive.
- “Designer babies” is a term used by the media. It describes an embryo chosen for specific characteristics or genetic traits.

## 7B 15b Family Matters

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Family A</b></p> <ul style="list-style-type: none"> <li>Mr and Mrs A are in their mid thirties and are unemployed.</li> <li>They come from large families and desperately want children of their own.</li> <li>They received one cycle of IVF on the NHS and spent their life savings on a second cycle. Both were unsuccessful.</li> <li>They cannot afford to pay for this treatment privately again.</li> </ul>                                              | <p><b>Family B</b></p> <ul style="list-style-type: none"> <li>Miss B is single and in her late forties.</li> <li>She is a successful business-woman.</li> <li>She always intended to have children, but never met the right man.</li> <li>She has tried insemination with donor sperm but did not conceive.</li> <li>Miss B would like IVF to try IVF treatment again with donated sperm.</li> </ul>     |
| <p><b>Family C</b></p> <ul style="list-style-type: none"> <li>Mr and Mrs C have been married for 2 years.</li> <li>Mr C is in his mid forties. Mrs C is thirty.</li> <li>He has one son from a previous marriage who lives with them.</li> <li>They have been trying for a child for 3 years, but with no success.</li> </ul>                                                                                                                                         | <p><b>Family E</b></p> <ul style="list-style-type: none"> <li>Mrs E is a 35 year old widow.</li> <li>Mr E died of cancer, but froze a sample of his sperm before he died.</li> <li>Mrs E has a good job that pays well.</li> <li>She is a smoker.</li> <li>Mrs E would like to have IVF with husband's frozen sperm to have the baby she has always wanted.</li> </ul>                                   |
| <p><b>Family D</b></p> <ul style="list-style-type: none"> <li>Mr and Mrs D are in their early twenties and have been married for five years.</li> <li>Mr D previously decided he did not want any children, so he had a vasectomy which prevents him having children.</li> <li>They now both want children.</li> <li>Mr D has tried, unsuccessfully, to have the vasectomy reversed.</li> <li>They would now like to have IVF treatment using donor sperm.</li> </ul> | <p><b>Family F</b></p> <ul style="list-style-type: none"> <li>Mrs F and Mrs F are a gay couple and have been together for ten years.</li> <li>They are in their mid thirties.</li> <li>Mrs F's male friend has offered to donate sperm so the couple can have a child.</li> <li>They would like to use IVF treatment to fertilise their eggs before implantation of embryos into one of them.</li> </ul> |

### Who should receive IVF treatment and why?

A, C and D: married couples, woman under 40, no biological children, and a medical reason (failed IVF / vasectomy). B (single, over 40), E (widow, no partner's consent) and F (gay couple) are not eligible in HK.

### Reflecting on the use of a reproductive technology

*With an increasing knowledge and understanding of the natural world including ourselves we are now in a position to manage not only our environment but also our own biology including our reproduction. This has consequences for individuals but also for society and even the world at large.*

*While many countries such as Japan and Germany worry about a falling birth rate and an ageing population, other countries such as China and India are trying to increase the life expectancy of their people and reduce birth rates.*

*While many couples use forms of contraception to prevent them from having children, more and more couples across the world find that they are unable to conceive children naturally and seek medical help.*

*The reproductive technologies / aids available are:*

- Barrier methods (condom, diaphragm)
- Contraceptive pill
- Sterilization (Male and female)
- Caesarean section
- Fertility treatments
- Surrogacy
- IVF

**Goal** – Your goal is to provide information on ONE reproductive technology.

**Role** – You work for the Hospital Authority. You are responsible for producing patient information leaflets.

**Audience** – Individuals who are interested in or considering using the reproductive technology.

**Situation** – Different reproductive technologies are becoming more common, depending on the need of the individuals. Before individuals decide to use these technologies they need to be informed on how it works and the positive and negative impacts associated.

**Product** – You will create a leaflet for a doctor's surgery on ONE reproductive technology / aid.

**Success criteria for assessment** –

- The leaflet should be a maximum of 2 sides of A4.
- The leaflet should include:
  - A brief introduction of the human problem that exists (e.g. overpopulation or underpopulation)
  - Scientific explanation of how the chosen reproductive technology works and will solve the problem.
  - Discussion on positive and negative impacts of this technology
  - Provide specific examples with useful data, statistics, graphs etc.
  - A conclusion including your judgement on how effective the technology would be.
- Write clearly and concisely. Scientific vocabulary should be applied consistently.
- The leaflet needs to include a clear bibliography, with at least five sources cited.

**Student Task Planning Notes:**

## 7B 16a Human reproduction review

- 1 Draw lines to match the words below with the correct meaning.  
The first one has been done for you.

## WORDS

Egg cell

Ejaculation

Fertilisation

Sexual intercourse

Sperm cell

## MEANINGS

Often called 'making love'

A sex cell made in the testes

Semen is released from the penis

The female sex cell

The joining of a sperm cell and an egg cell

· Lines matched: sperm cell -> the male sex cell; egg cell -> the female sex cell; fertilisation -> the nucleus of the sperm cell joining the nucleus of the egg cell; embryo -> the ball of cells that grows into a baby; uterus -> where the embryo grows.

- 2 Which of these sentences are about human reproduction? Tick the correct boxes.

1. The sperm cell and the egg cell join (**fuse**) outside the female.

☒ 2. The sperm cell and the egg cell fuse inside the oviduct.

3. The sperm cell and the egg cell fuse inside the sperm duct.

☒ 4. The fertilised egg cell divides into two.

☒ 5. A ball of cells that will grow into a baby is called an embryo.

☒ 6. The embryo grows inside the uterus.

7. The embryo grows inside the ovary.

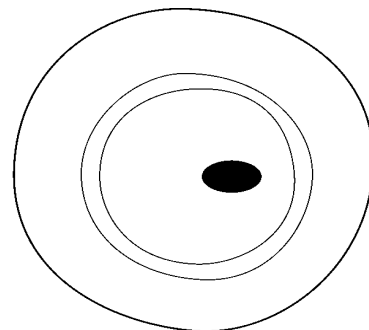
8. The embryo grows outside the female.

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

- 3 The nucleus of the egg cell contains half the instructions about how a new animal will look.  
The nucleus of the sperm cell contains the other half of the instructions.

On the diagrams below label the nucleus in the sperm cell and the nucleus in the egg cell.

· On the diagrams, draw a line to label the dark oval nucleus in the head of the sperm cell and the round nucleus in the middle of the egg cell.



**Questions:** If you have any questions about human reproduction, you may want to write them down below and show this to your teacher.

## 7B 16b Internal and External Reproduction

|                                                                                                                         |                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reptiles' eggs have leathery shells and contain lots of stored food                                                     | <p><b>Internal</b></p> <p>Frogspawn is a food source for other aquatic animals</p> <p>External - lots of eggs are eaten, so many must be laid; the eggs are fertilised in the water outside the frogs' bodies.</p> |
| Birds produce a much smaller number of eggs than fish or birds                                                          | <p><b>Internal</b></p> <p>Amphibians such as frogs lay lots of eggs in water</p> <p>External - frogs lay their eggs in water and the sperm fertilises them outside the female's body.</p>                          |
| <b>Internal</b><br>Birds' eggs have hard shells                                                                         | <p><b>Internal</b></p> <p>An elephant's fertilised egg grows inside the uterus</p>                                                                                                                                 |
| Female fish release eggs in very large numbers into the water. The eggs are then fertilized by the sperm from male fish | <p><b>External</b></p> <p>The sperm from the frog can swim to the eggs and fertilise them</p> <p>External - the sperm swims to the eggs in the water, so fertilisation happens outside the female's body.</p>      |
| Frogs eggs only contain a little stored food                                                                            | <p><b>External</b></p>                                                                                                                                                                                             |

Decide if these statements are talking about internal or external fertilization.

## 7B16c Animal Reproduction Factsheet

The table below gives information about how different animals reproduce and look after their young. First, look at this explanation of what the labels in the boxes mean:

- Type of animal – Whether the animal is an amphibian, bird, fish, mammal or reptile.
- Time for development – The length of time in days for an animal to develop from a fertilised egg cell until it is born or hatches.
- Number of young – The number of offspring that are usually born or hatch after the animals have mated.
- Aftercare – How long the parents look after their offspring.
- Fertilisation – Whether fertilisation is internal or external.
- Development – Whether the offspring develop inside their mothers, like humans (internal), or outside, like fish (external).
- Survival rate – The approximate percentage of offspring who will survive long enough to have offspring of their own.

|                      | <b>Black rat snake</b> | <b>Bullfrog</b> | <b>Cod</b> | <b>Gorilla</b> | <b>Grey squirrel</b> | <b>Grey wolf</b> |
|----------------------|------------------------|-----------------|------------|----------------|----------------------|------------------|
| Type of animal       | Reptile                | Amphibian       | Fish       | Mammal         | Mammal               | Mammal           |
| Time for development | 3 months               | 20 days         | 25 days    | 8.5 months     | 38 days              | 3 months         |
| Number of young      | 20                     | 5000            | 3000       | 1              | 5                    | 5                |
| Aftercare            | None                   | None            | None       | 2.5 years      | 13 weeks             | 1 year           |
| Fertilisation        | Internal               | External        | External   | Internal       | Internal             | Internal         |
| Development          | External               | External        | External   | Internal       | Internal             | Internal         |
| Survival rate        | 50%                    | 1%              | 2%         | 60%            | 20%                  | 40%              |

|                      | <b>Hare</b> | <b>Harvest mouse</b> | <b>King penguin</b> | <b>Human</b> | <b>Lion</b> | <b>Polar bear</b> |
|----------------------|-------------|----------------------|---------------------|--------------|-------------|-------------------|
| Type of animal       | Mammal      | Mammal               | Bird                | Mammal       | Mammal      | Mammal            |
| Time for development | 1.5 months  | 19 days              | 2 months            | 9 months     | 3 months    | 8 months          |
| Number of young      | 2           | 6                    | 2                   | 1            | 3           | 2                 |
| Aftercare            | 1 month     | 16 days              | 1 year              | 16 years     | 2.5 years   | 2.5 years         |
| Fertilisation        | Internal    | Internal             | Internal            | Internal     | Internal    | Internal          |
| Development          | Internal    | Internal             | External            | Internal     | Internal    | Internal          |
| Survival rate        | 30%         | 15%                  | 40%                 | 99%          | 50%         | 50%               |

|                      | <b>Rat</b> | <b>Salmon</b> | <b>Sea turtle</b> | <b>Stickleback</b> | <b>Tiger</b> |
|----------------------|------------|---------------|-------------------|--------------------|--------------|
| Type of animal       | Mammal     | Fish          | Reptile           | Fish               | Mammal       |
| Time for development | 22 days    | 2 months      | 2 months          | 7 days             | 3.5 months   |
| Number of young      | 7          | 200           | 150               | 20                 | 3            |
| Aftercare            | 2 weeks    | None          | None              | 1 week             | 2 years      |
| Fertilisation        | Internal   | External      | Internal          | External           | Internal     |
| Development          | Internal   | External      | External          | External           | Internal     |
| Survival rate        | 30%        | 15%           | 5%                | 30%                | 50%          |

## 7B 16d Animal Reproduction Communication Skills

Choose **2 different types of animals** (amphibians, reptiles, birds, fish, mammals)

Give the name of the animals that you have chosen (e.g. Siberian tiger and giant cobra).

**Research** how each of these animals reproduce.

You may want to consider some of the following;

- Time for development (The length of time between when the egg is fertilised until it is born/hatches)

- Number of young (How many babies/offspring are normally born/hatch).

- Aftercare (How long the parents look after the babies/offspring after they are born/hatch).

- Fertilisation (Does the sperm and the egg meet inside the mother's body or outside?)

- Development (Do the babies/offspring grow inside the mother (like humans) before they are born, or do they grow outside the mother (like fish)).

- Survival rate. How many of the babies/offspring normally survive to become adults?

- Which of these 2 animals do you feel has the most successful method of reproduction? Explain why.

Communicate your research in the form of an extended piece of writing (essay). You may include diagrams if you wish to help explain.

A good scientific essay includes the following:

- A Clear **Title**.

- An **Introduction**. A statement or paragraph that explains what you are writing about and some background information.

- A **discussion** of each of the points that you wish to raise, set out in clear paragraphs. This will make up about 80% of your essay and needs to include some of the following.

  - Good knowledge and understanding of the concepts covered.

  - Good use of scientific words, vocab and language.

  - Illustrations, pictures, diagrams or examples to help explain the points that you are making.

- A **Conclusion**. This is a summary of your findings or is a clear statement of what you found out.

- Bibliography**. Evidence that you have used more than one source for your research (text books, different internet web sites etc)

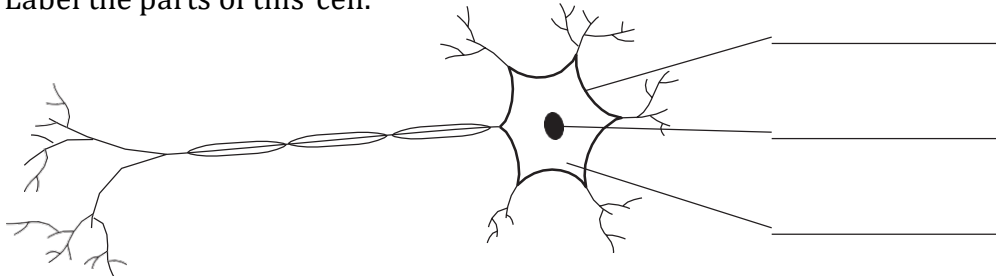
You will be assessing your **communication skills** on this piece of work. Some of the skills that you will need to show on this piece of work are on the 'Communication Skills Rubric'.

*The survival rate is the percentage of offspring that live to become adults. It is high (near 100%) where parents care for few young, and very low where thousands of eggs are laid.*

**Your essay should not be longer than 800 words.**

## 7B 17a Cells, Organs & Organ Systems Revision

1 a Label the parts of this cell.



b Which type of **tissue** does this cell form? \_\_\_\_\_

c Which **organ** might this tissue be found in? \_\_\_\_\_

d Which **organ system** would this organ be found in? \_\_\_\_\_

2 Which organ system:

a carries oxygen and food around the body? \_\_\_\_\_

b breaks down food? \_\_\_\_\_ The digestive system.

c takes oxygen out of the air? \_\_\_\_\_ The respiratory (breathing) system.

d carries messages around the body? \_\_\_\_\_ The nervous system.

3 In the following sentences, *one* organ is not correct. Put a cross through the wrong organ and write the name of the correct one in the space.

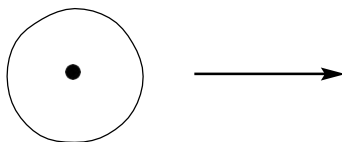
a The breathing system contains the lungs and intestines. \_\_\_\_\_

b The nervous system contains the brain and the little toe. \_\_\_\_\_ Cross out 'liver'; the correct organ is the heart - the circulatory system contains the heart and blood vessels.

c The circulatory system contains the liver and blood vessels. \_\_\_\_\_

4 The drawing shows an animal cell.

a Draw two or three more diagrams to show the steps taken when this cell splits into two.



b What is the name of this process? \_\_\_\_\_

c What are the two new cells called? \_\_\_\_\_ Cell division.

Daughter cells.

## 7B 17b Cells - Quick Quiz

On your answer sheet, write in or circle the correct letter for each question.

### 7Aa

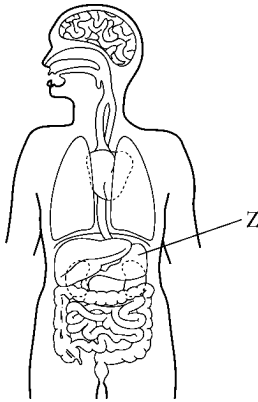
1 An organ is:

- A another name for a living thing.
- B a large part of an animal or plant that does a very important job.
- C the part of the body that makes sounds.
- D another name for a body.

2 Organs contain:

- A hearts.
- B tissues
- C glass.
- D newspaper.

3 In the picture, what is part 'Z'?



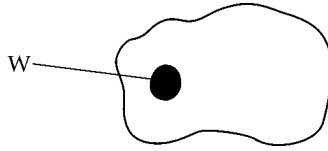
- A stomach
- B hand
- C kidney
- D lung

4 What are the kidneys used for?

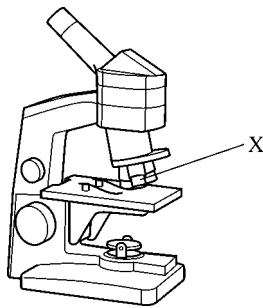
- A To get rid of waste from the body.
- B To help us to breathe.
- C To help us digest food.
- D To make blood.

## 7Ab

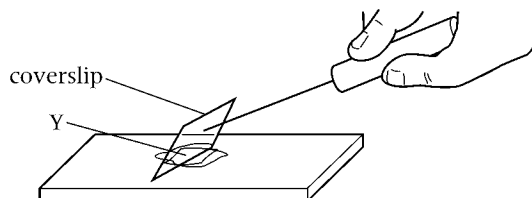
- 1 On this drawing of an animal cell, what is part labelled 'W'?



- A cell surface membrane  
B chloroplast  
C **nucleus**  
D heart
- 2 On the drawing of the animal cell, what does part 'W' do?  
A makes energy      B makes food  
C **controls the cell**      D holds the cell together
- 3 On the drawing of a microscope, what is the name of the part labelled 'X'?



- A stage  
B focusing wheel  
C **objective lens**  
D base
- 4 Look at the diagram of a slide being made. The object you want to look at ('Y') is called the:



- A **specimen**      B kingdom.  
C stage.      D focus.

**7Ac**

- 1 A tissue is:
- A a collection of organs helping each other.
  - B another name for an organ.
  - C a group of cells which are all different, all doing different jobs.
  - D a group of cells which are the same, all doing the same job.
- 2 A nerve cell has to carry messages around the body quickly. To help it do this it is:
- A short and square.
  - B very small.
  - C very long
  - D able to move around the body.
- 3 A root hair cell is adapted to its job because the root hair:
- A is hook-shaped to hold the plant in the ground.
  - B gives the cell a large surface area to help it absorb water.
  - C has many chloroplasts to help it make food.
  - D can move to help the plant move from place to place.

## 7B2 17c Reproduction Summary Sheets

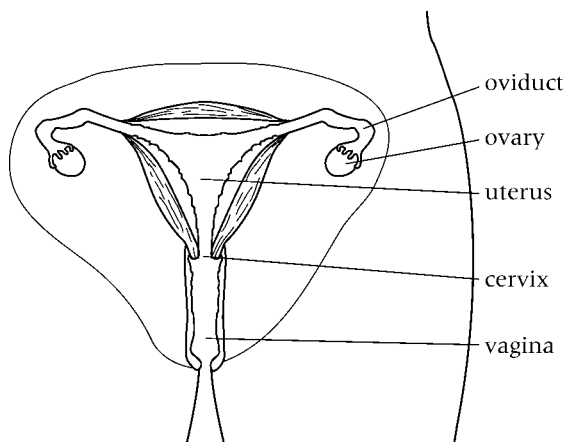
### Reproduction

**Reproduction** produces new living things (**offspring**). In **sexual reproduction** the nucleus in a male **sex cell** joins (**fuses**) with the nucleus in a female sex cell. This is called **fertilisation** and produces a **fertilised egg cell**. When fertilisation happens outside an animal's body it is called **external fertilisation**. Animals that use this method produce a lot of eggs since some will be eaten by other animals. Humans use **internal fertilisation**.

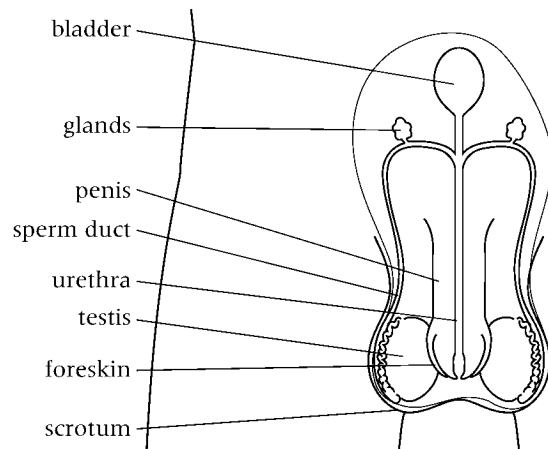
The fertilised egg cell grows into an **embryo** and the embryo eventually becomes a new living thing. Sexual reproduction needs two **parents**. The offspring from sexual reproduction are different from the parents; they are new **varieties**.

### *The human reproductive systems*

Humans have **reproductive organs** so that they can reproduce. The ovaries and testes produce sex cells.



Female – ovaries are where the female sex cells (**egg cells**) are produced.



Male – testes are where the male sex cells (**sperm cells**) are produced.

### *Puberty and adolescence*

The reproductive organs get bigger and start to make sex cells at **puberty**. This is a time when major physical changes occur in our bodies. These changes are caused by **sex hormones**.

#### Changes in boys

- voice deepens ('breaks')
- shoulders get wider
- hair grows under arms, on face and on chest
- pubic hair grows
- testes and penis get bigger
- testes start to make sperm cells
- body smell increases

#### Changes in girls

- underarm hair grows
- breasts develop
- ovaries start to release egg cells
- hips get wider
- pubic hair grows
- body smell increases

**Adolescence** is the time when puberty is occurring and emotional changes happen.

It starts between the ages of 10–15 and ends at about 18. The changes start sooner in girls. After puberty, men produce sperm cells for the rest of their lives. Women stop releasing egg cells at the age of 45–55. This is called the **menopause**.

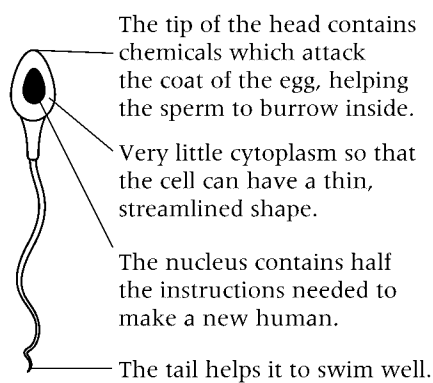
## The menstrual cycle

The **menstrual cycle** starts with **menstruation** (the loss of the uterus lining and some blood through the vagina). It takes 28–32 days for each cycle. About 14 days after menstruation starts, an egg cell is released from an ovary. This is called **ovulation**. If the egg cell is not fertilised, the uterus lining starts to break down and the cycle starts again.

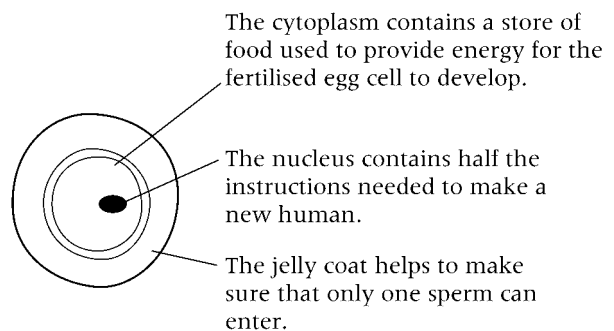
## Sex

The sperm cells enter the **vagina** during **sexual intercourse**. **Semen** (sperm cells mixed with special liquids from the **glands**) is forced out of the penis and into the top of the vagina. This is called **ejaculation**. The semen is moved into the top of the uterus and the sperm cells can swim down the oviducts.

Egg cells and sperm cells are **adapted** to their **functions**.



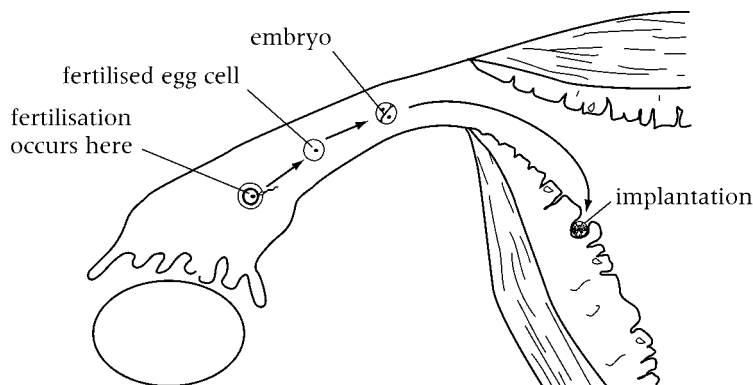
A sperm cell is much smaller than an egg cell.



The egg cell is about the size of a full stop.

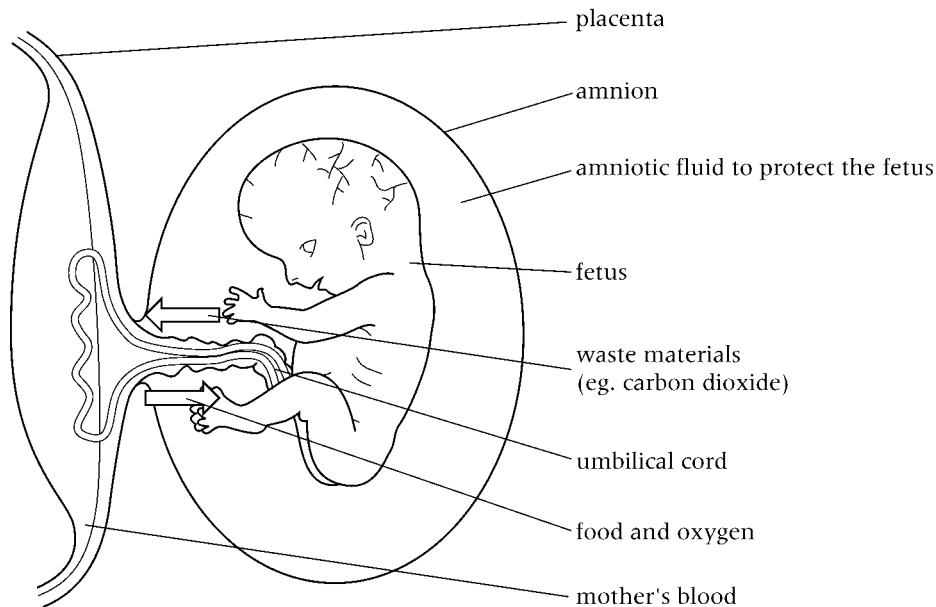
## Pregnancy

If the egg cell meets a sperm cell in an oviduct fertilisation can occur. The fertilised egg cell divides to form a ball of cells (an **embryo**). The embryo travels to the uterus where it sinks into the soft lining (**implantation**). The woman is now **pregnant**. Once it has developed all its organs (after about 10 weeks) it is called a **fetus**. It takes about 40 weeks (9 months) for a fertilised egg cell to grow into a baby ready to be born. This time is called the **gestation period**.



The fertilised egg cells of many animals grow and develop outside their parents. This is called **external development**. Humans use **internal development** and produce less offspring than animals using external development since the growing embryos are protected inside the mother.

While inside the uterus, the fetus is supplied with oxygen and food by the **placenta**. The placenta also gets rid of waste (especially carbon dioxide) from the fetus. The **cord** (or **umbilical cord**) connects the fetus to the placenta.



If a mother smokes, drinks too much alcohol or takes drugs while pregnant she might damage the baby. The baby might be **premature**.

### *Birth*

When the baby is ready to be born, the uterus starts **contractions** and the woman goes into **labour**. The muscles of the cervix relax. The baby is pushed out head first through the cervix and the vagina. After birth, the baby starts to breathe and the cord is cut. The scar left behind is the **navel**. After this the placenta is pushed out of the uterus. This is the **afterbirth**. The baby is fed on milk, often from the mother's breasts which contain **mammary glands** that produce milk. The milk contains **antibodies** which help destroy microbes that might cause a disease in the baby.

## 7B17d                      Reproduction - Quick Quiz

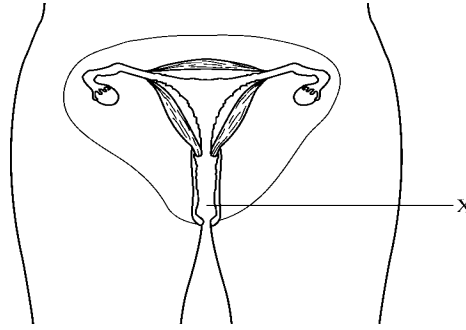
On your answer sheet, write in or circle the correct letter for each question.

### 7Ba

- 1 Fertilisation is when an egg cell joins with:  
A another egg cell.  
B a sperm cell.  
C a body cell.  
D root hair cells.
- 2 Salmon are fish which produce thousands of egg cells when they breed. One reason for this is that:  
A many of the egg cells get eaten by other animals.  
B the female salmon need to sort through their egg cells to find the biggest to use for fertilisation.  
C salmon breed on dry land and the mass of egg cells stops the ones in the middle drying out.  
D there is only one real egg cell, the others just help it get out of the female.
- 3 Before it is born, a calf is inside a cow's:  
A stomach.  
B uterus.  
C oviduct.  
D large intestine.
- 4 Tadpoles develop in water, not inside frogs. A calf develops inside a cow. One reason offspring develop inside their mothers is:  
A the mother finds it easier to walk with her offspring inside her.  
B the mother does not need to feed her offspring at first.  
C the offspring are prevented from moving and so they cannot run away.  
D the offspring are protected from other animals.

## 7Bb

1 What is the part labelled 'X' in the diagram?



A bladder

B cervix

C uterus

D vagina

2 One way that a sperm cell is adapted to its job is that:

A it dies quite quickly.

B it is hollow.

C it is surrounded by a liquid.

D it has a tail to allow it to swim.

3 An egg cell is much larger than a sperm cell because:

A it needs to be able to hide from sperm cells.

B it needs to be big to allow it to have a streamlined shape.

C it contains a store of food that can be used if it is fertilised.

D the nucleus in an egg cell is much larger than in a sperm cell.

4 Menopause is:

A the time when physical changes happen in the body.

B the time when men need to sleep.

C the time when a woman stops producing egg cells.

D the time when animals find a mate.

**7Bc**

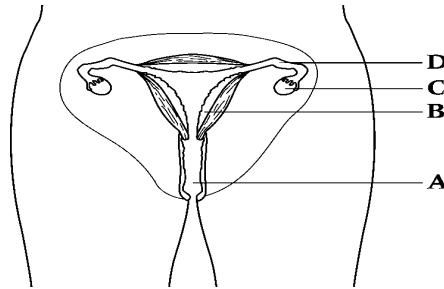
1 Fertilisation occurs in:

- A an ovary.                      B an oviduct  
C the scrotum.                D the penis.

2 A fertilised egg cell develops into an embryo by:

- A cell division                B puberty.  
C ovulation.                    D fertilisation.

3 Look at the diagram. Which letter shows where implantation happens?



4 The nucleus inside every sex cell contains information. How much of the information needed to make a new life does an egg cell carry?

- A 100%                      B 75%  
C 50%                      D 25%

The letter labelling the lining of the uterus - the embryo sinks into the soft lining (implantation).

**7Bd**

1 The loss of blood and the uterus lining in the menstrual cycle is called:

- A menstruation  
B ovulation.  
C fertilisation.  
D urination.

2 A developing baby is protected by:

- A a bag of fluid (amnion)  
B the vagina.  
C the ovary wall.  
D a cord.

3 The developing baby gets food and oxygen from:

- A the mother's stomach.  
B the placenta  
C the ovary.  
D the egg cell.

4 A premature baby is a baby that is:

- A born very late.  
B very active.  
C born if the parents are not married.  
D born small and very early.

**7Be**

1 Pregnancy lasts for:

- A nine days.                      B nine weeks.  
C nine months                      D ninety days.

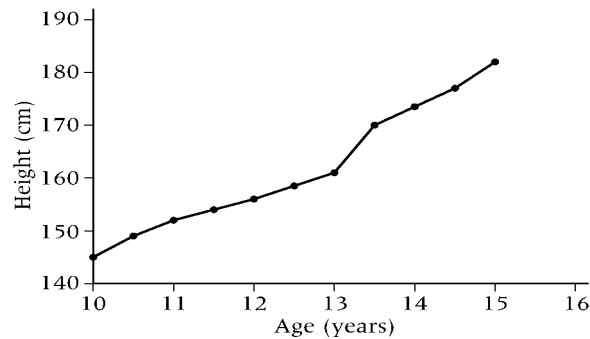
2 Mammary glands produce:

- A mashed up food.  
B milk  
C amniotic fluid.  
D sugar.

3 Puberty is caused by:

- A sex cells.  
B eating too much.  
C pubic hair starting to grow.  
D sex hormones

4 This graph shows the height of a boy between the ages of ten and fifteen.



Between which ages was he growing the fastest?

- A 10 and 11                      B 11 and 13  
C 13 and  $13\frac{1}{2}$                       D  $14\frac{1}{2}$  and 15

## Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p2: Cells have become adapted to one job: a sperm has a tail to swim, a nerve cell is long to carry impulses, a red blood cell has no nucleus to carry more oxygen and is a flat disc.

p2: Understanding fertilisation lets scientists use IVF and fertility hormones to raise fertility, and contraceptives or sterilisation to lower the chance of pregnancy.

p2: Consider how safe and effective the therapy is, the source of the stem cells (embryo, adult or iPS), risk of rejection and infection, side effects, cost, and ethical or religious views.

p2: Success rate, number of embryos implanted (multiple births), health risks to mother and baby, cost, what happens to unused embryos, and the ethical or religious beliefs of the couple.

p27: Draw the cheek cells at low and high magnification using a sharp pencil and continuous lines; label the parts you can identify, add a title and write the magnification.

p39: Systems such as the breathing, circulatory and digestive systems keep an individual alive; the reproductive system continues the species.

p62: The cut shows the thin skin, a layer of pale fat, pink muscle and the hard white bone, with tendons joining muscle to bone.

p63: Other tissues include bone (hard, white), blood (red liquid in vessels), nerves (thin white strings carrying signals) and the skin layers.

p108: A tissue - a group of similar cells all doing the same job (for example muscle tissue from muscle cells).

p108: An organ - a group of different tissues working together, for example the stomach (or a leaf in a plant).

p108: An organ system - a group of organs working together, for example the digestive system (or the shoot system in a plant).

p108: Cross out 'intestines'; the correct organ is the windpipe (trachea) - the intestines belong to the digestive system.

p108: Cross out 'the little toe'; the correct organ is the spinal cord - the nervous system contains the brain and spinal cord.

p108: Draw the cell dividing into two: the nucleus divides, then the cell splits into two identical cells (two more diagrams showing the stages).